DAMEN SHIPYARDS

DRAWINGLIST FOR YARDNo. 3030

Box 1

1 -	3029-000-00-001	B	General arrangement
2 -	3029-100-00-100	-	Dockingplan
3 -	3029-110-00-030	B	Tank arrangement
4 -	3029-110-00-200	F	Deck construction
5 -	3029-110-00-201	F	Bottom construction and cooling channels
6 -	3029-110-00-250	F	Long sections 250/750 1250 and 1750 from C.L.
7 -	3029-110-00-251	F	Long sections 2250/2750 3250 and 3750 from C.L.
8 -	3029-110-00-250	E	Cross sections aftship incl. bulkh. 7
9 -	3029-110-00-351	E	Cross sections Frame 8-9-10-11-12 and 13
10 -	3029-110-00-500	E	Bulwark and railing
11 -	3029-120-00-050	E	Construction superstructure above maindeck
12 -	3029-120-00-051	D	Construction superstructure above bridgedeck with funnel
13 -	3029-155-00-001	F	Arrangement window
14 -	3029-152-00-001	A	Pushbow
15 -	3029-163-00-001	A	Plan of anodes
16 -	3029-210-00-001	B	Engine arrangement
17 -	3029-210-00-700	B	Arrangement rudder-nozzle and sterntube
18. -	213-2110-0051	-	Composition propeller shaft ø 110 mm
19. -	3029-222-00-001	-	Arrangement steeringgear
20. -	3029-310-00-001	C	Bilge, deckwash and internal firefighting system
21 -	3029-321-00-001	-	Fuel oil service system
22 -	3029-322-00-001	D	Fuel oil cargo/filling system
23 -	3029-330-00-001	C	Coolingwater system

DAMEN SHIPYARDS

DRAWINGLIST FOR YARDNo. 3030

Box 1

24 -	3029-340-00-001	B	Freshwater cargo and sanitairy system
25 -	3029-350-00-001	B	Filling, sounding and deaeration system
26 -	000-350-01-001	S	Code list filling, sounding and deaeration scheme
27 -	3029-360-00-001	-	Lubrication system
28 -	3029-500-00-001	D	Arrangement maindeck
29 -	3030-570-00-001	-	Safety plan
30 -	3029-730-00-001	C	Topview engine room arrangement
31 -	3029-730-00-002	B	Cross sections engine room arrangement
32 -	3029-750-00-001	-	Arrangement accommodation/stores

DAMEN SHIPYARDS

INSTRUCTION MANUALS FOR YARDNo. 3030

Box 1

- Owner's manual for Caterpillar 3408 marine engine (SEBU 5454-02)
- Parts manual for Caterpillar 3408 marine engine (SEBP 1490-03)
- Drawing for Caterpillar 3408 drawingno. 12745-01
- Maintenance management schedules for Caterpillar marine engine
- Wiring diagram for Caterpillar 3408

DAMEN SHIPYARDS

INSTRUCTION MANUALS FOR YARDNo. 3030

Box 2

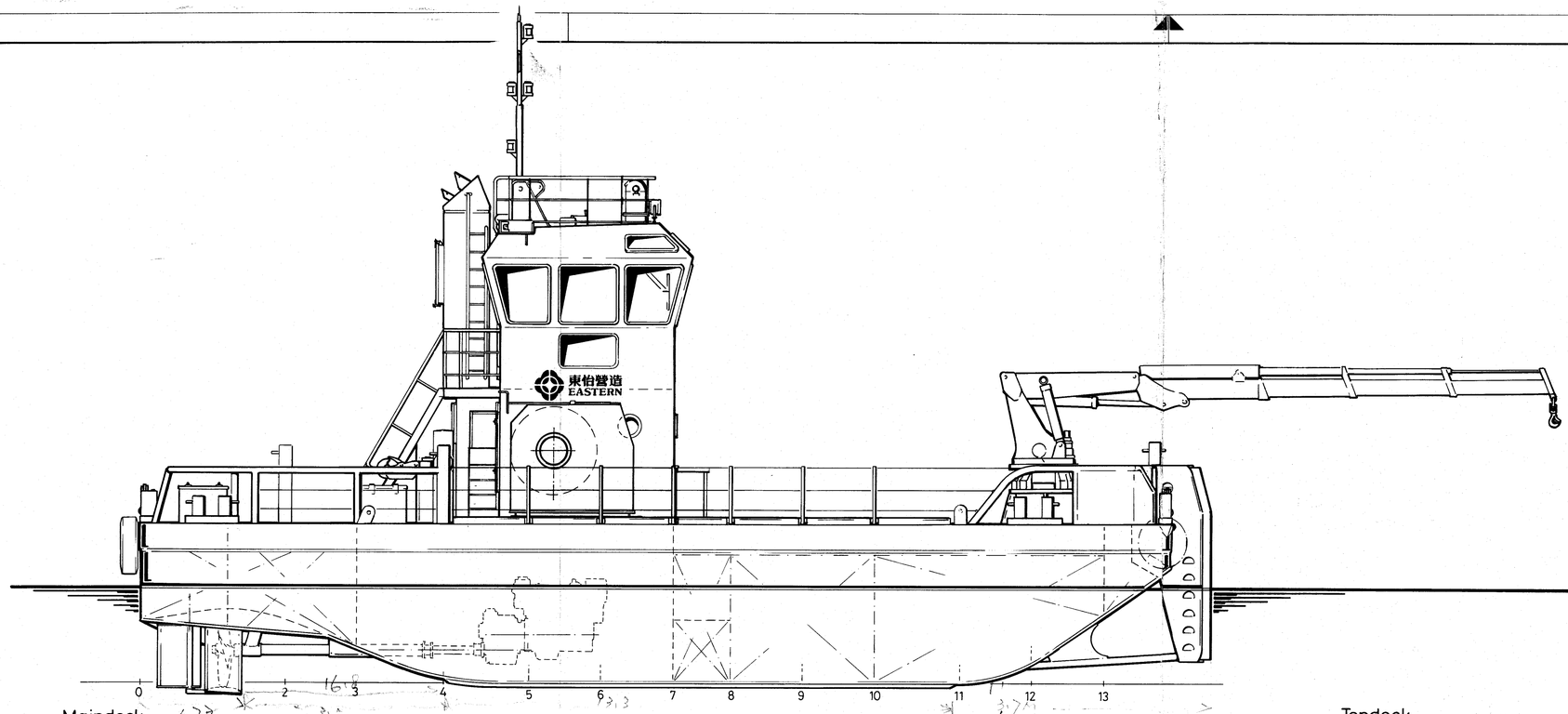
1. Instruction manual for Reintjes reverse reduction gear type WAF 262
2. Instruction for Sperry steeringgear installation
3. Instruction for de Waal propellershaft seals
4. Operator's handbook for lister petter type CD 4
5. Part book for lister petter type CD 4
6. Operation, service & maintenance manual for Stamford generator type UCM 224
7. Wiring diagram for lister petter generator set drawingno. 93038
8. Service manual for Trans-Hydro hydraulic system for crane and towing winch
9. Instruction and parts list for Sihi-pumps
   - type AKHK 5101 AN 031.01.0
   - type AKHK 5101 AN 081.22.0
   - type VW 20 S3 A.C.
   - type AOHA 3101 AN 135.01.0
10. Installation and maintenance instructions for Speck hydrophore pump type PM 15
11. Instruction for Harwood water separator
12. Instruction for Observator magnetic compass
13. Instruction for fiam sonabel signalhorn
14. Operating manual for skylla battery charger
15. List of inventory
16. Operator's manual for searchlight type HGS 351
17. EXXON marine lubrication chart
18. Instruction for Viking inflatable liferaft
19. Operator's manual for Furuno VHF radiotelephone model FM 7000
20. Operator's guide for Furuno VHF model FM 7000
21. Instruction manual for towing winch
22. Operator's handbook for Heila crane type 55/3S
23. Spare parts list for Heila crane type 55/3S

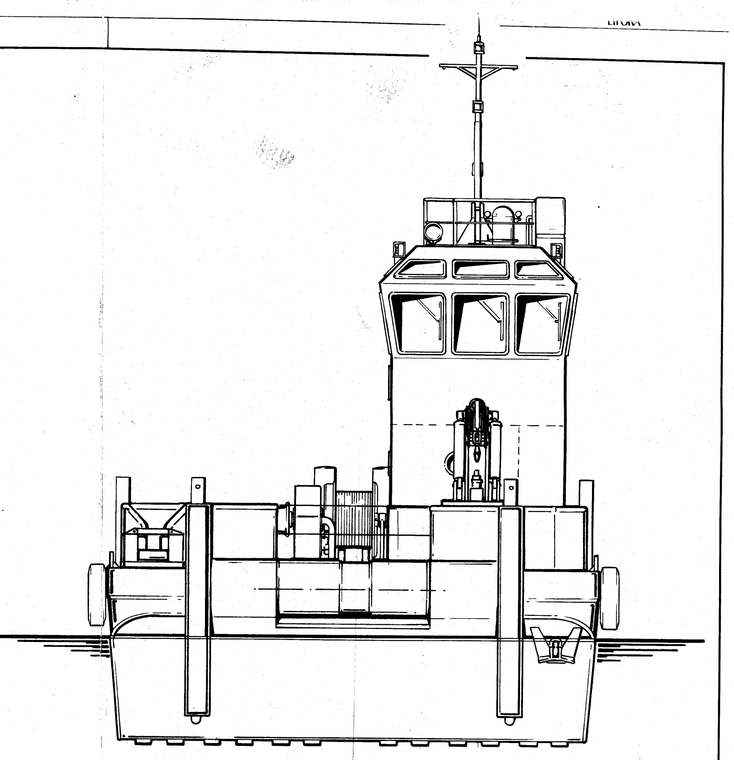
DAMEN SHIPYARDS

INSTRUCTION MANUALS FOR YARDNo. 3030

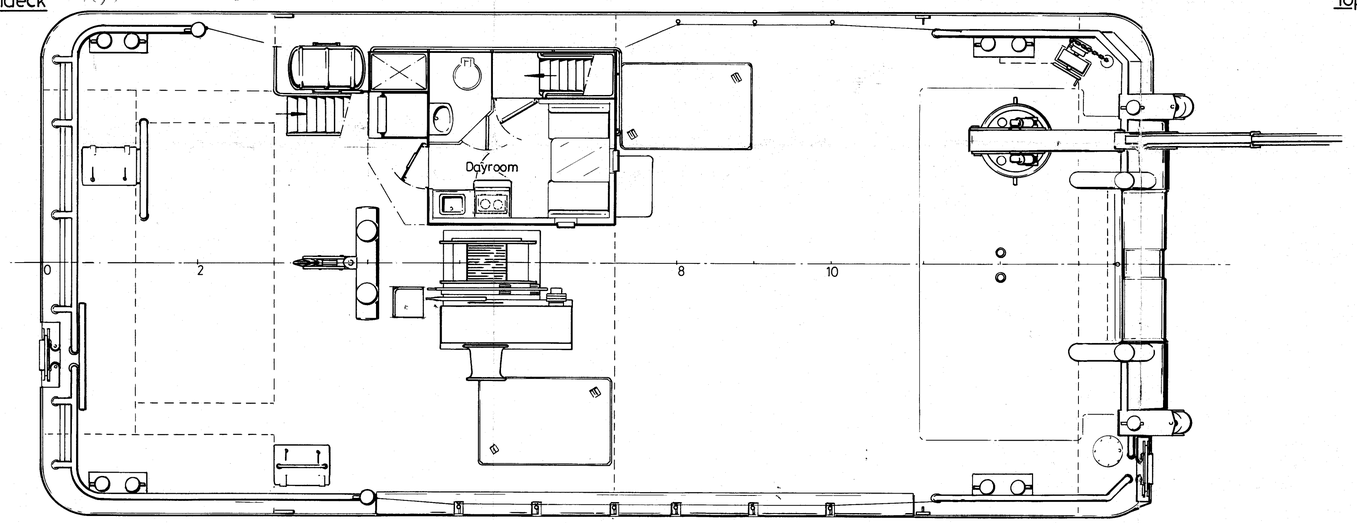
Box 2

24 Part list for Henderson MK V handpump
25 Instruction for Kobelt control system type 2048
26 - 1 Instruction for Mampaey towinghook type 8/15T
27 - 1 Electrical diagrams with documentation Elco
28 - 1 Spare part list Damen parts
29 - 1 Lubrication chart mobil

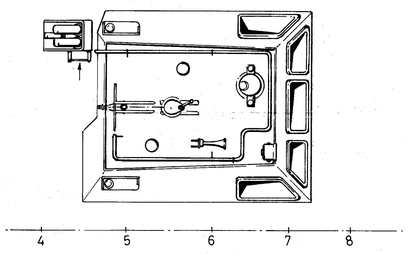




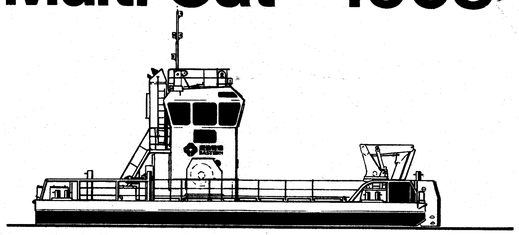
Maindeck



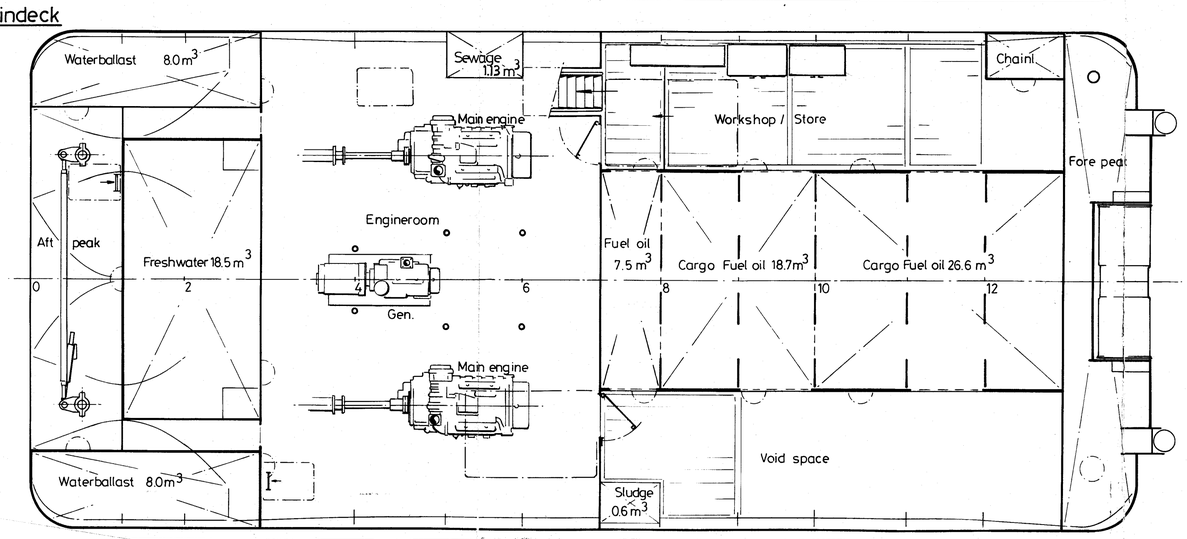
Topdeck



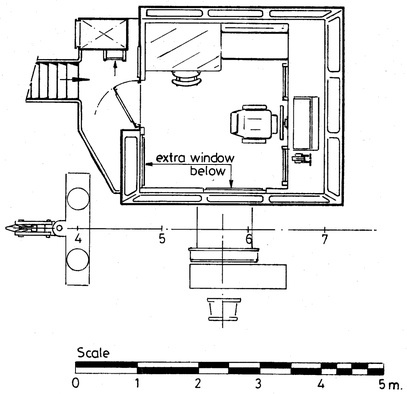
Damen Multi Cat 1908



Below Maindeck



Bridgedeck



Scale
0 1 2 3 4 5 m.

Principal dimensions:

Length o.a.	**18.70 m**
Beam o.a.	**8.06 m**
Depth at sides (at half length)	**2.75 m**
Draught aft (approx.)	**1.70 m**

inspected	date

rev.	date
A	10-01-94
B	28-01-94

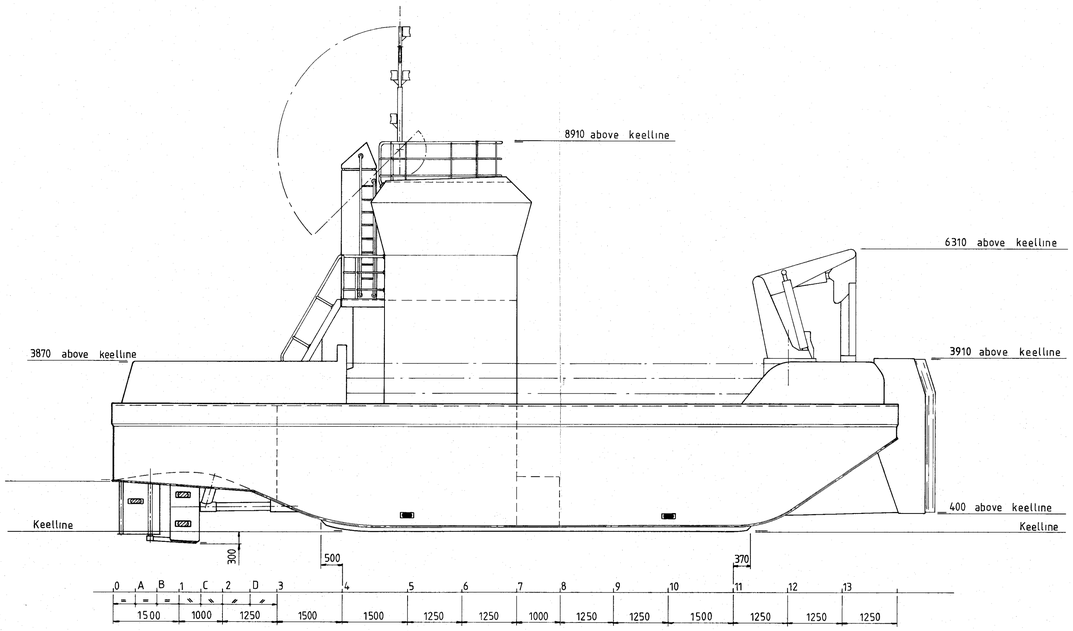
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem Holland
Tel.: 31.1830.39911
Telex: 25335 Damen NL
Telefax: 31.1830.32189

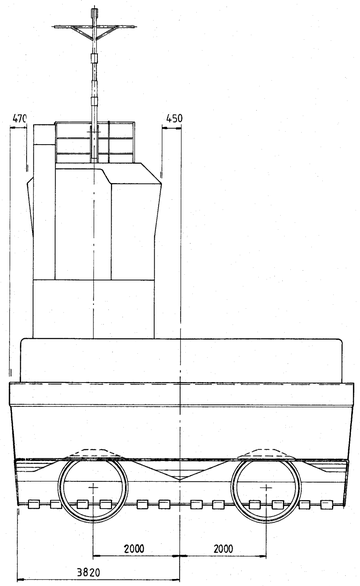
5-1-01

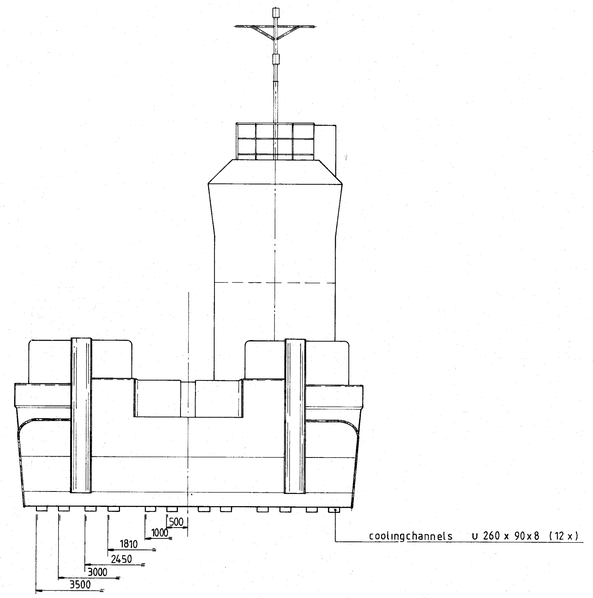
General arrangement

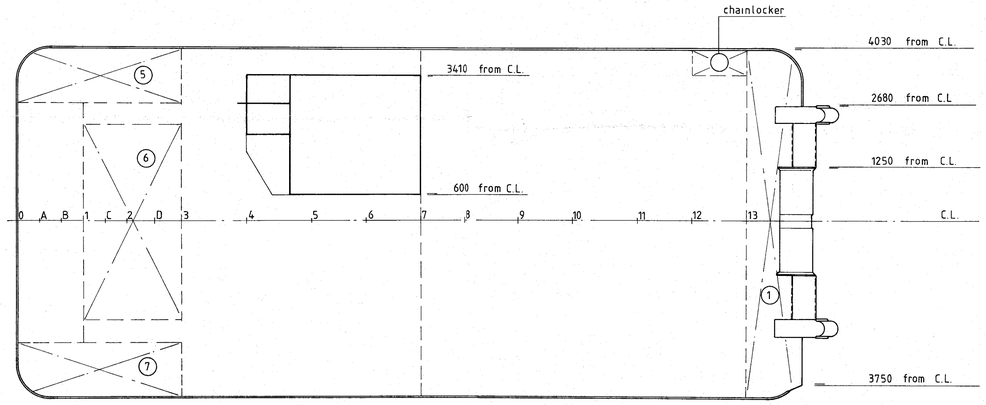
date 31-08-'93	drawn L. de J.	checked	scale 1:50
ident. to drwg. no.		derived fr. drwg. no. 185-0001/P.22.519A	
yard no. 3029-3030		drwg. no. 3029-000-00-001	

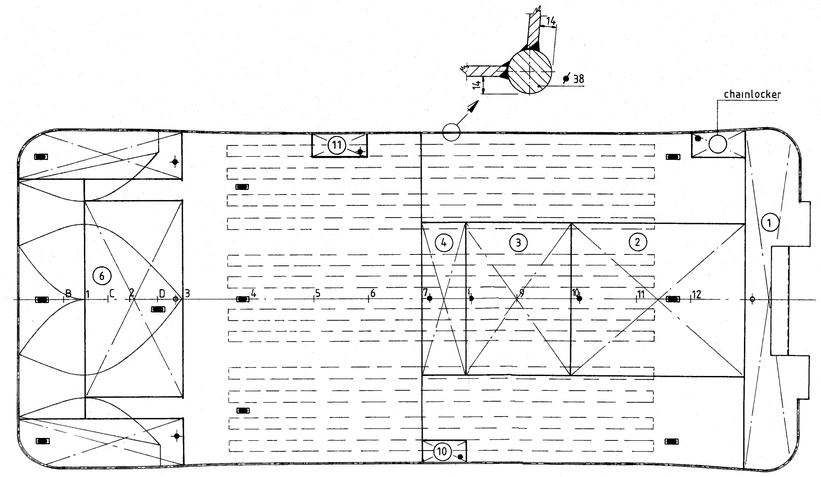
COPYRIGHT BY DAMEN SHIPYARDS











Anchor equipment :

1 anchor of 105 kg
1 spare anchor of 60 kg
with 190 m chain shortlink 16 mm Q2

Alum. anode AD 73 (14 x 5kg) welded on doubble plate.

Alum. anode AD 78 (12 X 2kg) welded on doubble plate

Plug 1 1/4' (square) freshwater

Plug 1 1/2' (hexagon) other contents

Principal dimensions

Length o.a.	:	18.70 m
Length mld.	:	18.00 m
Breath o.a.	:	8.60 m
Breath mld.	:	8.00 m
Depth mld.	:	2.75 m

Ref. drawings

Tank arr.	3029-110-00-030
Draugth marks	3029-159-00-001
Paint line	3029-161-00-001
Plan of anodes	3029-163-00-001

Tank no.	Contents
1	waterballast
2	cargo fuel oil
3	cargo fuel oil
4	fuel oil
5	waterballast
6	freshwater
7	waterballast
	void
	void
10	sludge
11	sewage

rev.	date	name	revision subject

inspected	date

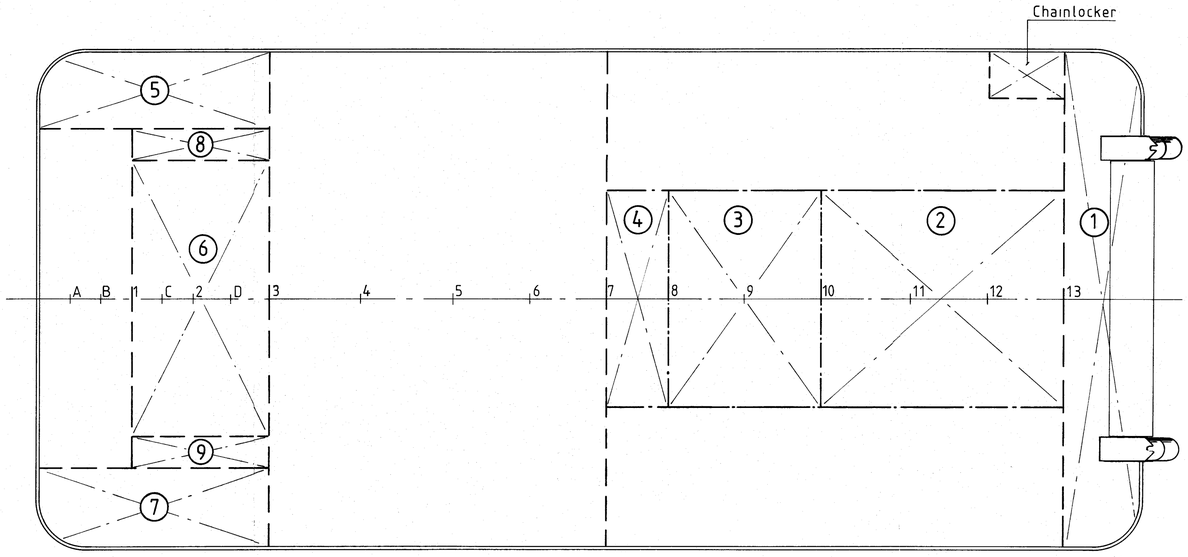
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
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Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

DOCKINGPLAN

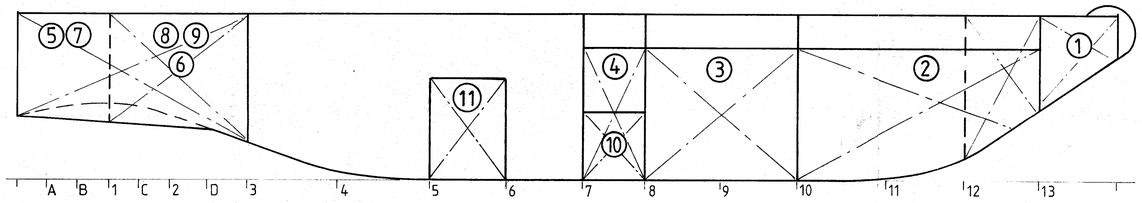
drawn	S.v.d. Linden	date 06-01-'94	scale 1:50
checked		dlont. to dwg. no.	
yard no.	3029/30	derived fr. dwg. no.	
COPYRIGHT BY DAMEN SHIPYARDS		dwg. no.	3029-100-00-100

0 5 10cm

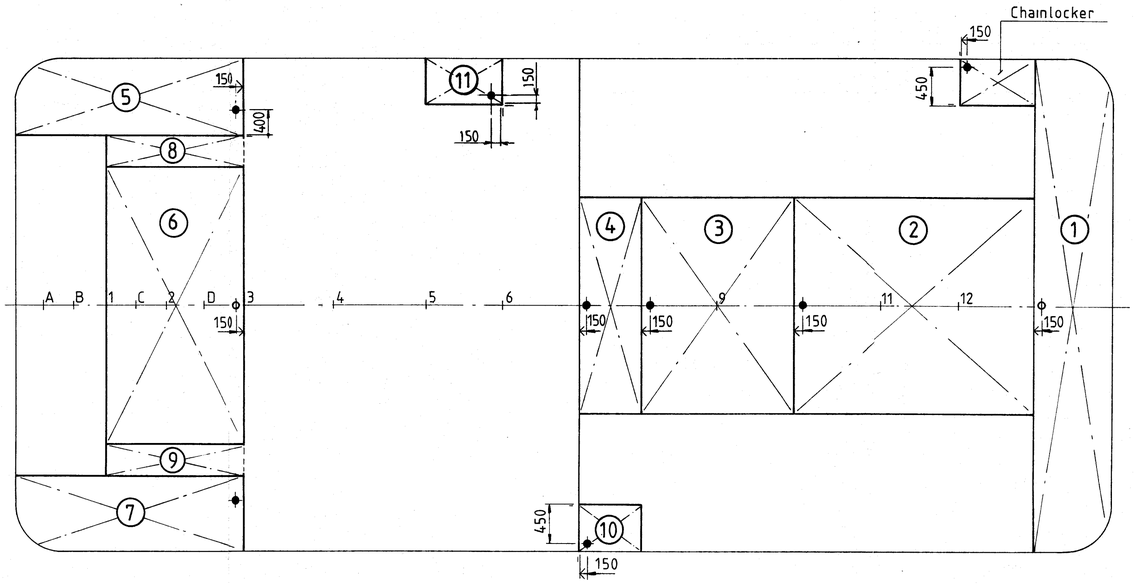
TOP VIEW



SIDE VIEW



BOTTOM VIEW



TANK	CONTENTS	VOLUME M^3
1	Waterballast	± 11.5
2	Cargo fuel oil	± 26.6
3	Cargo fuel oil	± 18.7
4	Fuel oil	± 7.5
5	Waterballast	± 8.0
6	Freshwater	± 18.5
7	Waterballast	± 8.0
8	Void	—
9	Void	—
10	Sludge	± 0.6
11	Sewage	± 1.13

o = 1¼" Drainplug art. 7412.0003 (Fresh-)water

● = 1½" Drainplug art. 7412.0004 Other spaces

rev.	date	name	revision subject
B	18-11-'93	F Zillig	Tank 11 added
A	06-04-'93	S.v.d.L	Remarks LLOYDS

inspected	date
LLOYDS	06-04-'93

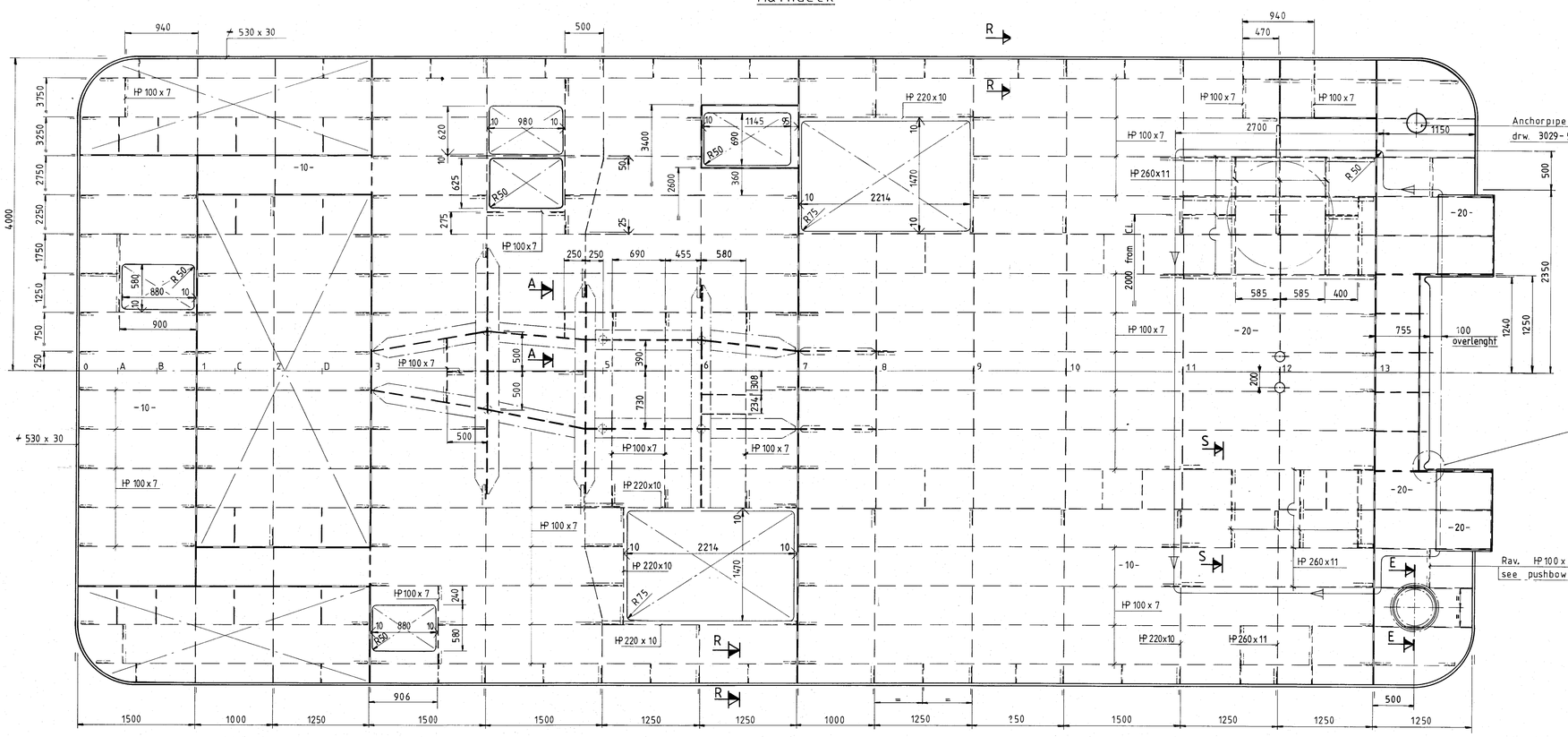
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

TANK ARRANGEMENT

drawn	S. v.d. Linden	date 10 - 03 - 93	scale 1 : 50
checked		ident. to drwg. no.	
yard no.	3029 / 30	derived fr. drwg. no.	
COPYRIGHT BY DAMEN SHIPYARDS		drwg. no. 3029 - 110 - 00 - 030	

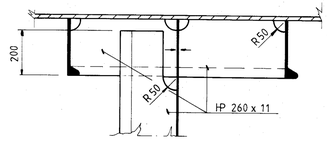
0cm 5 10cm

Maindeck

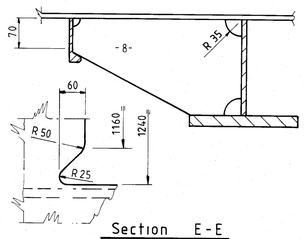


SECTION S-S

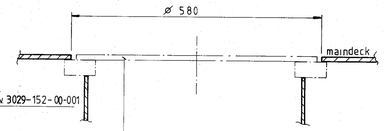
3 x on SB / 3 x on P.S.



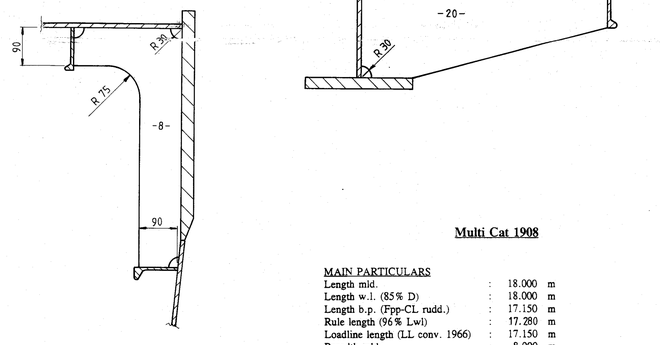
SECTION A-A



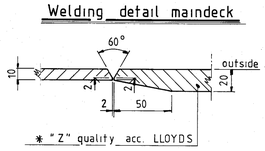
Section E-E



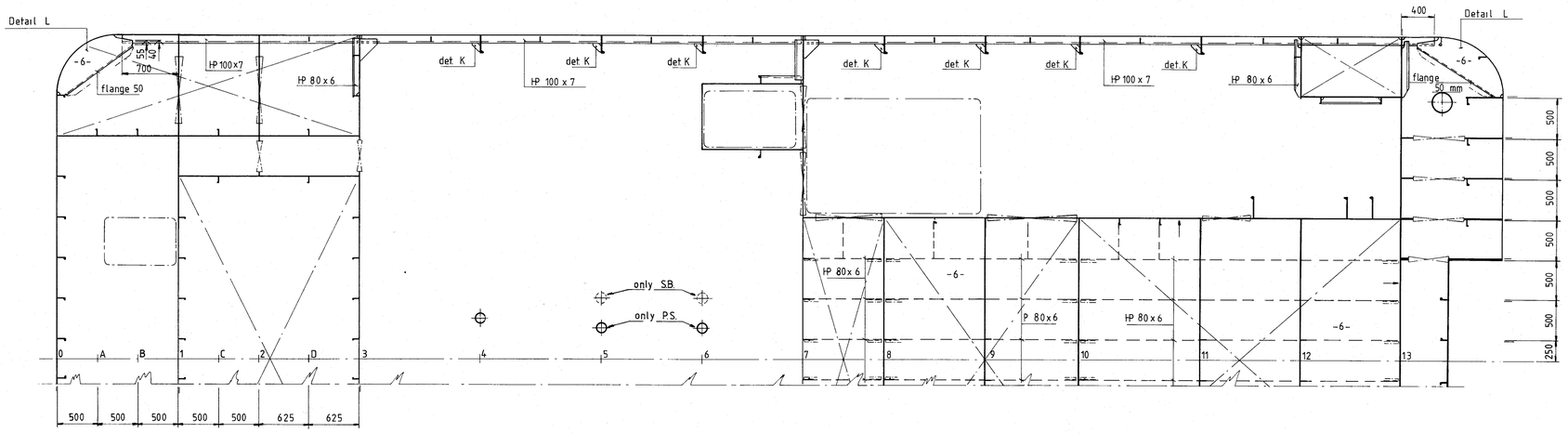
SECTION R-R



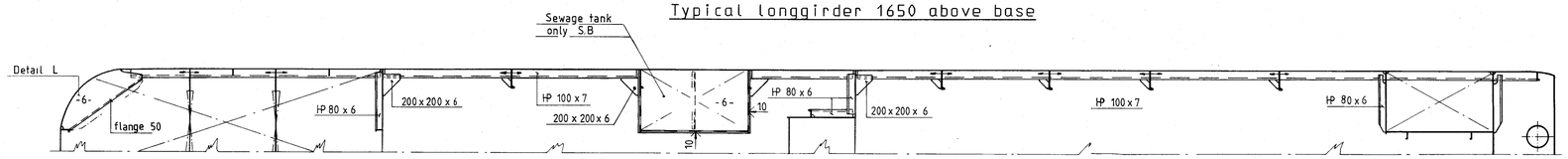
Welding detail maindeck



Tanktop 2200 above base



Typical longgirder 1650 above base



Multi Cat 1908

MAIN PARTICULARS

Length mld.	:	18.000	m
Length w.l. (85% D)	:	18.000	m
Length b.p. (Fpp-CL rudd.)	:	17.150	m
Rule length (96% Lwl)	:	17.280	m
Loadline length (LL conv. 1966)	:	17.150	m
Breadth mld.	:	8.000	m
Depth mld.	:	2.750	m
Draught (max.)	:	± 1.75	m (to hydr. base)
Draught (scantlings) (0.85 D)	:	2.334	m
Frame spacing (long.)	:	500	mm
Bollard pull (max.)	:	± 15	t
Speed (max.)	:	± 9.5	kn
Engine power (max.)	:	2x350	BkW
RPM	:	1800	RPM

LLOYD'S CLASSIFICATION
+ 100 A Specified Coastal Service

VERITAS CLASSIFICATION
I 3/3 (E) + Pontoon Coastal Waters

Notes:
- Deckload fr. 7-14 : 5 T/m² (Hatches 3 T/m²)
- Total deckload : 60 T

rev.	date	name	revision subject
F	19-11-93	F. Zilig	Frame 12
E	11-11-93	S.v.d.L.	Frame 5-6 / Sewage tank between fr 5 / 6
D	01-11-93	S.v.d.L.	Rav. cranefoundation / frame 13 (1650)
C	25-10-93	S.v.d.L.	Remarks B.V.
B	13-10-93	S.v.d.L.	Forepeak / vent. e.r. / frame 3 / flush hatch
A	23-04-93	S.v.d.L.	Remarks LLOYDS

inspected	date
LLOYDS	23-04-93
B.V.	25-10-93

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Telefax: 31.1830.32189

Deck construction

drawn: S.v.d.Linden	date: 23-02-93	scale: 1:25	
checked:	ident. to dwg. no.:		
yard no.: 3029/30	derived fr. dwg. no.:		

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dwg. no.: 3029-110-00-200

1350 above base

Gratings
drw 3029 - 510 - 00 - 001

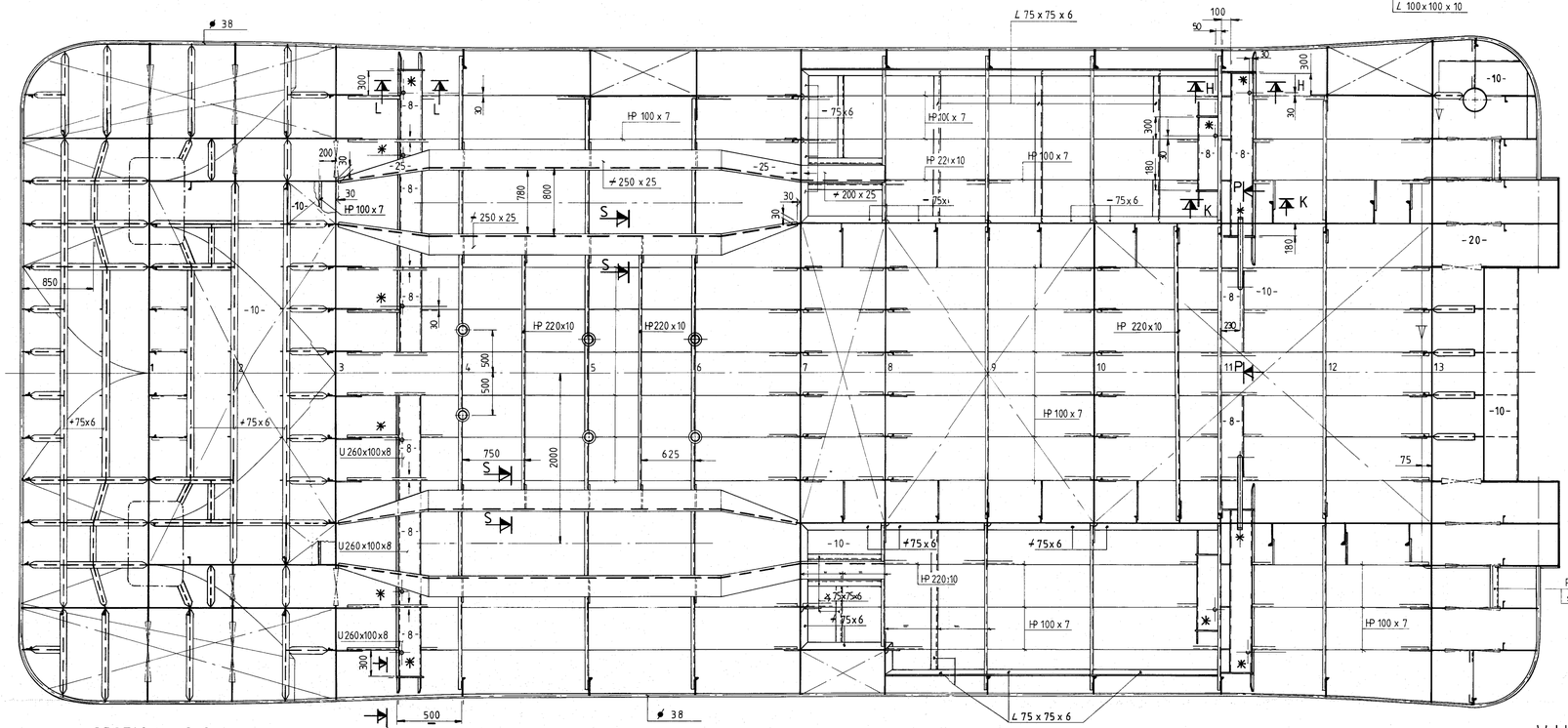
12 13

+ 80 x 10 + 80 x 10

L 100 x 100 x 10

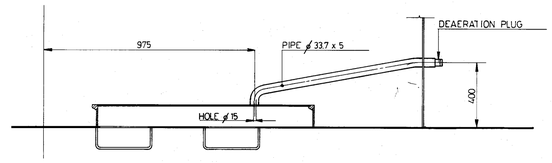
BOTTOM

BOTTOMPLATING 10 mm

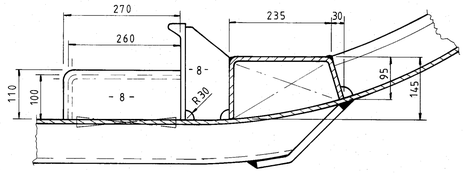


SECTION P_P

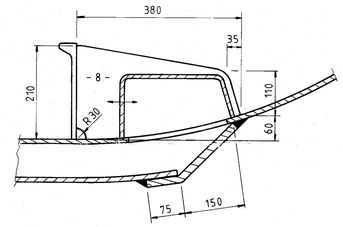
PS DRAWN, S.B OPPOSITE HAND



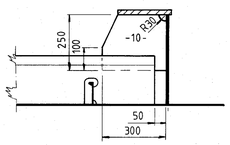
SECTION K-K

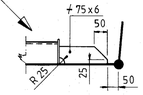


SECTION H-H

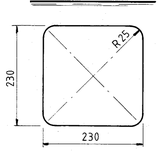


SECTION S-S

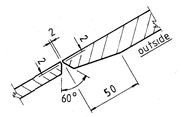




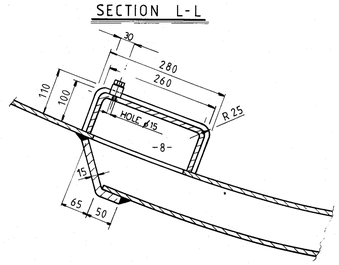
Hole in bottom



Welding detail bottom

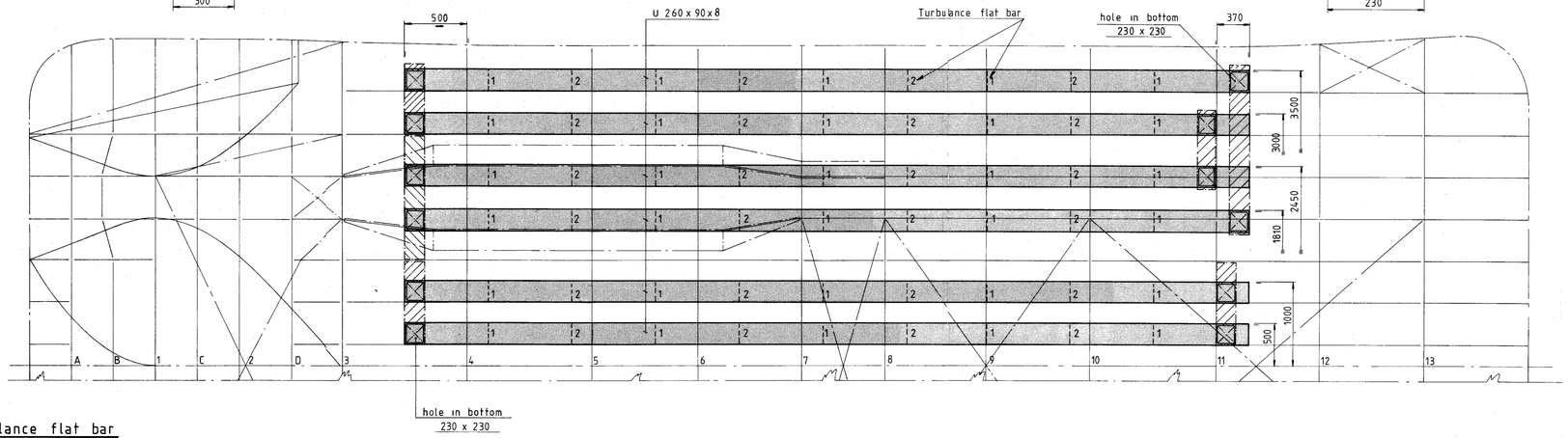


SECTION L-L



COOLING CHANNELS

PS drawn SB opposite hand

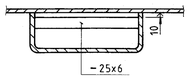


Remark

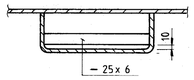
* TOTAL QUANTITY OF DEAERATION PLUGS ACC. DRW 300_1185_0001(ART.NO.1185.0001) : 12*

Turbulance flat bar

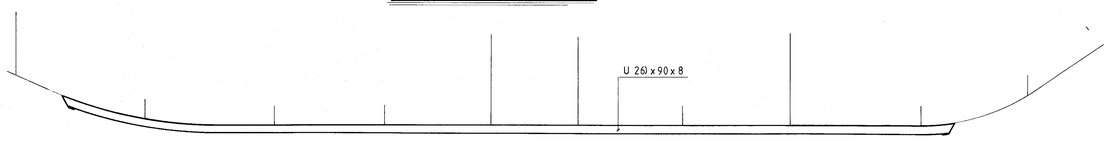
Type 1



Type 2



TYPICAL LONGSECTION



revision	date	name	revision subject
F	5-1-'94	EA.v.L	Additional floorgirders
E	12-11-'93	S.v.d.L	Sewage tank between fr. 5-6
D	01-10-'93	S.v.d.L	Rav. crane foundation
C	21-10-'93	S.v.d.L	Crossover aft ship / frame 4
B	13-10-'93	S.v.d.L	Forepeak / frame C-4 / flatbar void space
A	23-04-'93	S.v.d.L	Remarks LLOYDS

inspected	date
LLOYDS	23-04-'93
B.V.	25-10-'93

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Aveningen West 20
P.O. Box 1
4200 AA Gorinchem Holland
Tel: 31.1830.39911*
Telex: 25335 Damen NL
Telefax: 31.1830.32189

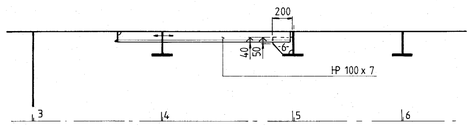
BOTTOM CONSTRUCTION AND COOLING CHANNELS

date 05-03-'93	drawn S.v.d.L	checked	scale 1:25
ident. to dwg. no.		derived fr. dwg. no.	
yard no. 3029 30		dwg. no. 3029 - 110 - 00 - 201	

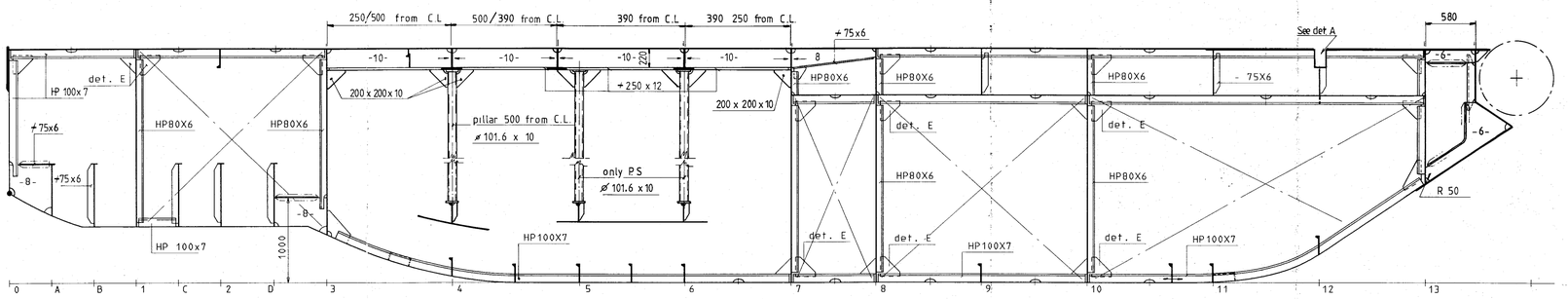
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0cm 5 10cm

Long section on C.L.

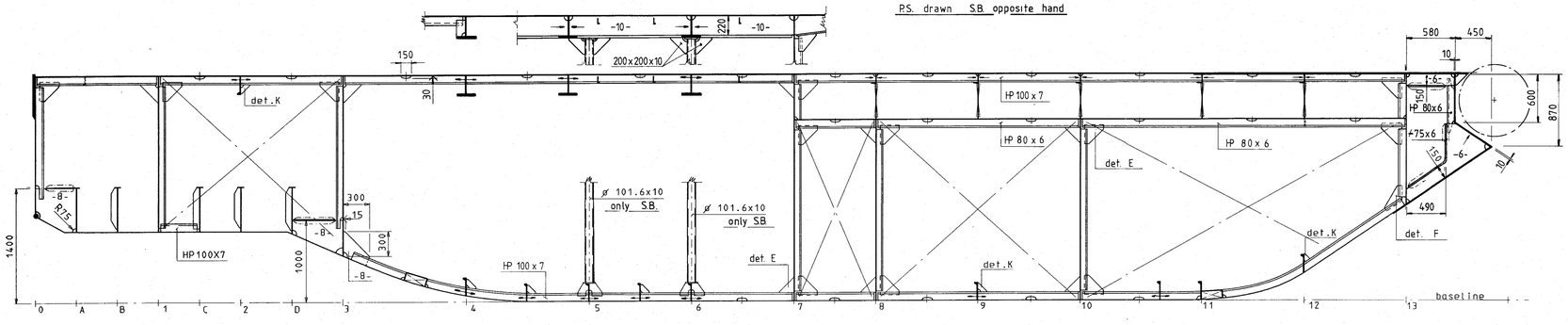


Long. section 250 from C.L.

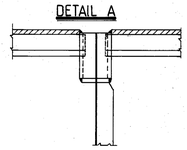


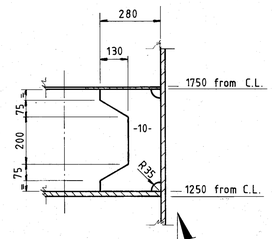
Long. section 750 from C.L.

P.S. drawn S.B. opposite hand

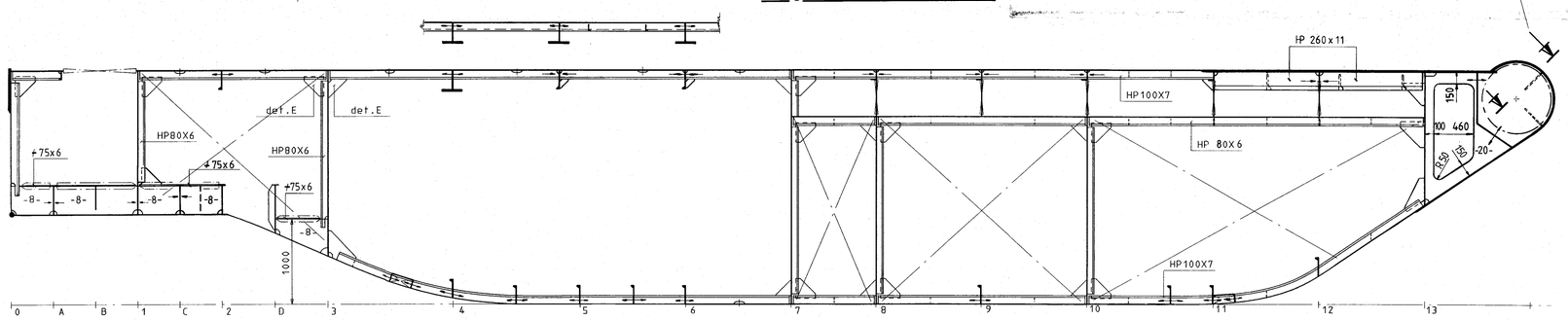


DETAIL A

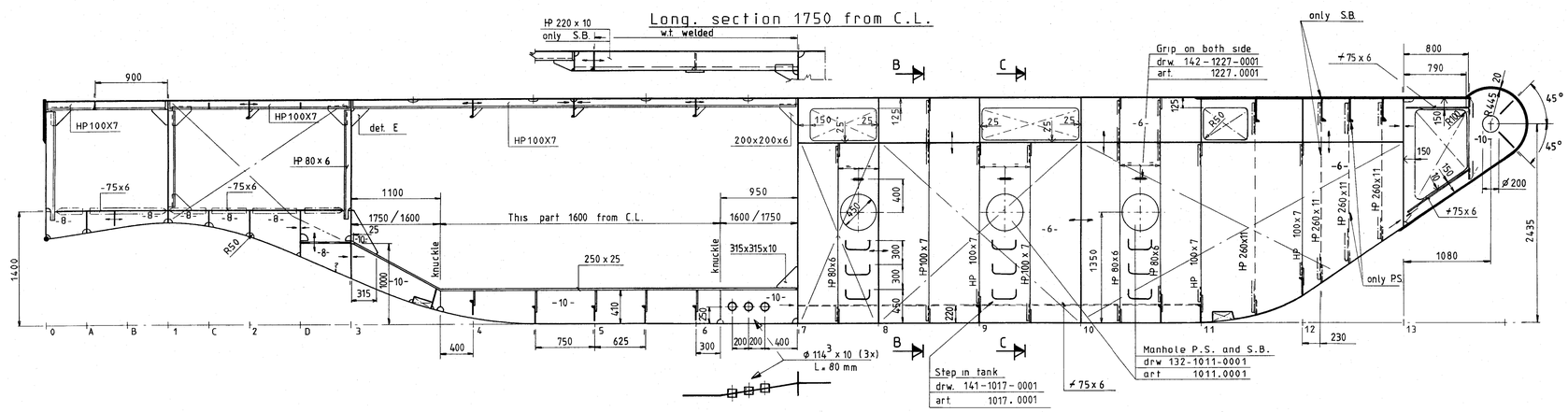




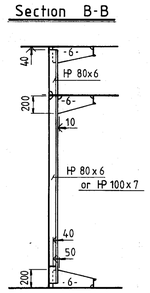
Long. section 1250 from C.L.



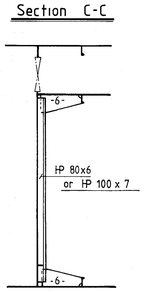
Long. section 1750 from C.L.



Section B-B



Section C-C



rev.	date	name	revision subject
F	19-11-'93	F.A.v.L	Detail A added
E	12-11-'93	S.v.d.L	Frame 3-5-6
D	01-11-'93	S.v.d.L	Rev. crane foundation
C	25-10-'93	S.v.d.L	Remarks B.V.
B	13-10-'93	S.v.d.L.	Forepeak / 1750 s.b.
A	23-04-'93	S.v.d.L	Remarks LLOYDS

inspected	date
LLOYDS	23-04-'93
B.V.	25-10-'93

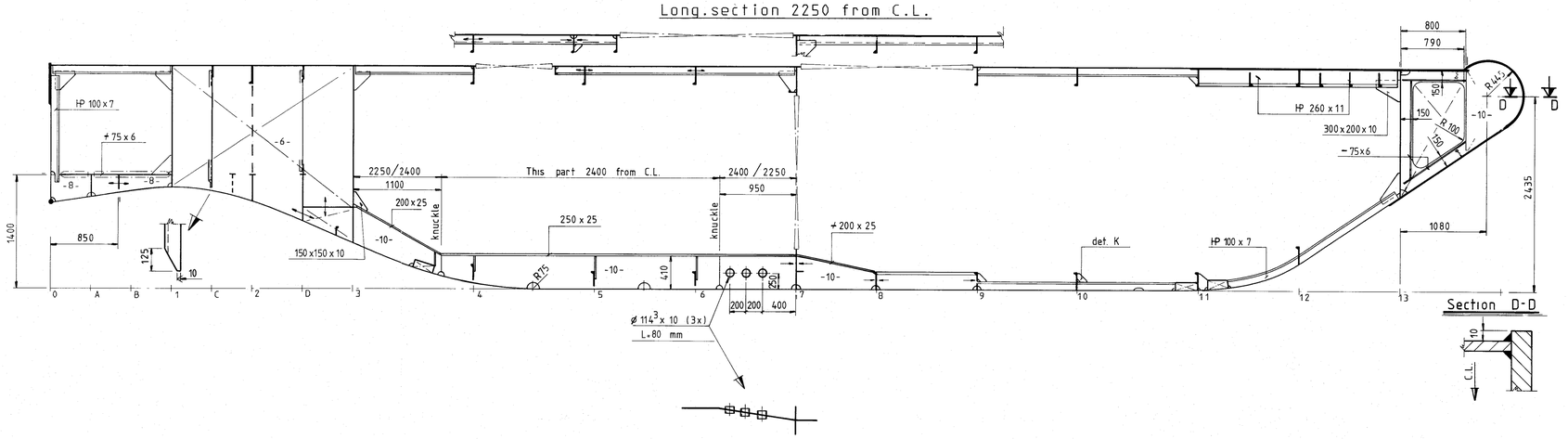
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32109

Long. sections 250/750 1250 and 1750 from C.L.

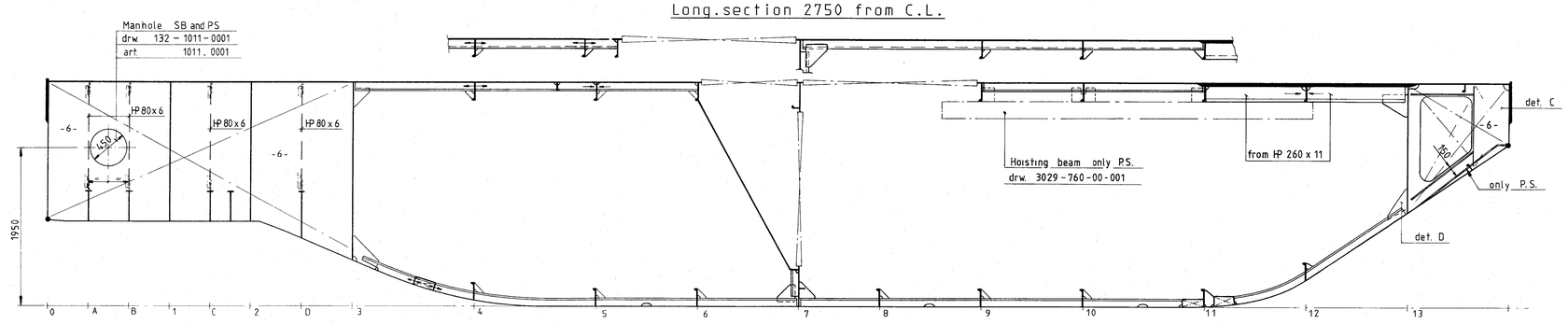
drawn:	S.v.d.Linden	date: 23-02-'93	scale: 1:25
checked:		Ident. to drwg. no.:	
yard no.:	3029/30	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS		drwg. no.: 3029 - 110 - 00 - 250	

0cm 5 10cm

Long.section 2250 from C.L.

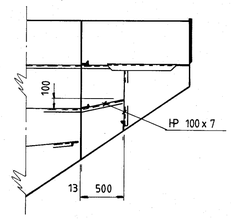


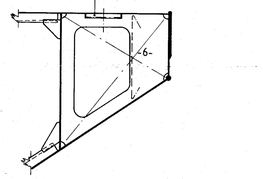
Long.section 2750 from C.L.



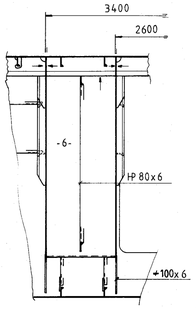
SHELL

P.S. is drawn

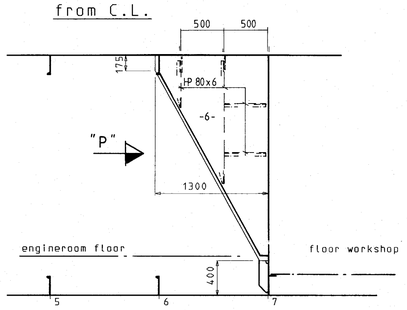




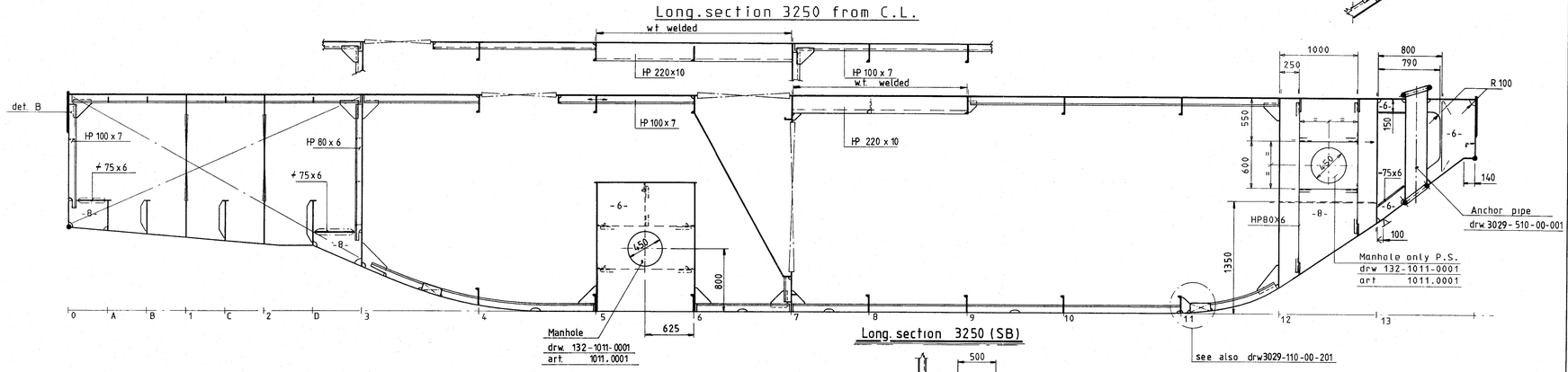
VIEW "P"



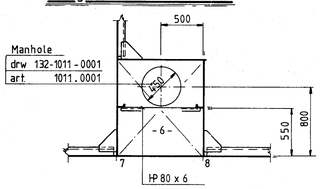
Long.section 2600 and 3400 from C.L.



Long.section 3250 from C.L.

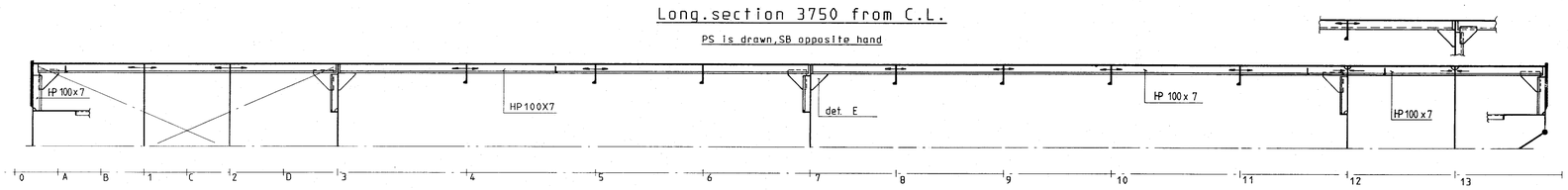


Long.section 3250 (SB)



Long.section 3750 from C.L.

PS is drawn, SB opposite hand



rev.	date	name	revision subject
F	10-11-93	F.A.v.L	Welding hole in long. bottom girder 2250 from ℄ at fr. 3 removed
E	01-11-93	S.v.d.L	Shell frame 13
D	21-10-93	S.v.d.L	Frame 4 (2250 from c.l.)
C	20-10-93	S.v.d.L	Hoisting beam
G	12-11-93	S.v.d.L.	Frame 5-6 / Sewage tank between fr. 5/6

Inspected	date
LLOYDS	23-04-93
B.V.	25-10-93

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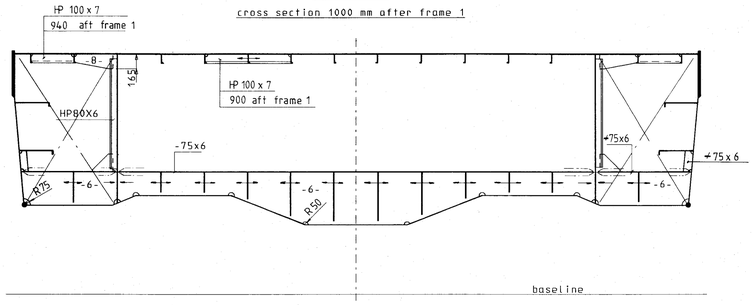
Long.sections 2250/2750 3250 and 3750 from C.L.

drawn: S.v.d.Linden	date: 23-02-93	scale: 1:25
checked:	ident. to dwg. no.:	
yard no.: 3029 / 30	derived fr. dwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	dwg. no.: 3029 - 110 - 00 - 251	

0cm 5 10cm

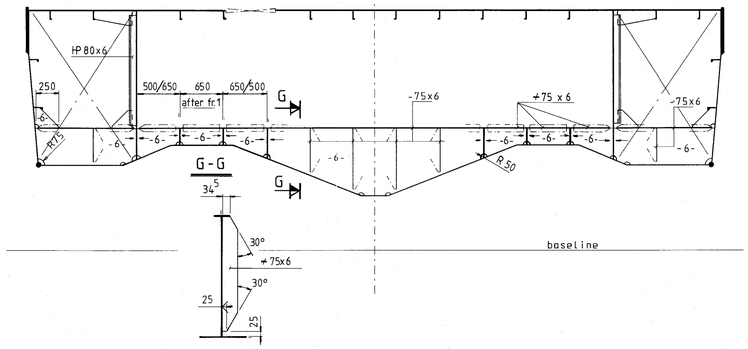
Frame A

cross section 1000 mm after frame 1

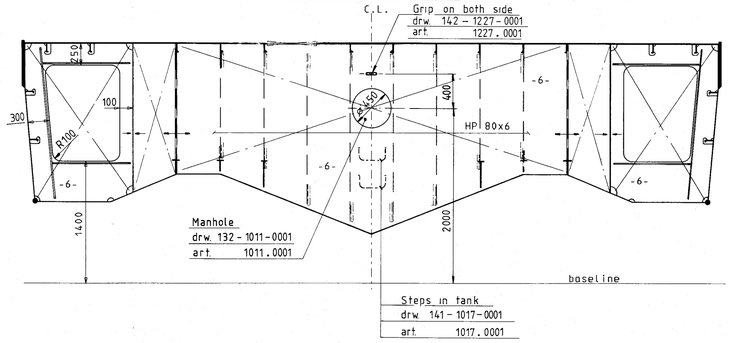


Frame B

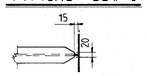
cross section 500 mm after frame 1



Bulkhead frame 1

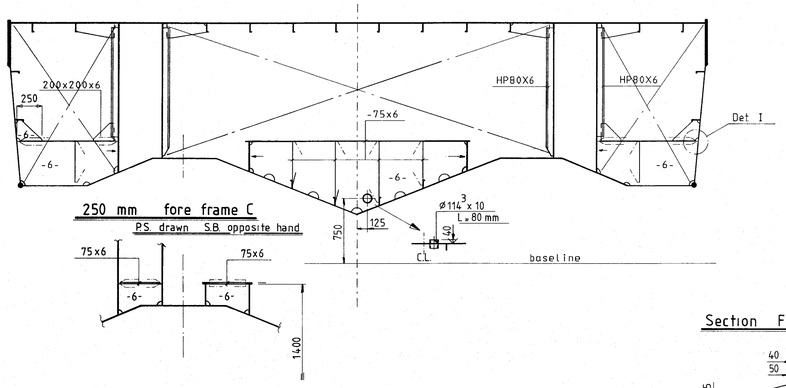


TYPICAL DET I

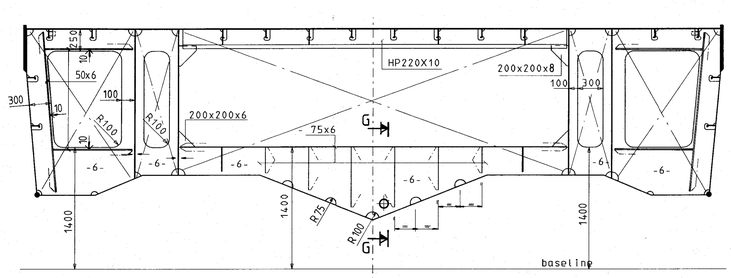


Frame C

cross section 500 mm before frame 1

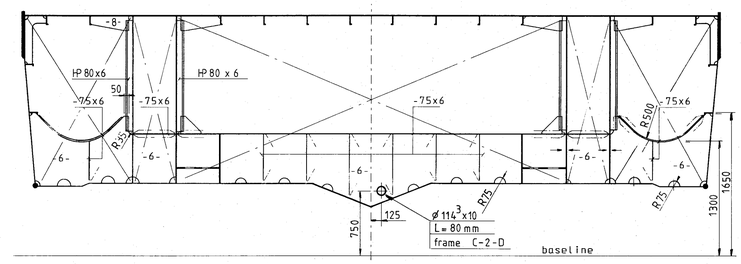


Frame 2

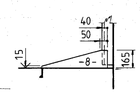


Frame D

cross section 625 mm before frame 2

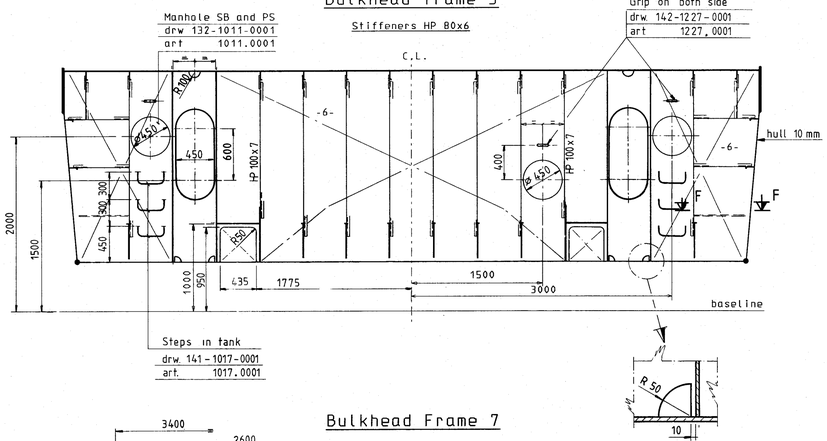


Section F-F

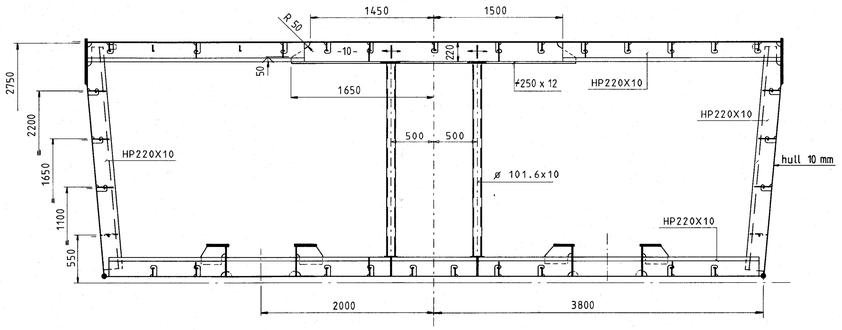


Bulkhead frame 3

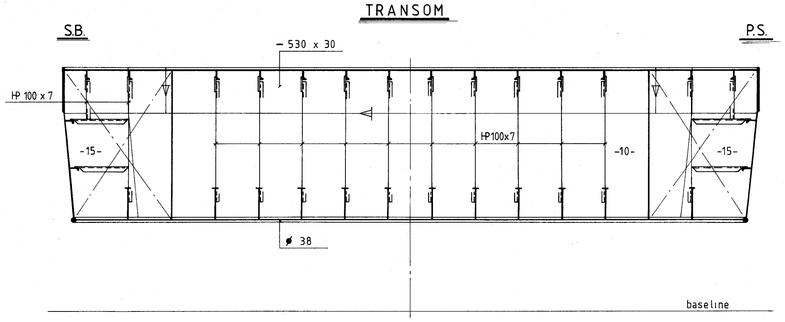
Stiffeners HP 80x6



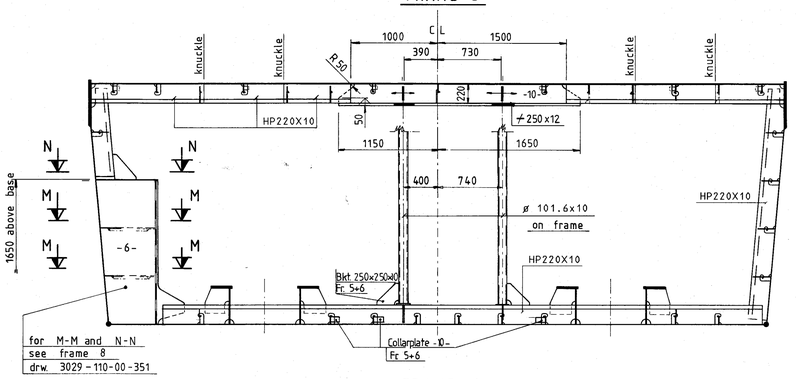
Frame 4



TRANSOM

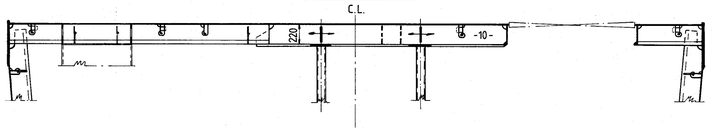


FRAME 5



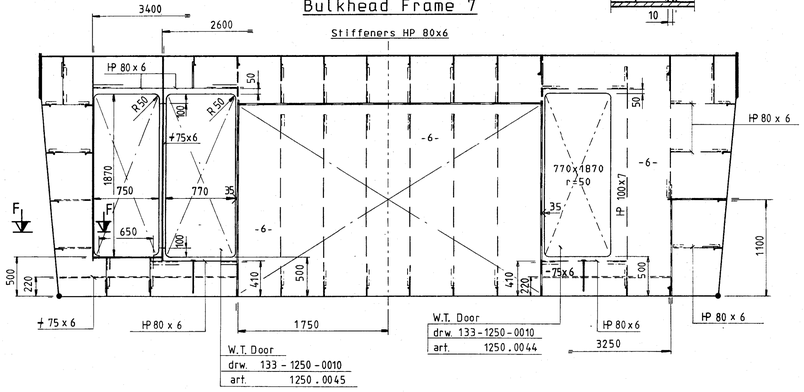
FRAME 6

see also frame 5



Bulkhead Frame 7

Stiffeners HP 80x6



E	19 11 93	FAv.L	Remark BV winchfoundation
D	12 11 93	S.v.d.L	Frame 5 - 6 / Sewage tank between fr. 5/6
C	21-10-93	S.v.d.L	Frame 4
B	12-10-93	S.v.d.L	Frame 3-4-7-C
A	23-04-93	S.v.d.L	Remarks LLOYDS
rev.	date	name	revision subject

inspected	date
LLOYDS	23-04-93
B.V.	25-10-93

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Cross sections Aftship incl. Bulkh. 7

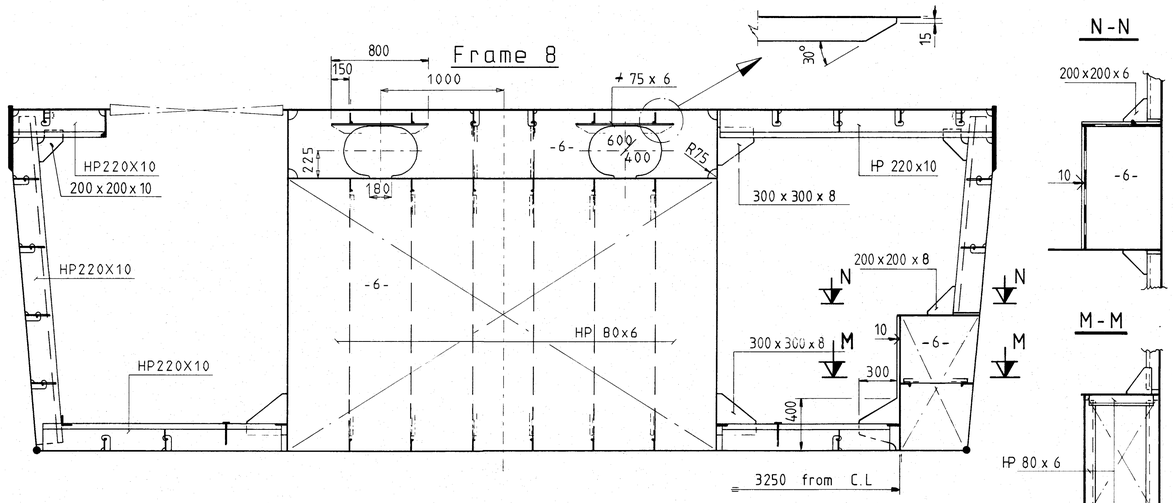
drawn:	S.v.d.Linden	date: 23-02-93	scale: 1:25
checked:		Ident. to drwg. no.:	
yard no.:	3029 / 30	derived fr. drwg. no.:	

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drwg. no.: 3029-110-00-350

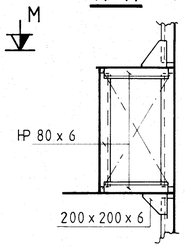
0cm 5 10cm

Frame 8

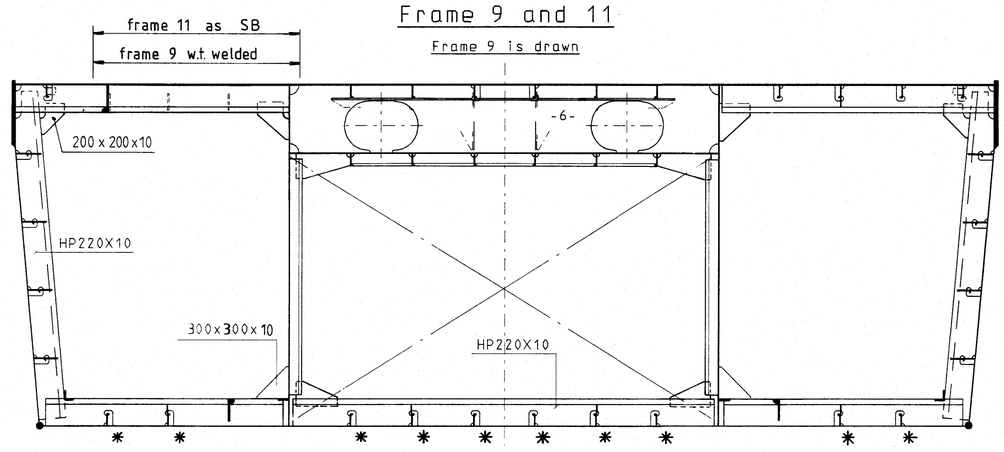


N-N

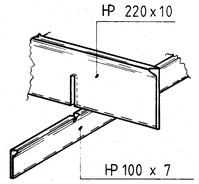
M-M



Frame 9 and 11



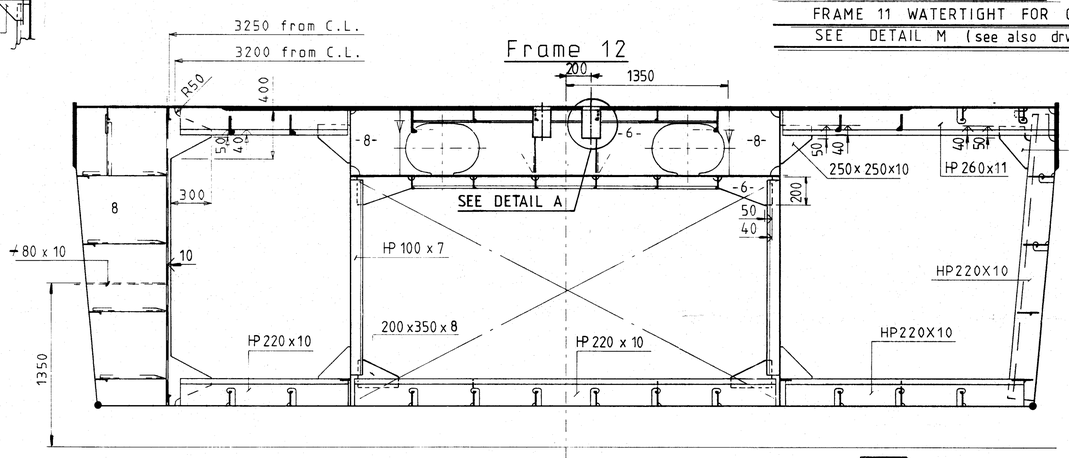
DETAIL M



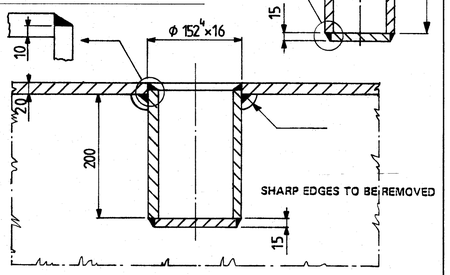
✱ NOTCHES ONLY FRAME 9
FRAME 11 WATERTIGHT FOR COOLING CHANNEL
SEE DETAIL M (see also drw 3029-110-00-201)

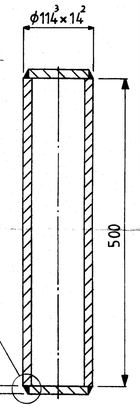
Frame 10

Frame 12

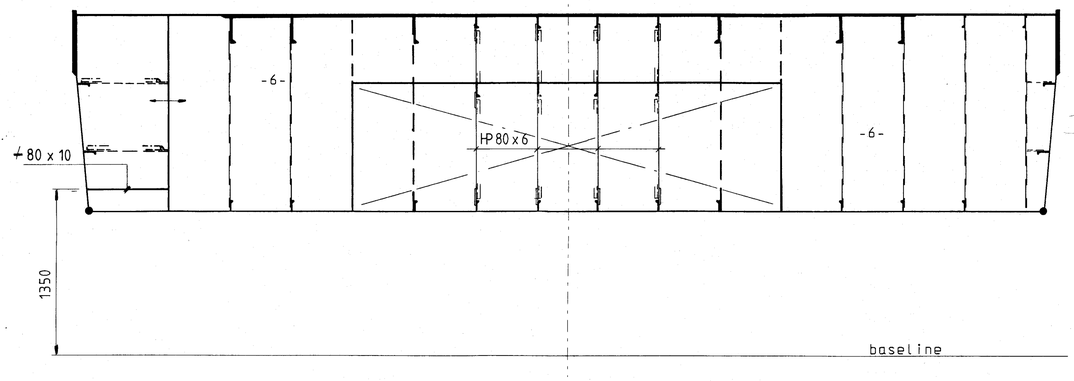


DETAIL A

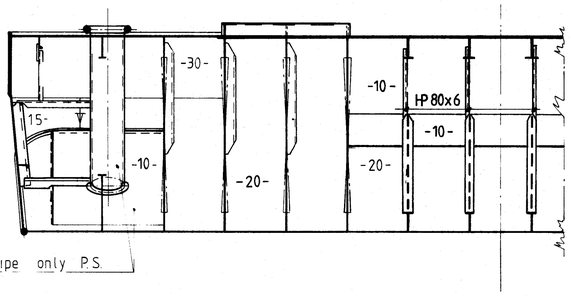




Bulkhead Frame 13



STEM PLATE



Anchor pipe only P.S.

rev.	date	name	revision subject
E	19-11-'93	F.Zillig	Detail A added
D	01-11-'93	S.v.d.L.	Frame 13 / Stemplate
C	25-10-'93	S.v.d.L	Remarks B.V.
B	13-10-'93	S.v.d.L	Stem plate
A	23-04-'93	S.v.d.L	Remarks LLOYDS

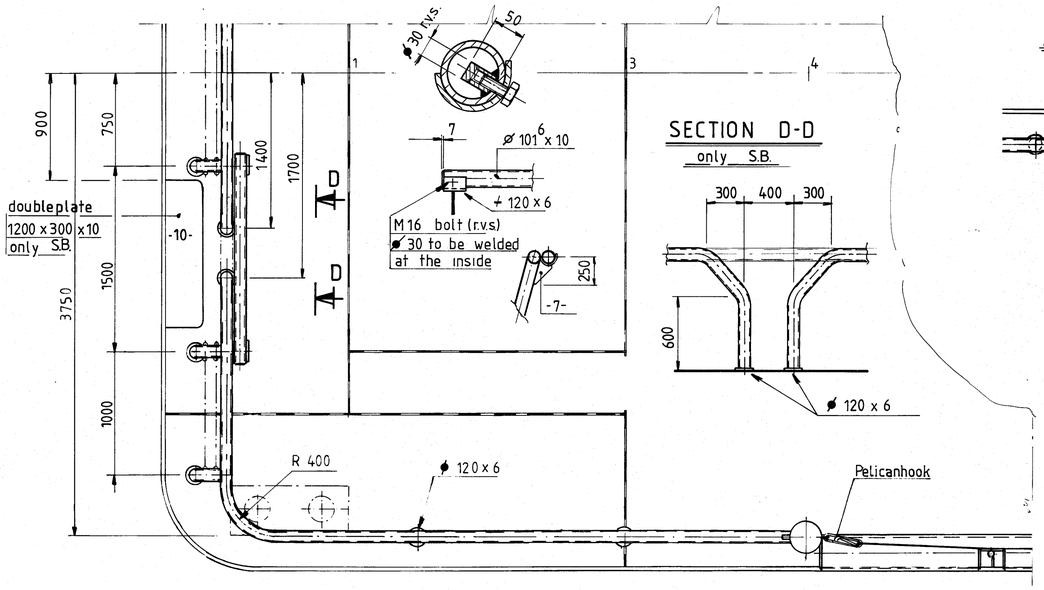
inspected	date
LLOYDS	23-04-'93
B.V.	25-10-'93

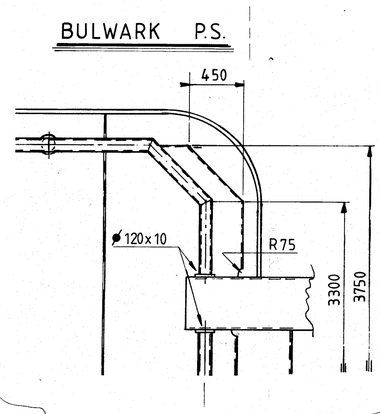
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32109

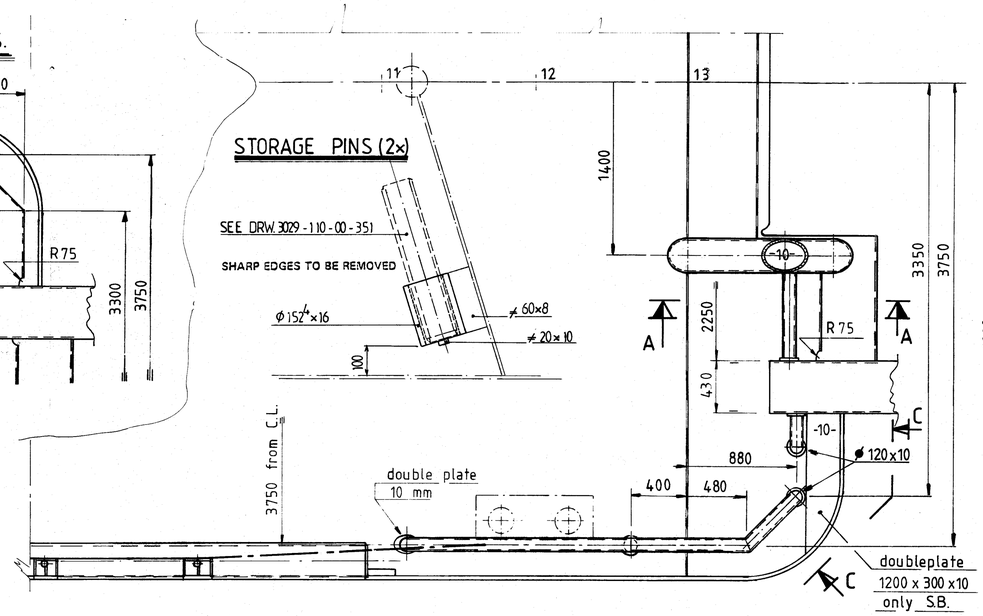
Cross sections frame 8-9-10-11-12 and 13

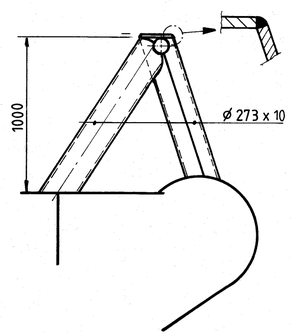
drawn: S.v.d.Linden	date: 23-02-93	scale: 1:25
checked:	Ident. to drwg. no.:	
yard no.:	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-110-00-351	

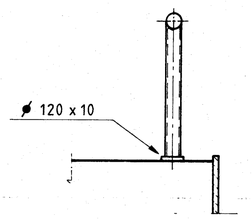
0cm 5 10cm

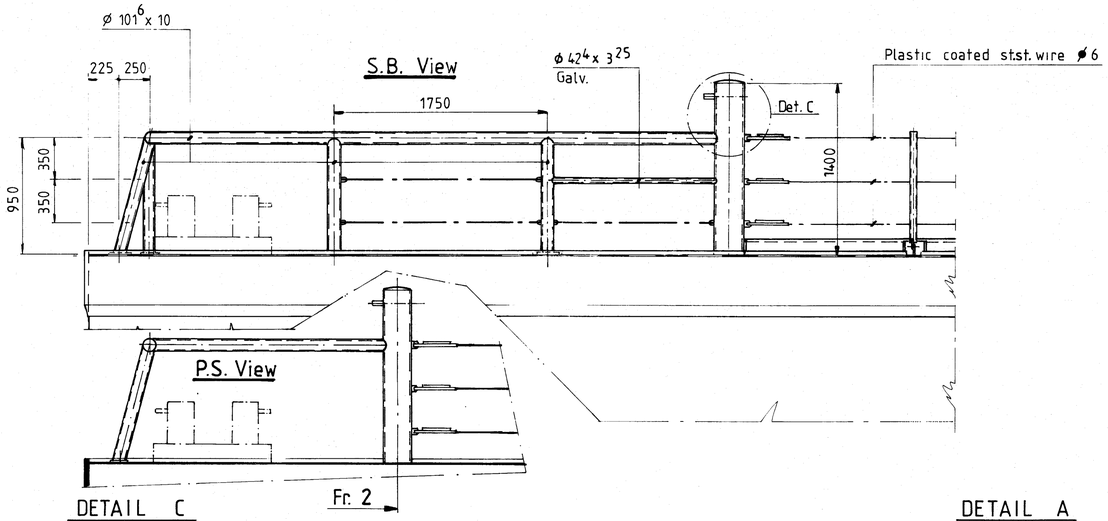


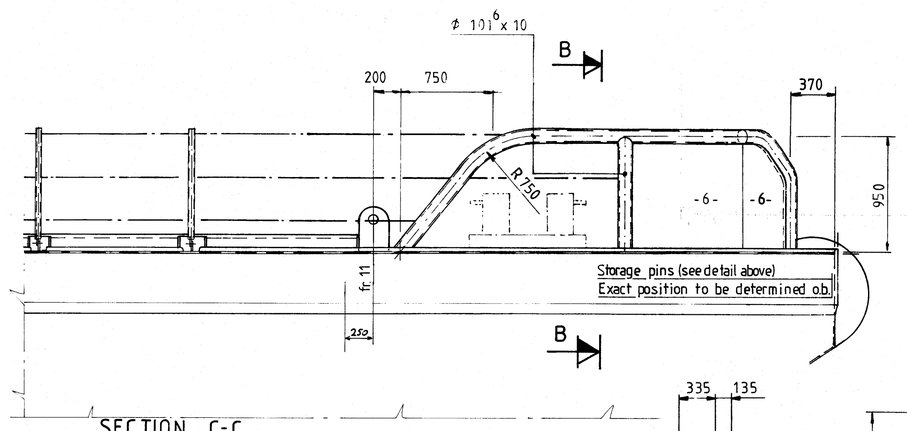


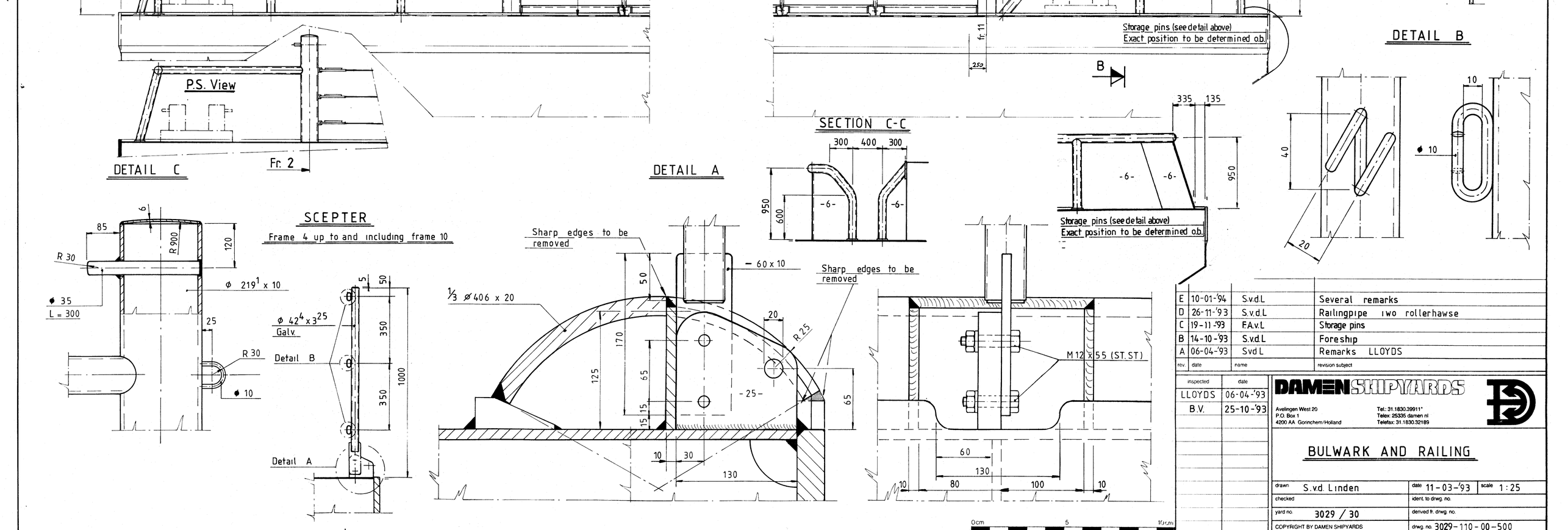












rev.	date	name	revision subject
E	10-01-'94	S.v.d.L	Several remarks
D	26-11-'93	S.v.d.L	Railingpipe two rollerhawse
C	19-11-'93	F.A.v.L	Storage pins
B	14-10-'93	S.v.d.L	Foreship
A	06-04-'93	S.v.d.L	Remarks LLOYDS

inspected	date
LLOYDS	06-04-'93
B.V.	25-10-'93

DAMEN SHIPYARDS
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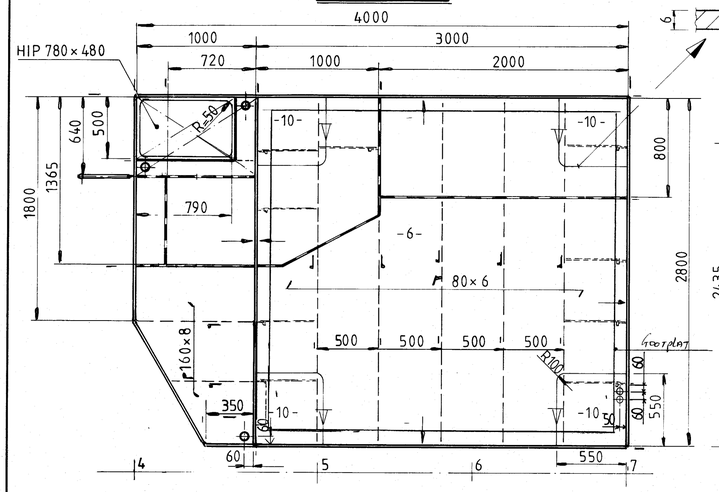
BULWARK AND RAILING

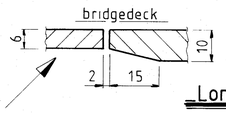
drawn	S.v.d. Linden	date	11-03-'93
checked		scale	1:25
yard no.	3029 / 30	ident. to drwg. no.	
		derived fr. drwg. no.	

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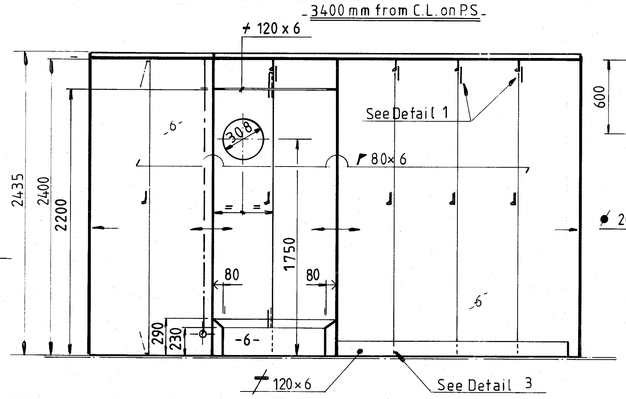
drwg. no. 3029-110-00-500

Bridgedeck

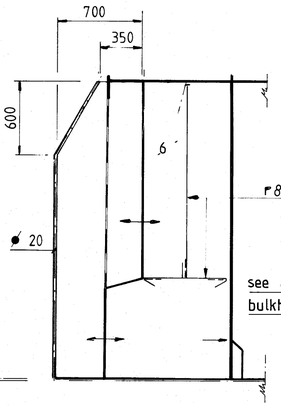




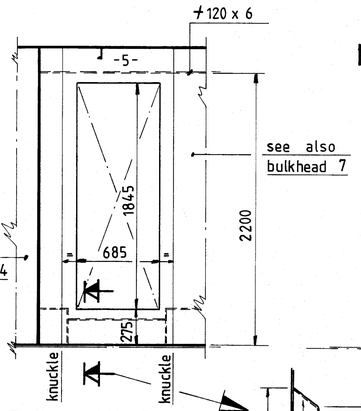
Longitudinal Bulkhead Nr: 1



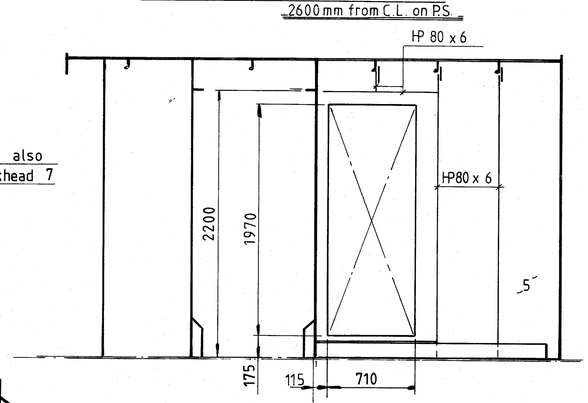
Bulkhead Nr: 2



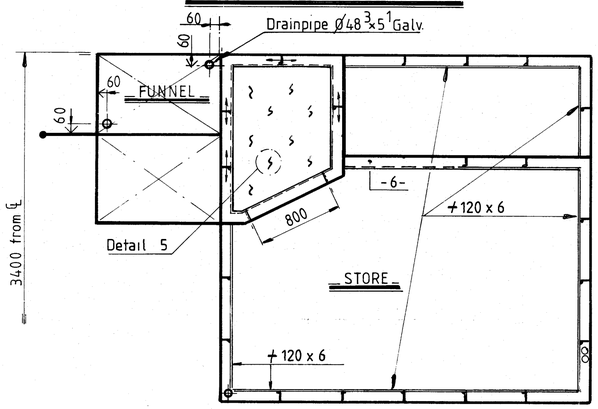
Bulkhead Nr: 12



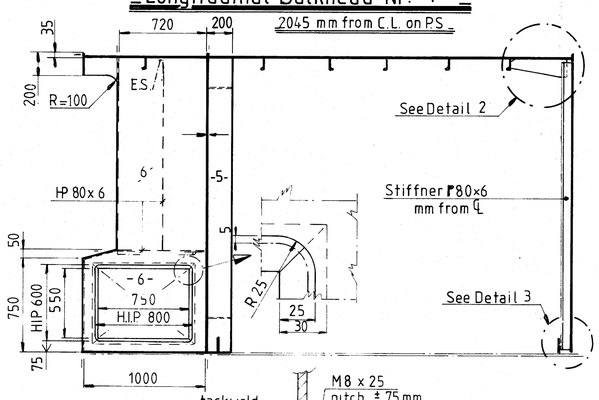
Longitudinal Bulkhead Nr: 3



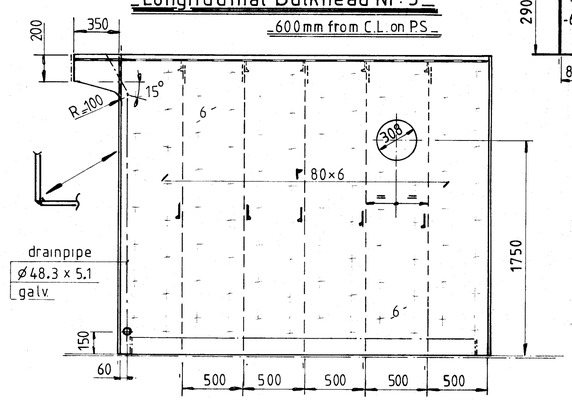
Section about Deckhouse

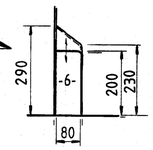


Longitudinal Bulkhead Nr: 4

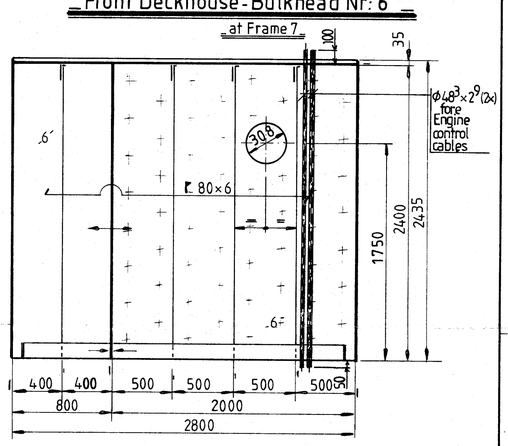


Longitudinal Bulkhead Nr: 5

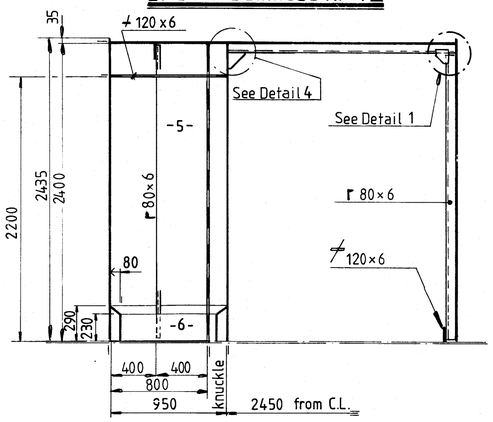




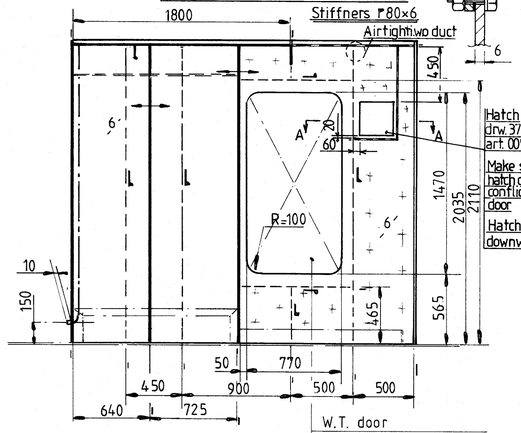
Front Deckhouse-Bulkhead Nr: 6



Transv. Bulkhead Nr: 7



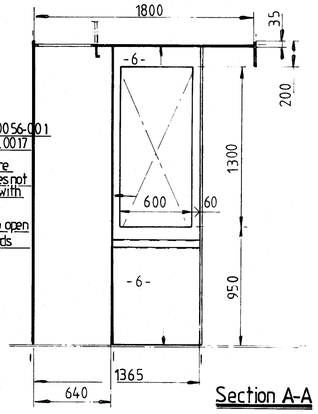
Transv. Bulkhead Nr: 8



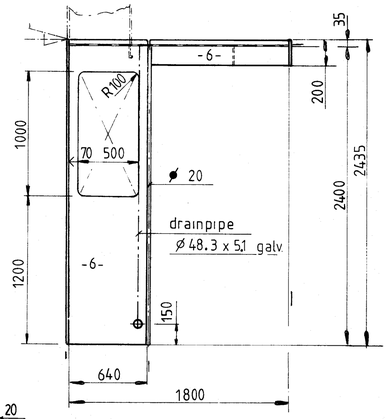
W.T. door

drw.	133 - 1250 - 0010	
art.	1250 . 0046	

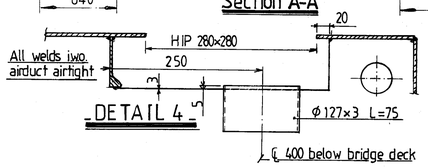
Transv. Bulkhead Nr: 9



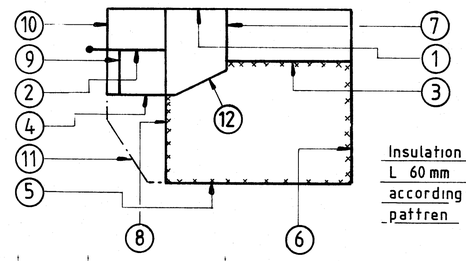
Transv. Bulkhead Nr: 10



Section A-A

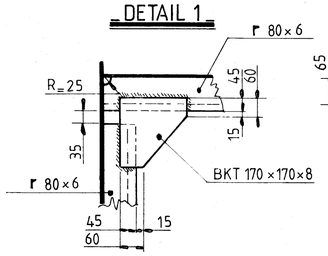


NOTE:
FRAMESPACING MAINDECK 1250mm

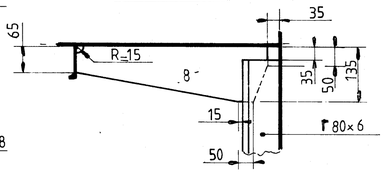


Insulation pins Ø 2½
L 60 mm to be welded
according to indicated
pattren

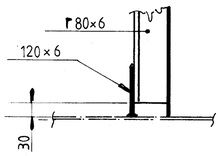
DETAIL 1



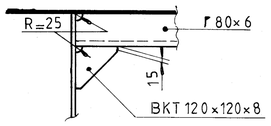
DETAIL 2



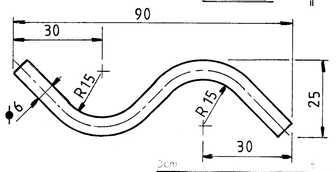
DETAIL 3

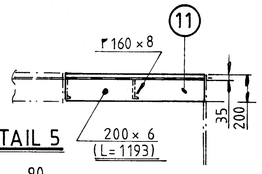


DETAIL 4



DETAIL 5





E	10-01-'94	S.v.d.L	Blh. 3 flat 6 mm
D	29-10-'93	Fv.Leersum	Hatch+duct extractor fan / pipes for engine control cables
C	18-10-'93	S.v.d.L	Side light bulkhead 1 / Door bhd. 3
B	14-10-'93	S.v.d.L	Wet spaces / bulkhead 12
rev.	date	name	revision subject

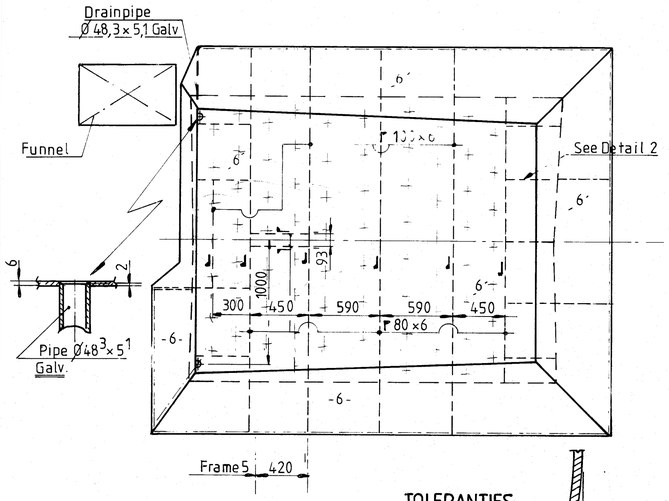
inspected	date
LLOYDS	09-04-'93

DAMEN SHIPYARDS
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P.O. Box 1
4200 AA Gorinchem Holland
Tel. 31 1830 39911*
Telex 25335 damen nl
Telefax 31 1830 32189

CONSTRUCTION SUPERSTRUCTURE
above Maindeck

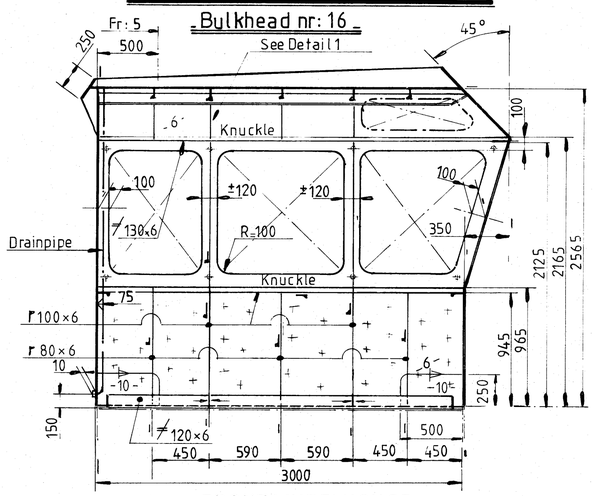
drawn	Idelenburg	date 05-03-93	scale 1:25
checked		ident to drwg no	
yard no	3029/30	derived fr drwg no	
COPYRIGHT BY DAMEN SHIPYARDS		drwg no	3029-120-00-050

VIEW ON TOPDECK



SIDE WHEELHOUSE 3400 from ℄ PS.

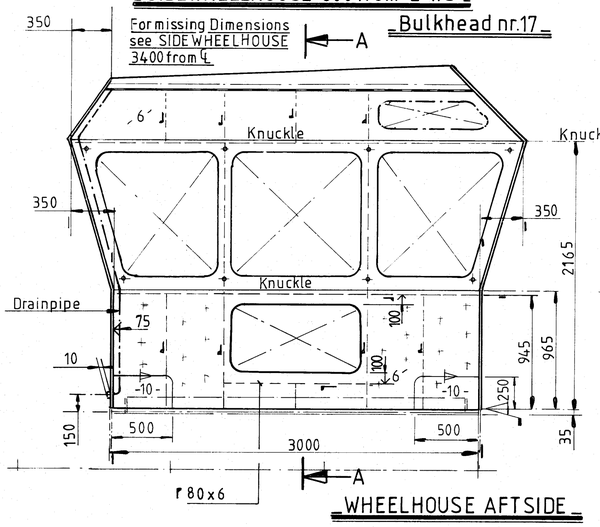
Bulkhead nr: 16



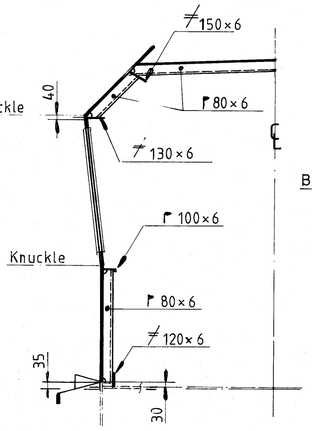
SIDEWHEELHOUSE 600 from ℄ PS

For missing Dimensions see SIDE WHEELHOUSE 3400 from ℄

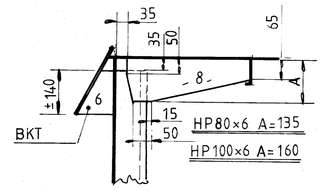
Bulkhead nr.17



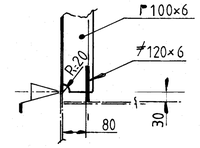
Section A-A



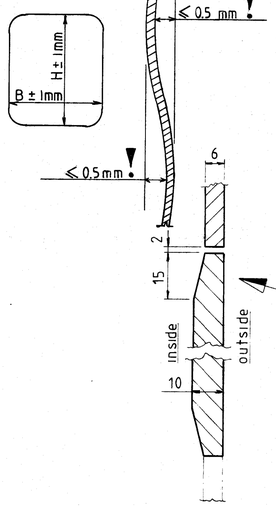
DETAIL 3



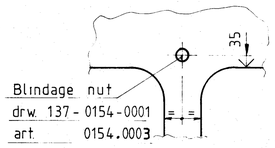
DETAIL 4



TOLERANTIES RAAMGATEN

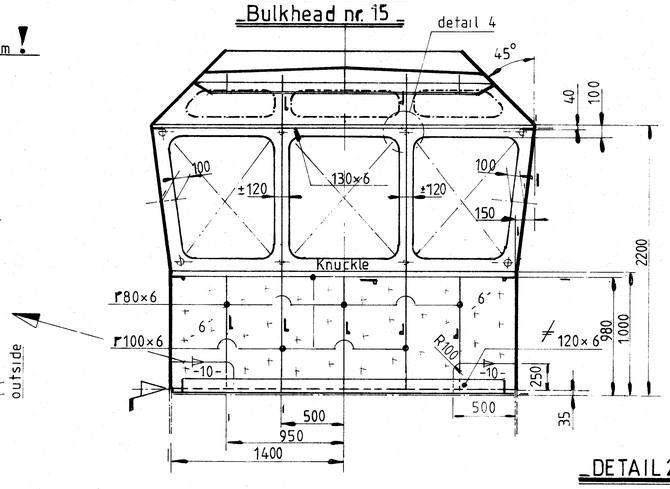


DETAIL 4



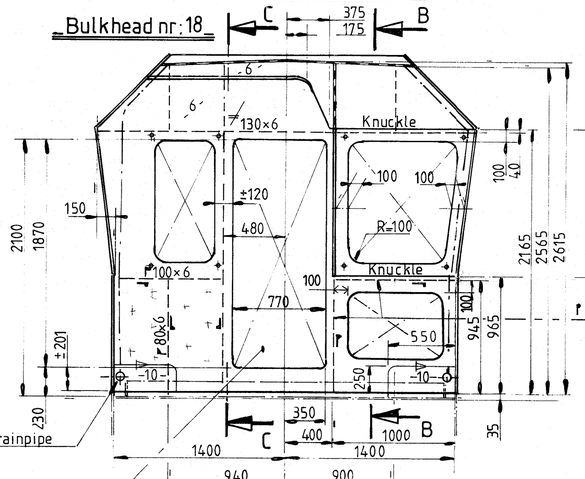
FRONT WHEELHOUSE

Bulkhead nr. 15



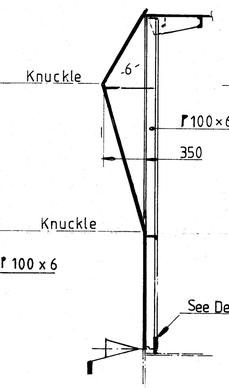
WHEELHOUSE AFTSIDE

Bulkhead nr: 18

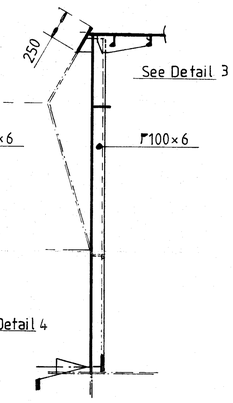


W.T. Aluminium door with steel frame
drw. 133 - 1250 - 0001
atr. 1250 . 0022

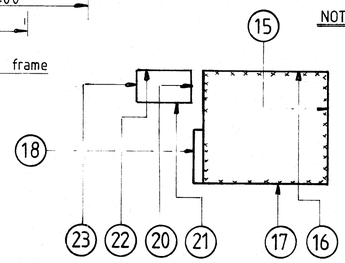
B-B



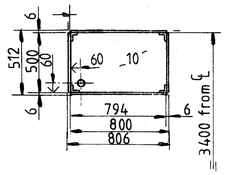
C-C



NOTE
FRAMESPACING MAINDECK 1250 mm

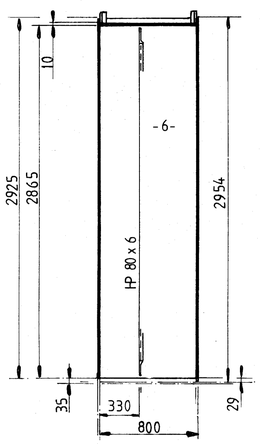


Insulation pins ⌀ 2 ½
L 60 mm to be welded according to indicated pattren

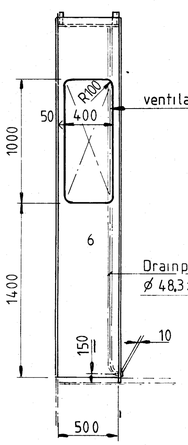


BULKHEADS FUNNEL

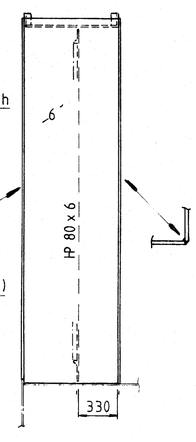
BULKHEAD Nr: 22



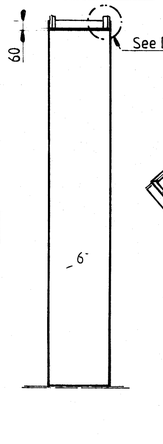
BULKHEAD Nr: 23



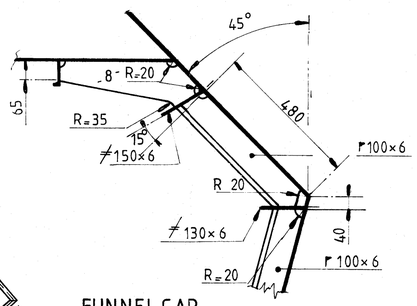
BULKHEAD Nr: 21



BULKHEAD Nr: 20

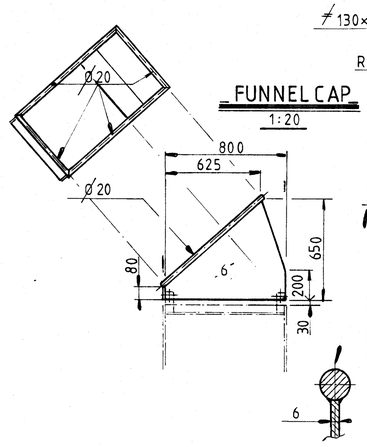


DETAIL 2

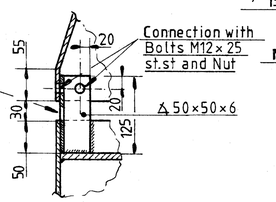


FUNNEL CAP

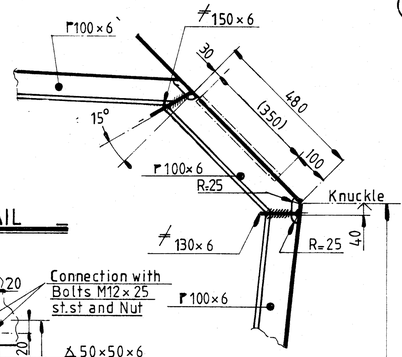
1:20



DETAIL



DETAIL 1



D	10-01-'94	S.v.d.L	Window side bulkhead
C	22-11-'93	F. Zillig	Art. Nr. Blindage Nut Acc. Spec.
B	18-11-'93	FA.v.L	Toleranties raamgaten
A	16-04-'93	A.M. Dolman	Remarks LLOYDS
rev.	date	name	revision subject

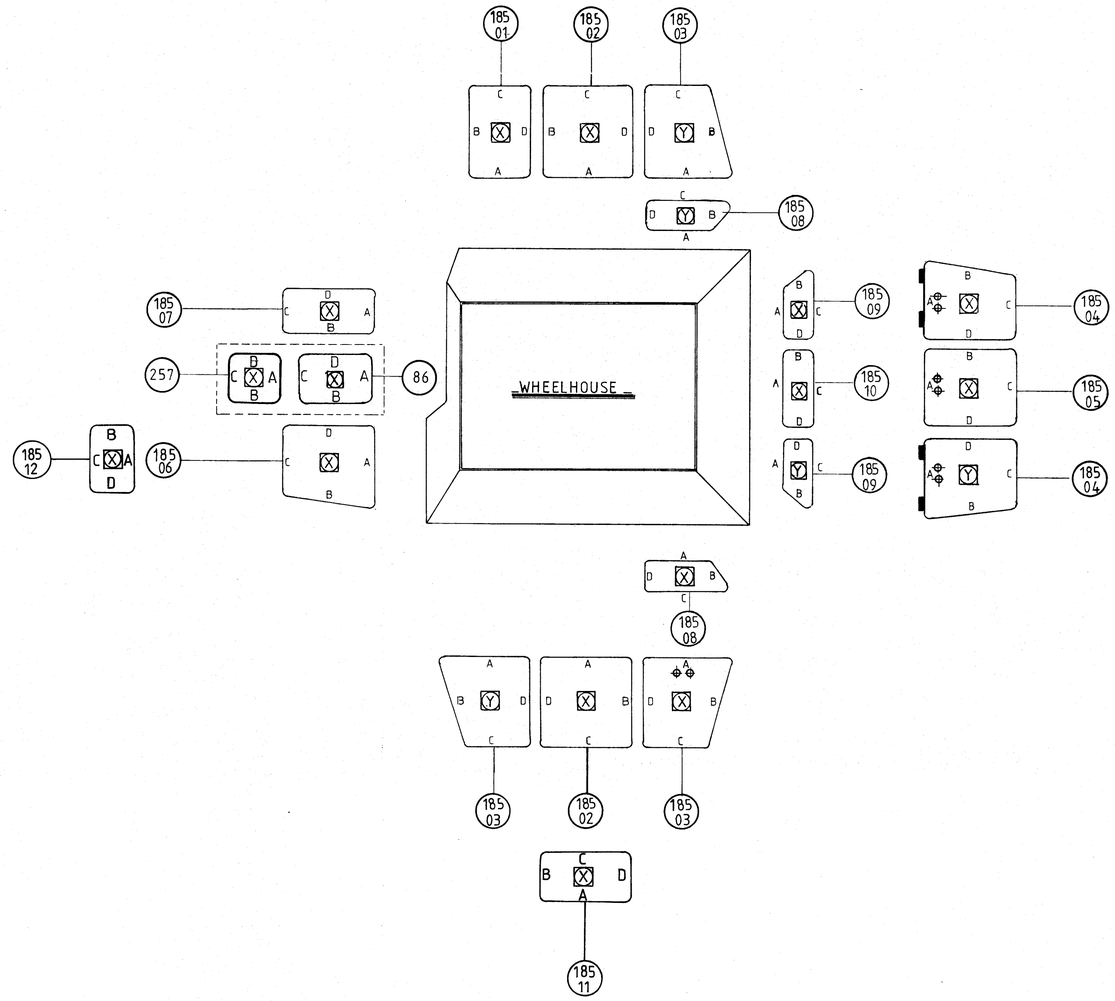
inspected	date
LLOYDS B.V.	09-04-'93 15-11-'93

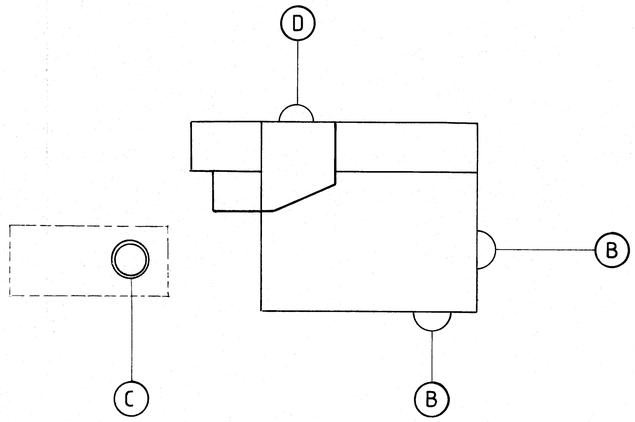
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem Holland
Tel. 31 1830 39911*
Telex 25335 damen nl
Telefax 31 1830 32189

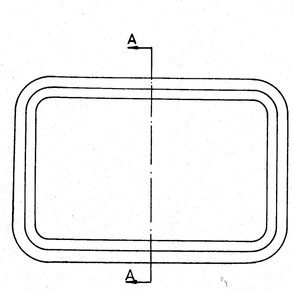
CONSTRUCTION SUPERSTRUCTURE above Bridgedeck with FUNNEL

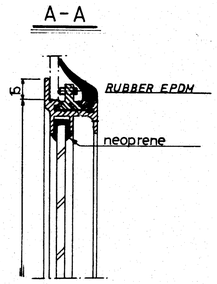
drawn: Idelenburg	date 10-03-93	scale 1:25	
checked	ident to drwg. no.		
yard no. 3029 / 30	derived fr. drwg. no.		
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no. 3029-120-00-051		

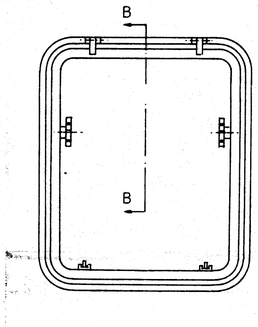
0cm 5 10cm

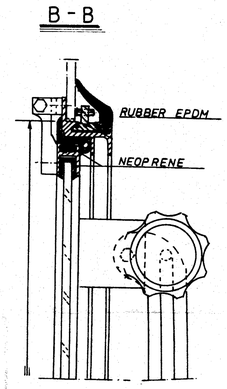












Template	Quant.	Dimensions Hole in plate H×B	Number of Storm Covers	Glass mm	Remarks
185 - 01	1	1020 × 750		8	
185 - 02	2	1020 × 1060		10	
185 - 03	1 + 2	1020 × 1060/770		9	
185 - 04	1 + 1	1050× 875/750		10	
185 - 05	1	1050× 875		10	
185 - 06	1	1050× 925/800		10	
185 - 07	1	1000× 500		8	
185 - 08	1+1	350 × 760× 1010		6	Tinted glass
185 - 09	1+1	350 × 570/820		6	Tinted glass
185 - 10	1	350 × 875.		6	Tinted glass
185 11	1	1060 x 550		10	
185 - 12	1	800 x 550		10	
86	1	800 x 400		6	
257	1	500 x 400		6	

✳ - For L.R.S. required glass thickness minimal

NOTES: - For L.R.S. Coastal Service.
- For all windows aluminium frame AK+
- Glass to be Toughened Safety Glass

For window templates see drw. 3029 - 135 - 00 - 002

Symbol	Description	Dim. Hole in pl.	Glass mm
B	Side light opening type with steel deadlight flange L = 200 mm	ø 308	8
C	Bullseye fixed type with steel deadlight PAN 2188	ø 230	8
⊖⊖	Window wiper		
▬▬	Opening window of the hinged type on the top		
D	Side light opening type with deadlight frosted glass (flange L = 80 mm)	ø 308	12

0cm 5 10cm

rev	date	name	revision subject
F	10-01-'94	S.v.d.L	Window 185 - 11 /12
E	11-11-93	FA.v.L	Mould 185-8 PS correctly defined
D	10-11-'93	FA.v.L	Item C
C	18-10-'93	S v d L	Sidelight D / Sky windows tinted
B	12-10-'93	S.v.d.L	Porthole (B)
A	16-04-'93	A.M. Dolman	Remarks LLOYDS

inspected	date
LLOYDS	09-04-'93

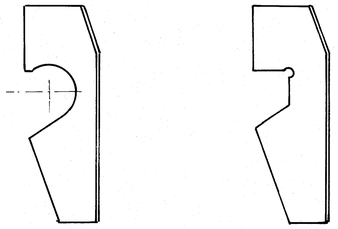
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

ARRANGEMENT WINDOWS

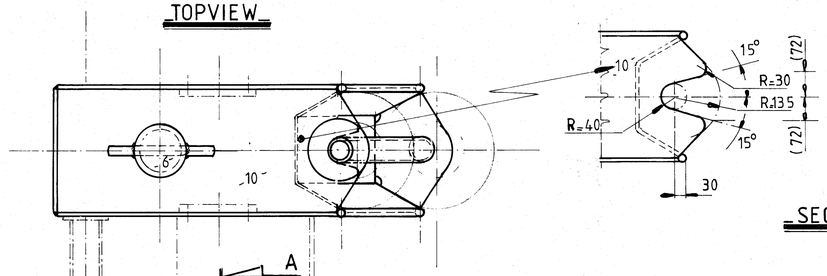
drawn Idelenburg	date 10-03-93	scale %
checked	ident. to drwg. no.	
yard no.	derived fr. drwg. no.	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no. 3029 - 135-00-001	

Inside
Plate Nr. 1

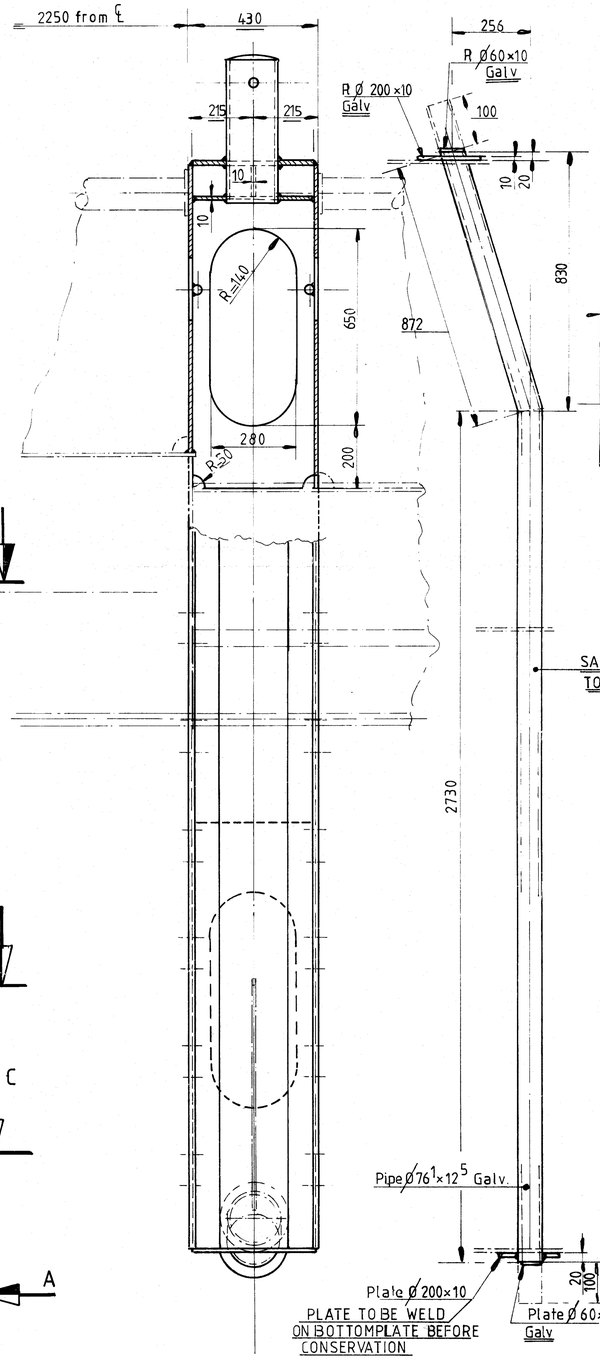
Outside
Plate Nr. 2



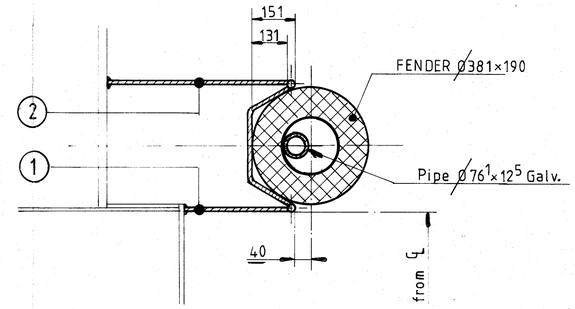
TOPVIEW



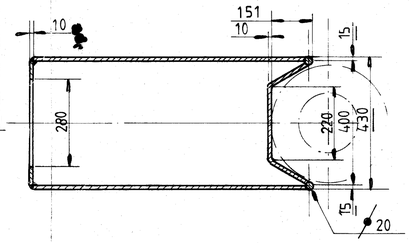
SECTION A-A



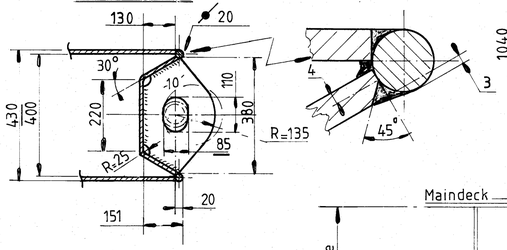
SECTION D-D



SECTION B-B

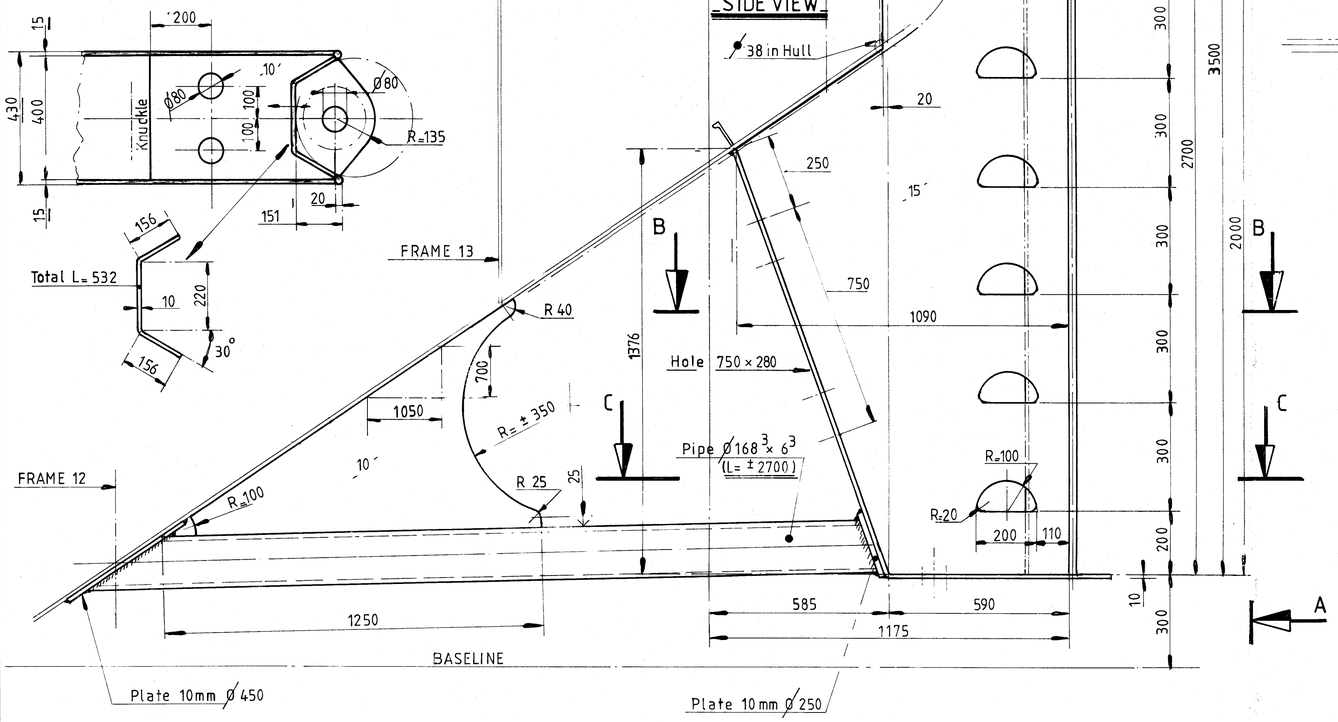


DETAIL A



SECTION C-C

SIDE VIEW



Total L= 532

SHARP EDGES TO BE REMOVED

Note!
P.S. Pushbow drawn.
S.B. Pushbow oppositehand.

rev.	date	name	revision subject
A	16-04-'93	A.M. Dolman	Remarks LLOYDS

inspected	date
LLOYDS B.V.	09-04-'93 25-10-'93

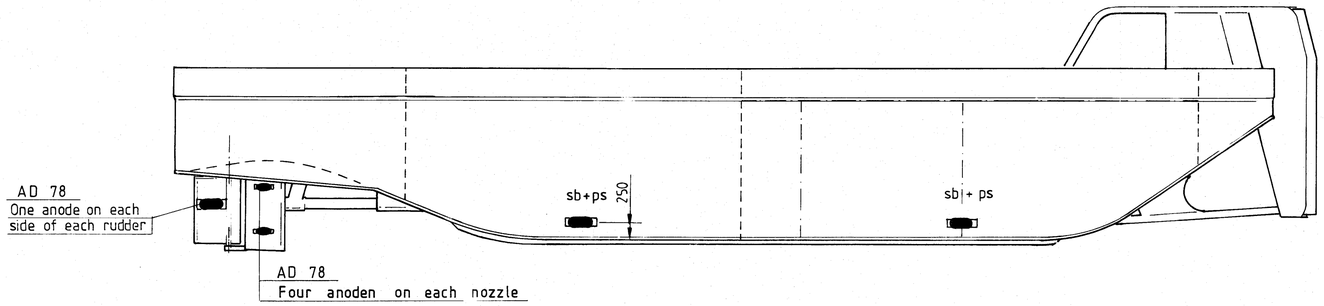
DAMEN SHIPYARDS
Avelingen West 20
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5-1-14

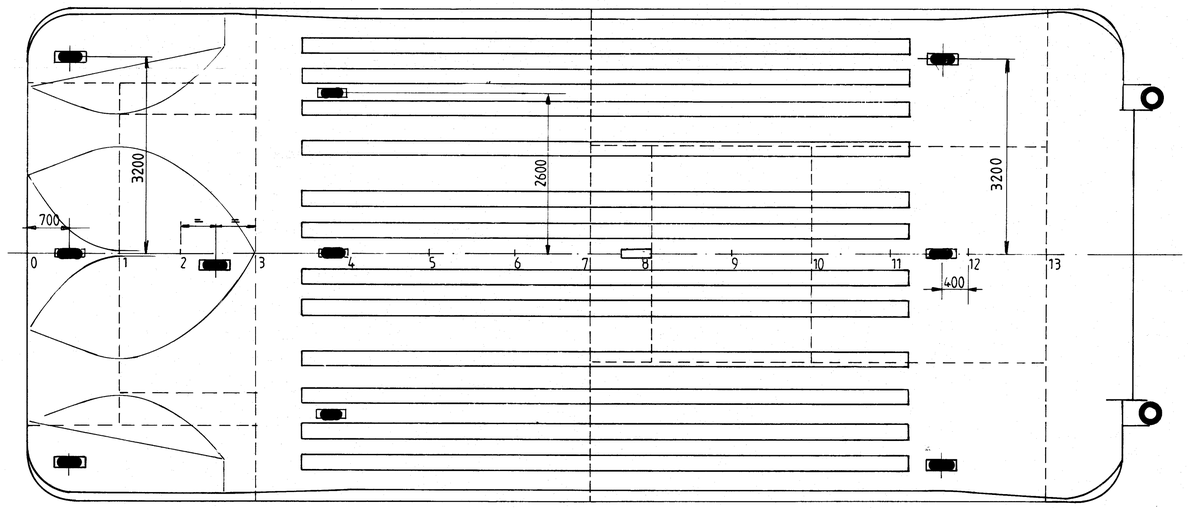
PUSHBOW

drawn: Idelenburg	date: 12-03-93	scale: 1:10 1:1	
checked:	dient to drwg no:		
yard no: 3029/30	derived fr drwg no:		
COPYRIGHT BY DAMEN SHIPYARDS	drwg no: 3029-152-00-001		

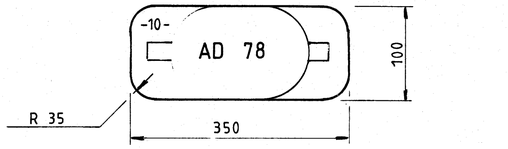
SIDE VIEW



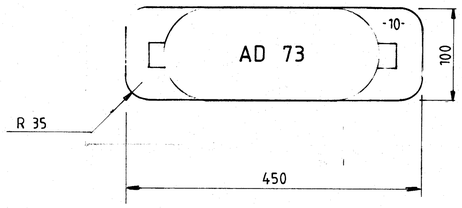
BOTTOM VIEW



ANODE ON NOZZLE AND RUDDER



ANODE ON HULL



AD 73	14 x	5 kg =	60 kg
AD 78	12 x	2 kg =	24 kg
			84 kg for each ship

SHARP EDGES TO BE REMOVED

0cm 5 10cm

rev.	date	name	revision subject
A	27-12-'93	S.v.d.L.	Alu. anode

inspected | date

DAMEN SHIPYARDS

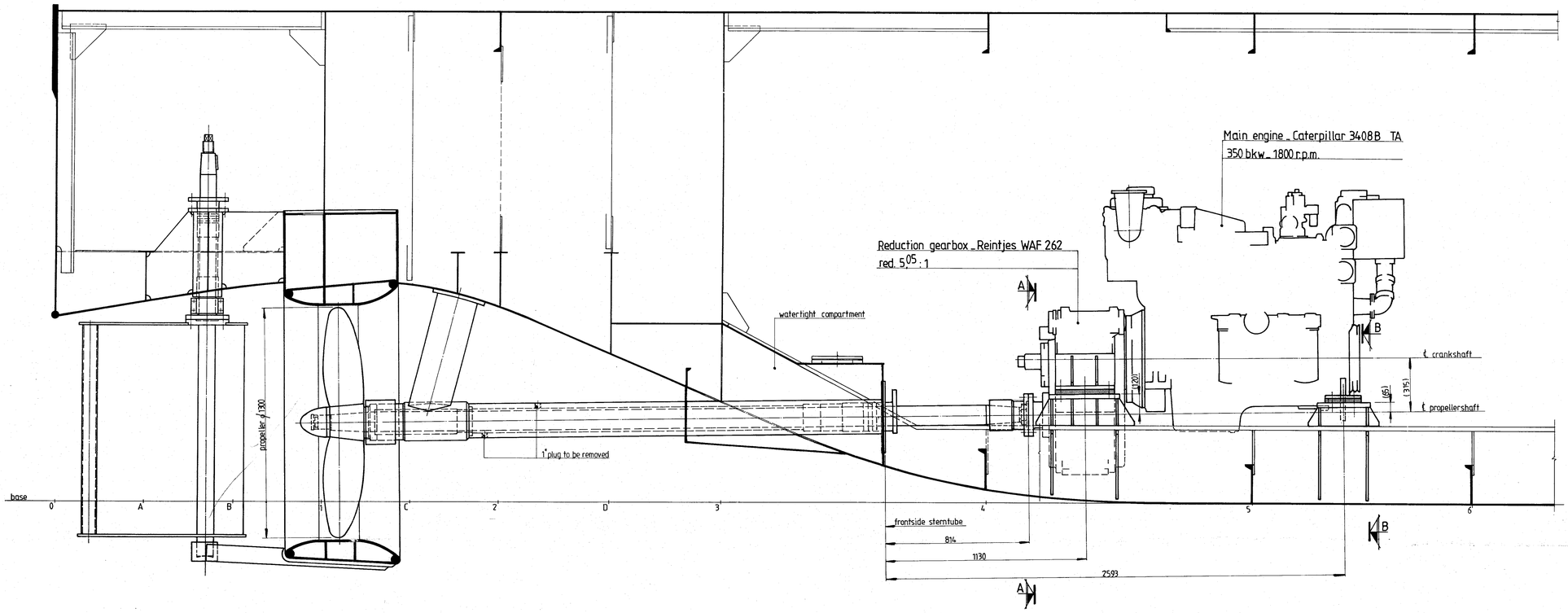
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem Holland

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Telex 25335 damen nl
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PLAN OF ANODES

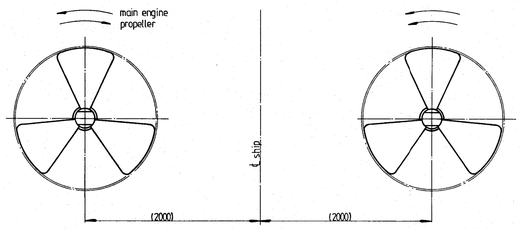
drawn A.M. DOLMAN		date 23-03-'93	scale 1 : 50
checked		dwnl to dwg no	
yard no 3029/30		derived fr dwg no	
COPYRIGHT BY DAMEN SHIPYARDS		dwg no 3029-163-00-001	

Longitudinal section 2000 mm from ℄

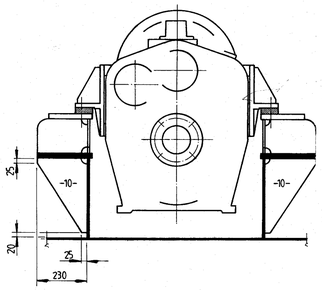


Direction of propellers

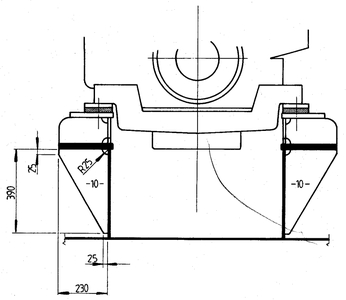
(seen to fore ship)



Section A-A



Section B-B



Tiller for emergency steering	2222.0006	222..2222..0001
Arrangement steeringgear		3029.222..00.001
Casting area		000..210..01..010
Mounting details main engine on foundation brackets		000..211..01..004
Mounting details reduction gear on foundation brackets		000..212..01..002
Arrangement and details foundation brackets		3029..211 ..00..001
Arrangement rudder-nozzle and sternfube		3029..210..00..700
Composition rudder installation	2140.0060	221..2141..0015
Composition propellershaft installation	2110.0211	213..2110..0051
Description	Article nr.	Drawing nr.

Reference drawing

rev.	date	name	revision subject
B	09..11..93	J.L.	Article nr. added
A	08..11..93	J.L.	Watertight compartment added (remark B.V.)

inspected	date
B.V.	27..10..93

DAMEN SHIPYARDS
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Engine arrangement

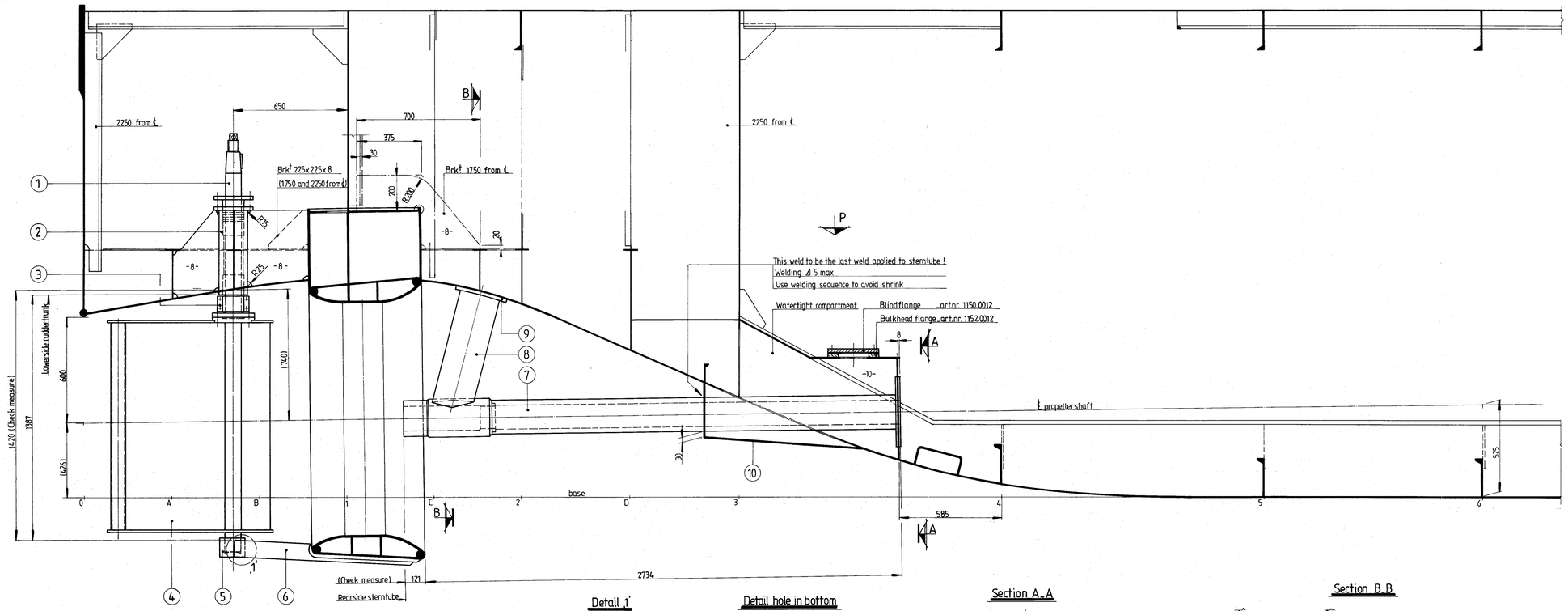
Main engine : Caterpillar 3408B TA
Reduction gearbox : Reintjes WAF 262

MultiCat 1908

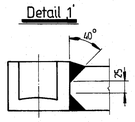
drawn: J.Leeuwis	date: 15..10..93	scale: 1:10	
checked	ident. to dwg. no.		
yard no.: 3029/3030	derived fr. dwg. no.		
COPYRIGHT BY DAMEN SHIPYARDS	dwg. no.: 3029..210..00..001		

0 cm 5 10 cm

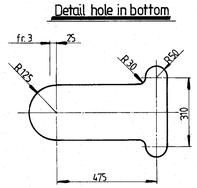
Longitudinal section 2000 mm from ℄



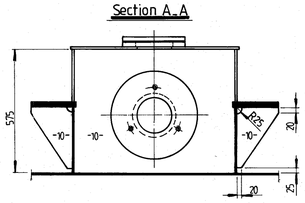
Detail 1'



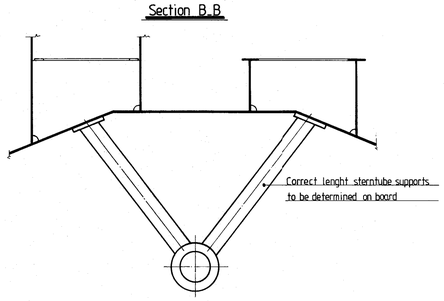
Detail hole in bottom



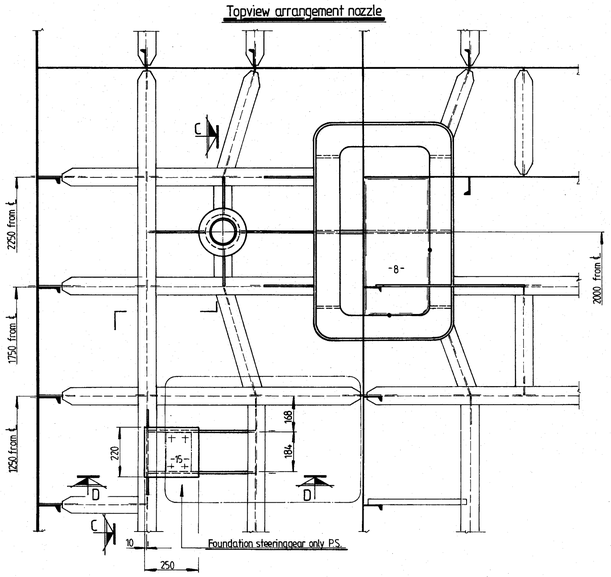
Section A-A



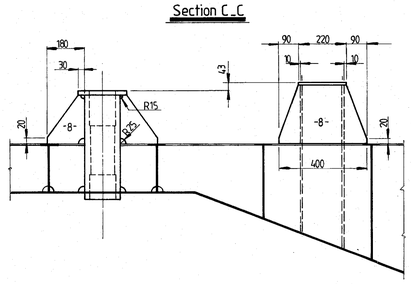
Section B-B



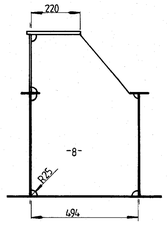
Topview arrangement nozzle



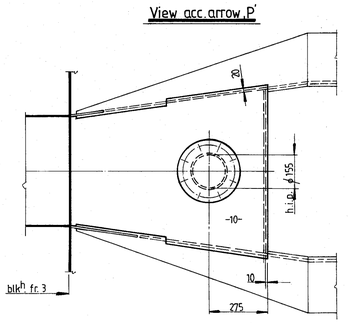
Section C-C



Section D-D



View acc. arrow P'



item	Description	Drawing nr.	Art.nr.
10	Bossing	Z13..1005..0001	1005.0006
9	Reinforcement plate	Z13..0070..0001	0070.0001
8	Sterntube support	Z13..2108.0001	2108.0003
7	Sterntube	Z13..2112..0009	2112.0028
6	Nozzle	Z13..2120..0021	2120.0021
5	Stepbearing	Z21..2144..0006	2144.0006
4	Rudder	Z21..2146..0016	2161.0056
3	Safety clamp	Z21..2145..0003	2145.0003
2	Rudderstrunk	Z21..2143..0005	2143.0006
1	Rudderstock	Z21..2142..0006	2142.0007
	Composition rudder installation	Z21..2141..0015	2141.0056

Reference drawings

rev.	date	name	revision subject
B	10..11..93	J.L.	Brkt 1750 from ℄
A	08..11..93	J.L.	Remarks B.V.

inspected	date
B.V.	27..10..93

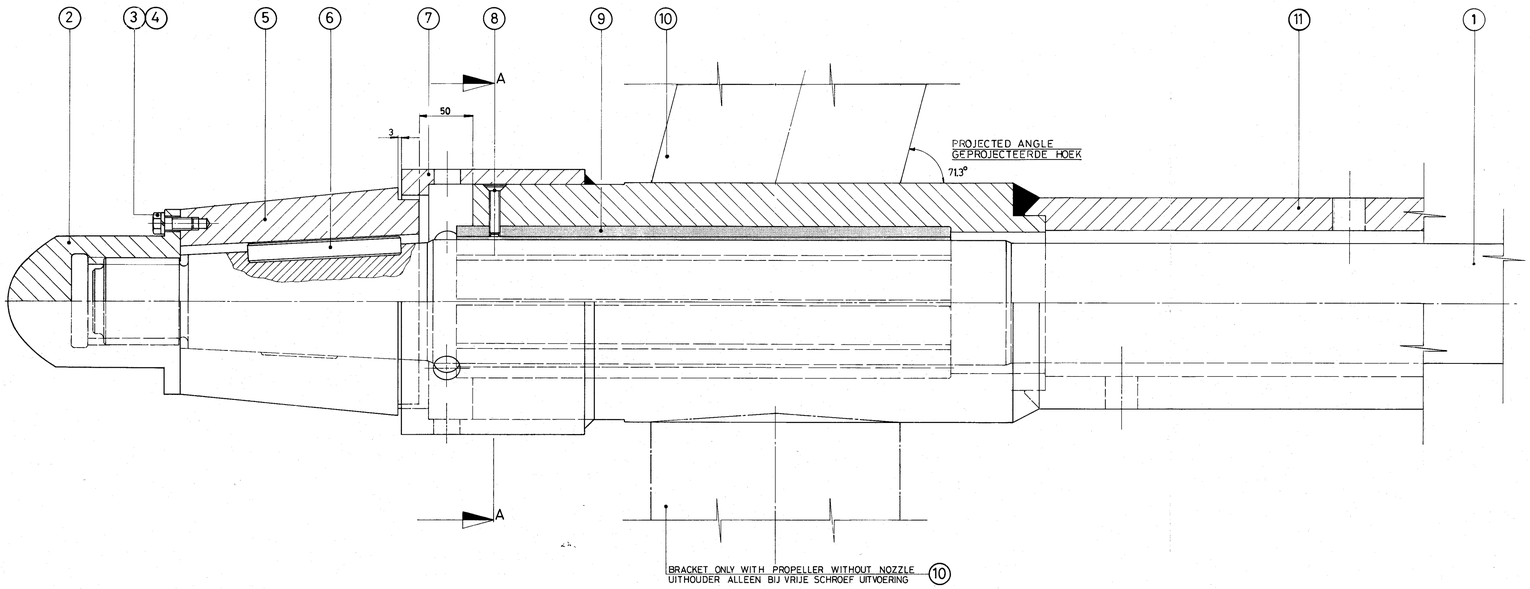
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

Arrangement rudder, nozzle and sterntube

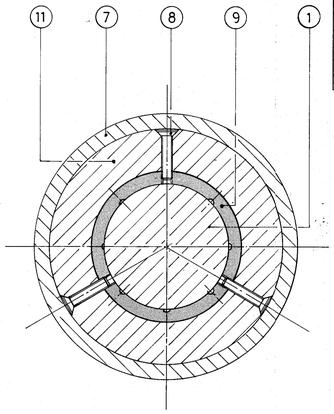
drawn J.Leeuwis | date 15..10..93 | scale 1:10
checked | client. to drwg. no.
yard no. 3029/3030 | derived fr. drwg. no.
COPYRIGHT BY DAMEN SHIPYARDS
drwg. no. 3029..210..00..700

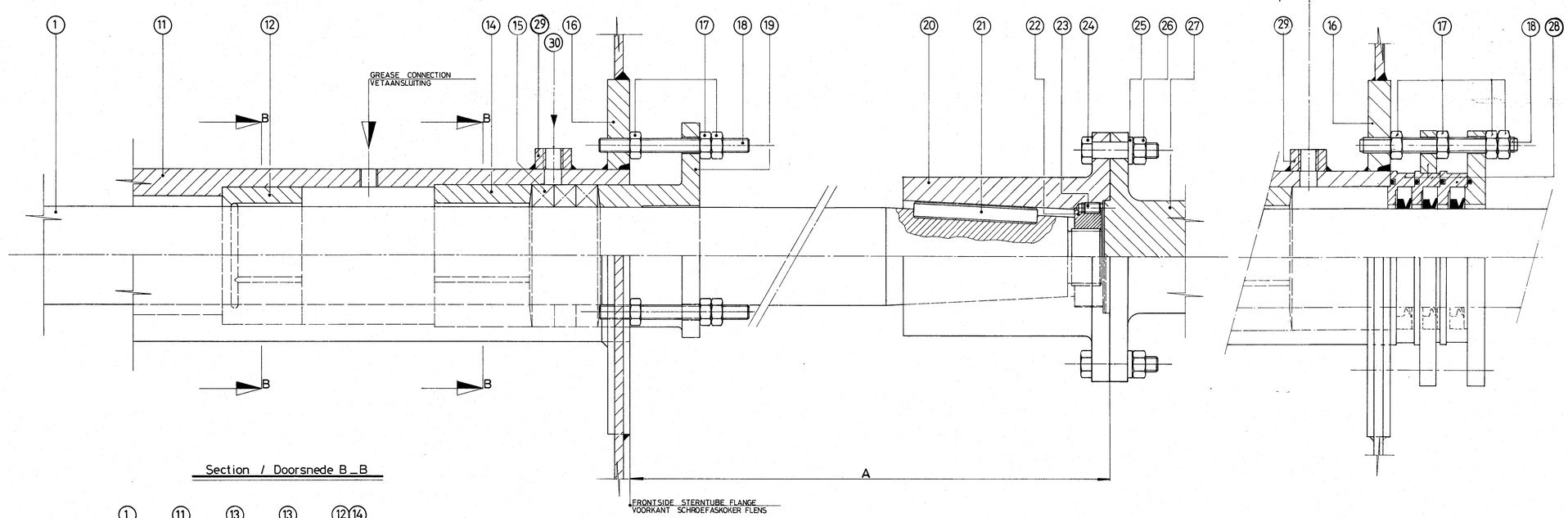
0 cm 5 10 cm

Art.nr.	Composition Type	A	Propeller shaft Drw213_2111_0010 Type	L	Sterntube Drw213_2112_0009 Type	L	Nr. of struts.	Forward seal.
2110.0210							2	Gland seal.
2110.0248							2	de Waal seal.
	110	999	110	4192	110	2855		
2110.0207							3	Gland seal.
2110.0249							3	de Waal seal.
2110.0211							2	Gland seal.
2110.0250							2	de Waal seal.
	111	814	111	4007	110	2855		
2110.0208							3	Gland seal.
2110.0251							3	de Waal seal.
2110.0212							2	Gland seal.
2110.0252							2	de Waal seal.
	112	902	112	4640	111	3400		
2110.0209							3	Gland seal.
2110.0253							3	de Waal seal.

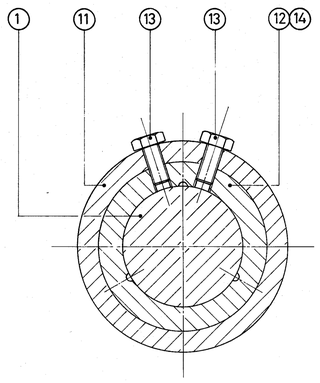


Section / Doorsnede A_A





Section / Doorsnede B_B

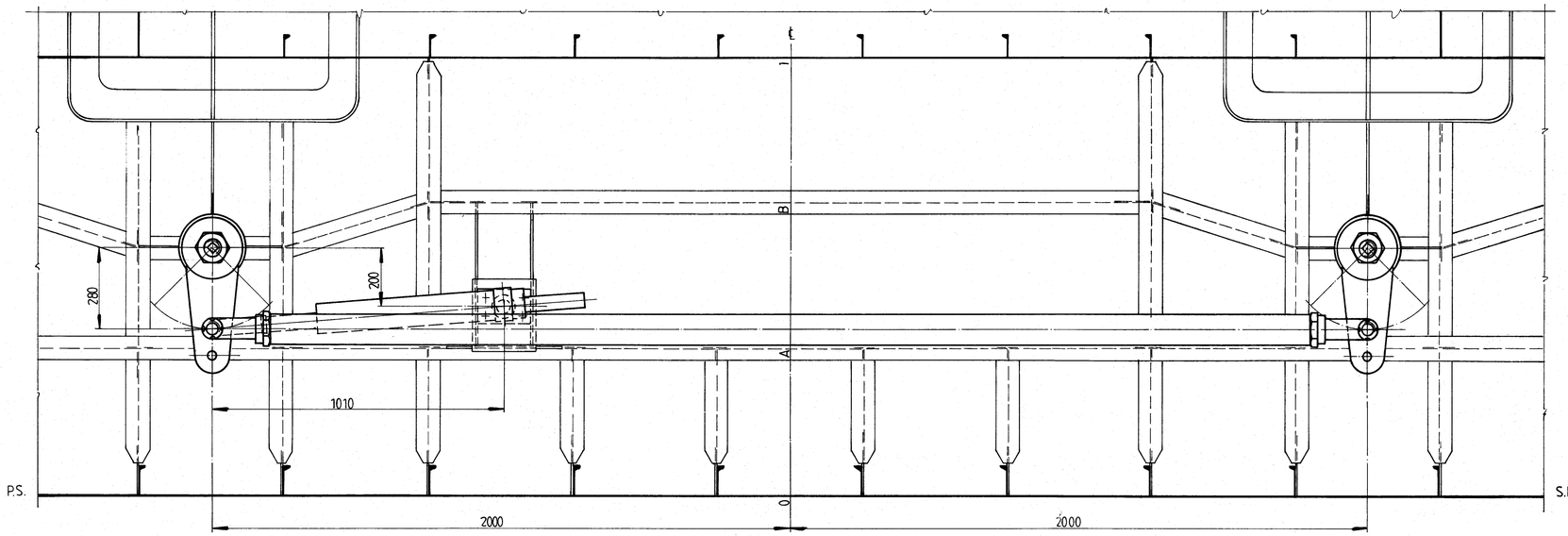


ITEM ST.NR.	27	26	25	24	20
	REINTJES WAF 262	TWIN DISC MG 516 H.R. 4,52:1	REINTJES WAF 360-460-540	TWIN DISC MG 509 ABOVE 3,83:1	TWIN DISC MG 514 ABOVE 4,13:1
DEL. REINTJES		1B 4208 UNF 7/8"-14	DEL. REINTJES		1B 4207 UNF 3½"-16
NOT APPL.		5 P 8249	NOT APPL.		A 21 DIN 125
DEL. REINTJES		6 F 7036 UNF 7/8"-14 L=3"	DEL. REINTJES		2B 1712 UNF 3½"-16 L=24"
		DRW 213_2116_0010 TYPE 110 Artnr 2116.0015		DRW 213_2116_0008 Artnr 2116.0014 TYPE 110	DRW 213_2116_0003 Artnr 2116.0012 TYPE 110
		DRW 213_2116_0011 Artnr 2116_0016 TYPE 110			DRW 213_2116_0007 Artnr 2116.0013 TYPE 110

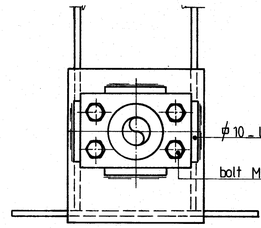
ITEM	QUAN.	DESCRIPTION	MAT.	DIMENSIONS / REMARKS
ST.NR.	AANT.	BESCHRIJVING	MAT.	AFMETINGEN / OPMERKINGEN
30	1	PLUG	ST	GAS 3/4" ART NR 0612.0129
29	1	SOCKET	ST	3/4" ART NR 1180.0002
28	1	DE WAAL SEAL		ART NR 2117.0022
27	1	RED.GEAR KEERKOPPELING		SEE TABLE ZIE TABEL
26	8	NUT MOER	ST	SEE TABLE ZIE TABEL
25	8	WASHER SLUITRING	ST	SEE TABLE ZIE TABEL
24	8	BOLT BOUT	C 22	SEE TABLE ZIE TABEL
23	1	LOCKING BOLT BORGBOUT		M 12 x 20 DIN 916
22	1	FLANGE NUT FLENSMOER	C 22	DRW 213_2115_0002 Art nr 2115_0008
21	1	KEY SPIE	ST 60_1k	DRW 200_1009_0001 Art nr 1009_0004
20	1	PROPELLER SHAFT FLANGE SCHROEFASFLENS		SEE NOTE ZIE NOOT
19	1	GLAND PAKKINGDRUKKER	CAST IRON GIET IJZER	DRW 213_2114_0004 (POS 4) ART NR 2117.0004
18	3	THREADEND DRAADEIND	ST	M 16 x 180
17	9/12	NUT MOER	BRASS MESSING	M 16 DIN 934
16	1	STERNTUBE FLANGE SCHROEFASKOKER FLENS	ST 42	DRW 213_2112_0009 (POS 4) TEK.
15	–	GREASE PACKING VETPAKKING		1" x 1000
14	1	BEARING (FORWARD) LAGER (VOOR)	RG 10	DRW 213_2114_0004 (POS 3) ART NR 2114.0004
13	4	LOCKING BOLT BORGBOUT	S.ST A4 R.V.S.	M 16 x 35 DIN 933
12	1	BEARING (INTERMEDIATE) LAGER (TUSSEN)	RG 10	DRW 213_2114_0004 (POS 2) ART NR 2114.0015
11	1	STERNTUBE SCHROEFASKOKER	ST	DRW213_2112_0009 (POS 2) TEK.
10	2/3	BRACKET UITHOUDER	ST 42	DRW213_2112_0009 (POS 3) TEK.
9	1	BEARING (AFT) LAGER (ACHTER)	RUBBER	DRW 213_2114_0004 (POS 1) ART.NR 2114.0022
8	3	LOCKING BOLT BORGBOUT	S.ST A4 R.V.S.	M 10 x 40 DIN 963
7	1	ROPE GUARD TROSBESCHERMER	ST 35	DRW 213_2112_0009 (POS 1) ART NR 2113_0004
6	1	KEY SPIE	S.ST 431 R.V.S.431	DRW 213_1009_0001 ART.NR 1009.0015
5	1	PROPELLER BOSS SCHROEFNAAF	BRONZE BRONS	DRW 000_213_01_002 TEK.
4	1	SPRINGWASHER VEERRING	S.ST A4 R.V.S.	A 12.2 DIN 127
3	1	BOLT BOUT	S.ST A4 R.V.S.	M 12 x 30 DIN 931
2	1	COLAR NUT KRAAGMOER	BRONZE RG 10	DRW 213_2111_0010 (POS 4) ART.NR 2115.0007
1	1	PROPELLER SHAFT SCHROEFAS	W.ST 14057	DRW 213_2111_0010 (POS 1)

gekeurd	datum	wijziging	datum
B.V. (F)	16_02_1989		

Topview



Detail mounting steeringgear on foundation



bolt M16x70 - 8.8 - DIN 931 with double nut M16 - 8 - DIN 934

Steeringgear : Sperry - type 2NS.10.45.1
Rudder torque 4 KNm
Rudder angle 2x45°

rev.	date	name	revision subject

inspected	date

DAMEN SHIPYARDS

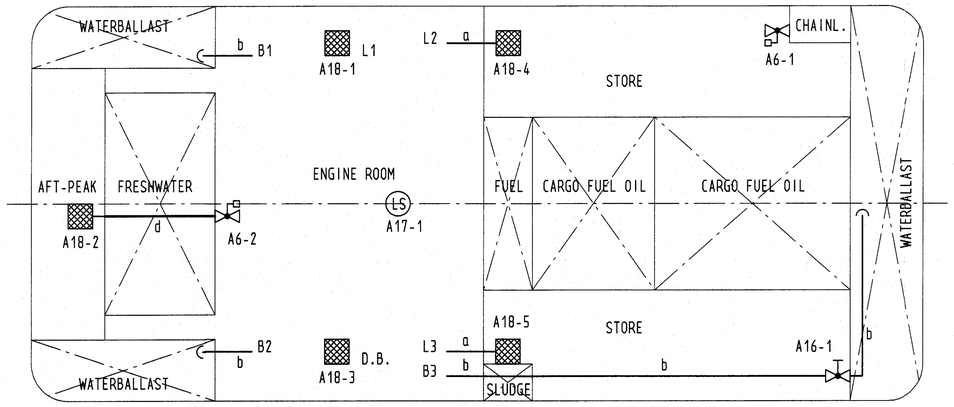
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland

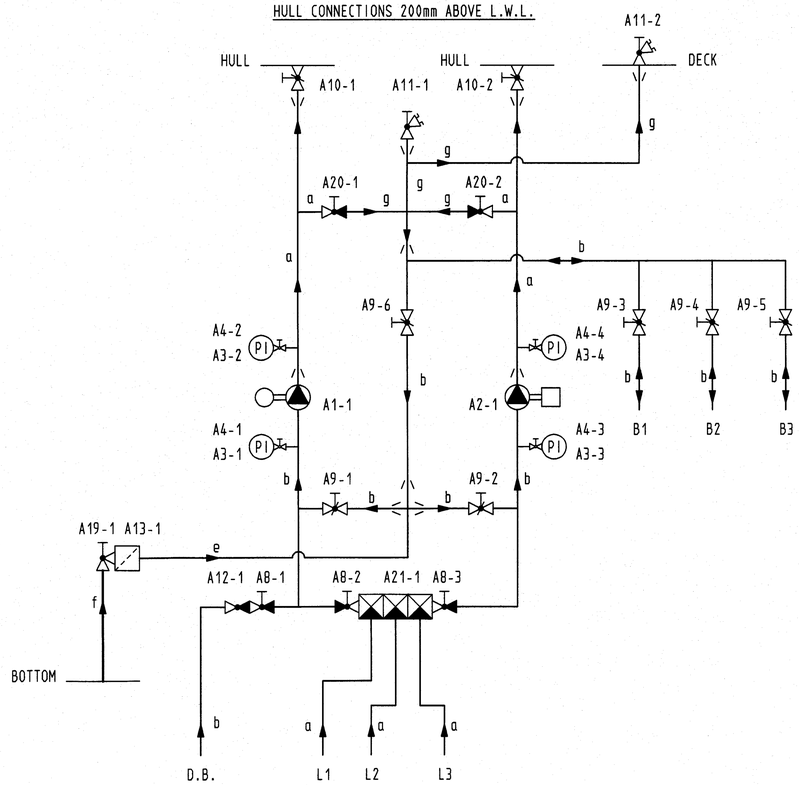
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

Arrangement steeringgear

drawn	date	scale
J.Leeuwis	22.10.'93	
checked	ident. to drwg. no.	
yard no. 3029/3030	derived fr. drwg. no.	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no. 3029.222.00.001	

0cm 5 10cm





PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
310-a	48.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
310-b	60.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
310-d	60.3 x 10 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
310-e	88.9 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
310-f	88.9 x 10 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
310-g	48.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	HOT DIP ZINC COATED

NOTES:
- ALL ECONOSTO VALVES DRILLED ACCORDING TO DIN 2501-PN10.
- VALVES IN THE ENGINE ROOM SHALL BE PLACED IN AN ACCESSIBLE POSITION, AND GROUPED IF POSSIBLE.
- PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE.
- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- PIPING SHALL BE INSTALLED TENSION-FREE.
- HOT DIP ZINC COATED ACCORDING TO NEN 1275 & 1332.
- 1/2' DRAIN PLUGS SHALL BE FITTED AT LOWEST POINTS OF ALL LINES.

ITEM	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS
310-A21	1	SDNR DISTRIBUTION BOX	440LK	3x DN 40 PN10, GG25	CAST IRON	ECON	
310-A20	2	SDNR VALVE	268LK	DN 40 PN16, drilled PN10, GG25/RG5, STRAIGHT	CAST IRON	ECON	
310-A19	1	SD VALVE	1271	DN 80 PN16, drilled PN10, RG5, ANGLED	BRONZE	ECON	APPROVED
310-A18	5	SUCTION BASKET		Perforation:3mm Zinc coated acc.NEN1275	STEEL		
310-A17	1	LEVEL SWITCH	LS-270E	NC, -20/+100 DEG., 10 BAR IP 67	STEEL	GEMS	HIGH LEVEL ALARM
310-A16	1	SD VALVE	1270	DN 50 PN16, drilled PN10, RG5, STRAIGHT	BRONZE	ECON	APPROVED B.V.
310-A13	1	STRAINER	1187	DN 80 PN 2.5, GG25, drilled PN10, STRAIGHT	CAST IRON	ECON	
310-A12	1	NR VALVE	102	DN 50 PN16, drilled PN10, GG25, ANGLED, horizontal	CAST IRON	ECON	
310-A11	2	FIRE HYDRANT	907/920	50x66mm, drPN10, RG5, 120 deg, incl. closure PN10	BRONZE	ECON	
310-A10	2	BUTTERFLY VALVE	4622 + gearbox	DN 50 PN 16, drilled PN 10 body GGG40 valve	NOD.CAST IRON	ECON	APPROVED
310-A9	6	BUTTERFLY VALVE	5722 + gearbox	DN 50 PN 16 body GG25 valve Alum/bronze w.	CAST IRON	ECON	
310-A8	3	SDNR VALVE	268LK	DN 50 PN16, drilled PN10, GG25/RG5, STRAIGHT	CAST IRON	ECON	
310-A6	2	SELFCLOSING VALVE	268ZS	50mm, drPN10, GG25/RG5 STRAIGHT PN16	CAST IRON	ECON	
310-A4	4	PRESSURE INDICATOR	382	G 1/4inch, -1.0/+5.0 bar, D=63mm	BRASS	ECON	
310-A3	4	PRESSURE INDICATOR VALVE	369	1/4inch, plug type, pmax= 6bar	BRASS	ECON	
310-A2	1	PUMP SET MECH DRIVEN	VW 20 S3 A.c.	G 2inch/G 2inch, 24m3/h, 1 bar pulley 1xB d=125	BRONZE/C.I.		
310-A1	1	PUMP SET ELECTR DRIVEN	VW 20 S3 A.c.	VW 20 S3A c, 2 inch, 35 m3/h, 1.0bar, 60Hz/3 kW,	BRONZE/C.I.	SIHI	440V/60HZ

rev.	date	name	revision subject
C	04-01-94	H.J. Cornege'	System updated
B	02-12-93	W.F.L.van Roon	ITEM A7 DELETED, A19 ADDED
A	19-11-93	W.F.L.van Roon	ITEM A1 CHANGED FROM 50 TO 60 Hz

inspected	date
B.V.	-

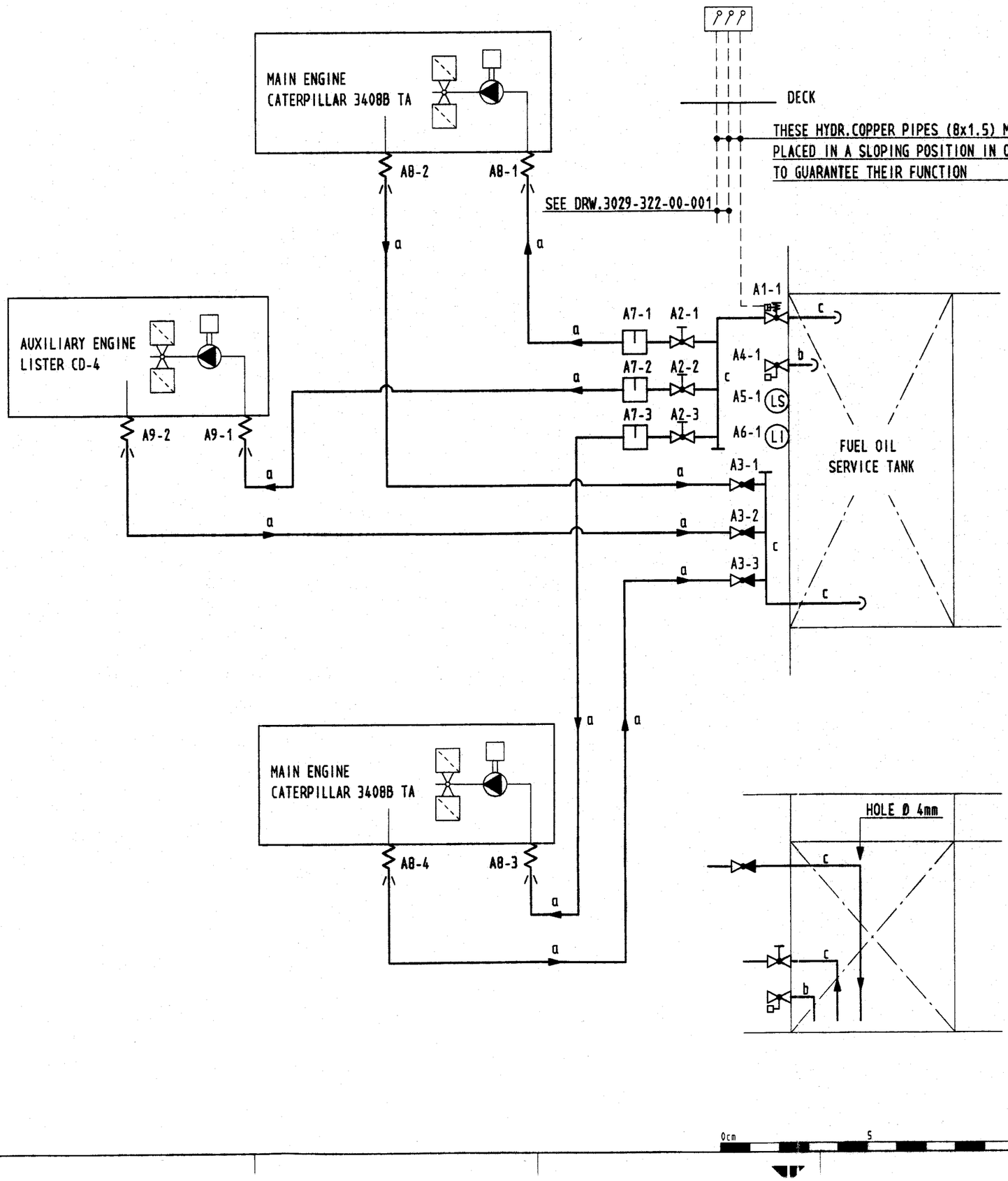
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 dame nl
Telefax: 31.1830.32189

BILGE, DECKWASH AND INTERNAL FIRE-FIGHTING SYSTEM

drawn: J.LEEUWIS	date: 26-10-93	scale:
checked:	ident. to drwg. no.:	
yard no.: 3029-3030	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-310-00-001	

0cm 5 10cm

PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
321-a	16 x 2 SEAML.PREC.DIN 2391 St.35.4 NBK	St 35.4 NBK	
321-b	33.7 x 2.6 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
321-c	60.3 x 2.9 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	



NOTE:- ALL PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE

- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- ALL PIPING SHALL BE INSTALLED TENSION-FREE.
- ALL PIPING SHALL BE CLEAN BEFORE INSTALLING.
- ALL ECONOSTO VALVES DRILLED ACCORDING TO DIN 2501-PN10.

	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS
321-A9	2	FLEXIBLE CONNECTION			REINF.RUBBER		APPROVED TYPE, DEL.D.B.R.
321-A8	4	FLEXIBLE CONNECTION			REINF.RUBBER		APPROVED TYPE, DEL.GEVEKE
321-A7	3	WATERSEPARATOR	5-80	405L/h, G 1/2Inch		HARWO	
321-A6	1	LEVEL INDICATOR CPL		1/2Inch selfcl. upper/T/Lower part		ECON	
321-A5	1	LEVEL SWITCH	S1A-1/F1	FLANGE 92x92, max 130 deg.C, type S1A-1/F1,		MOBRE	
321-A4	1	SELFCLOSING VALVE	574	DN 25, drilled PN10, ANGLED, weight at right	STEEL	ECON	
321-A3	3	NR VALVE	505BR	1/2Inch, RG5, STRAIGHT T	BRONZE	ECON	
321-A2	3	SD VALVE	1275	1/2Inch, PN16, RG5, , STRAIGHT, SEC.VALVEHEAD	BRONZE	ECON	
321-A1	1	SOS SD VALVE	100/417H	50mm, drPN10, C22, STRAIGHT PN40	STEEL	ECON	

rev.	date	name	revision subject

Inspected	date

DAMEN SHIPYARDS

Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland

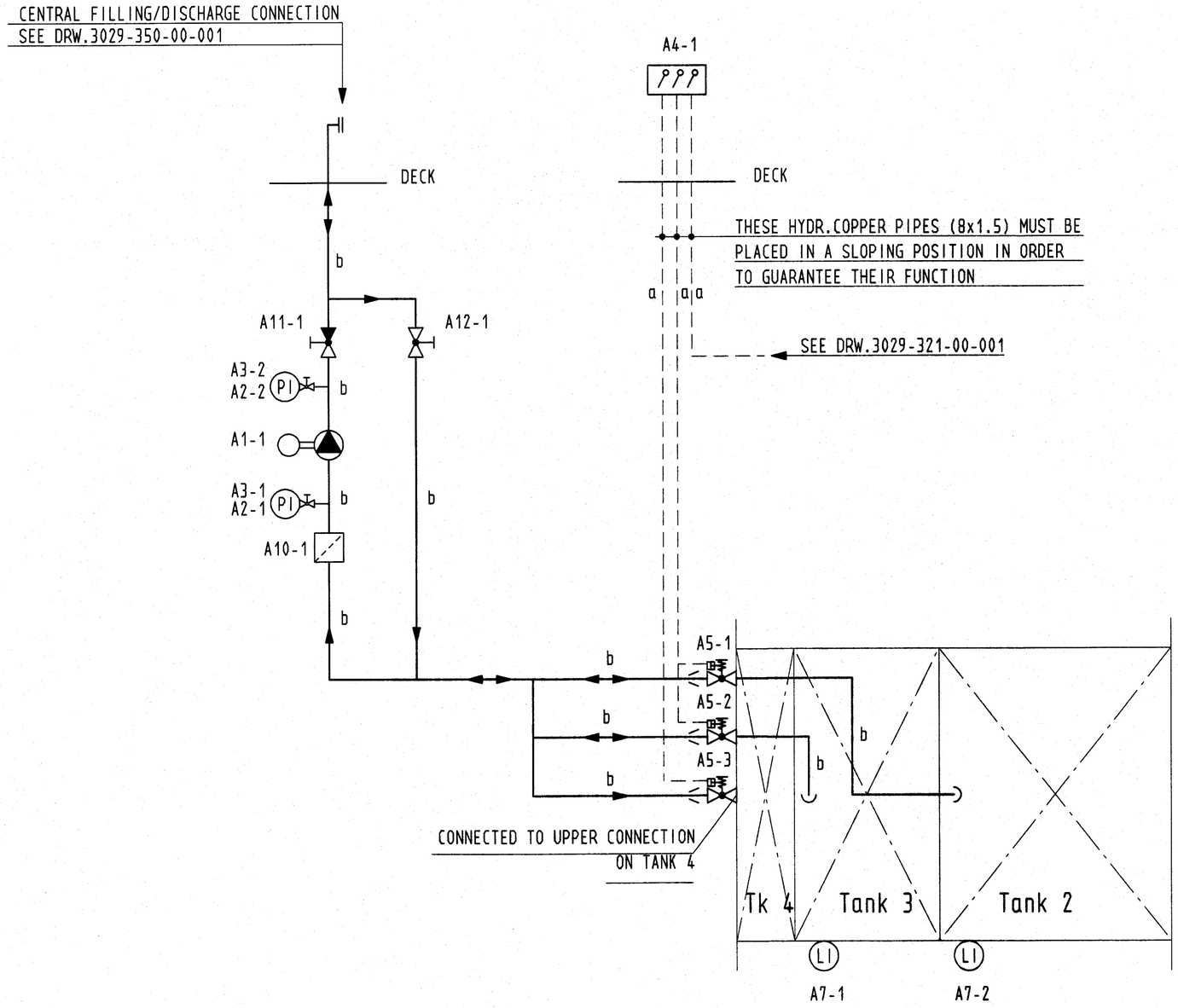
Tel.: 31.1830.39911*
Telex: 25335 dame nl
Telefax: 31.1830.32189

FUEL OIL SERVICE SYSTEM

drawn: J.LEEUWIS	date: 27-10-93	scale:
checked:	ident. to drwg. no.:	
yard no.: 3029-3030	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-321-00-001	

0cm 5 10cm

PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
322-a	8 x 1.5 DIN 1754 LRS and BV wall thickness	COPPER	
322-b	60.3 x 2.9 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	



NOTE:- ALL PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE
- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- ALL PIPING SHALL BE INSTALLED TENSION-FREE.
- ALL PIPING SHALL BE CLEAN BEFORE INSTALLING.
- ALL ECONOSTO VALVES DRILLED ACCORDING TO DIN 2501-PN10.
- EMERGENCY STOPS OF E-DRIVEN PUMPS MUST BE PLACED OUTSIDE THE ENGINE ROOM
- N.C. = NORMALLY CLOSED

ITEM	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS
322-A12	1	SD VALVE	275	DN 50 PN16, drilled PN10, RG5, STRAIGHT,	BRONZE	ECON	
322-A11	1	SDNR VALVE	275LK	DN 50 PN16, drilled PN10, RG5, STRAIGHT,	BRONZE	ECON	
322-A10	1	STRAINER	1019	DN 50 PN16, GG25, drilled PN10	CAST IRON	ECON	
322-A7	2	LEVEL INDICATOR CPL		1/2inch selfcl. upper/T/lower part		ECON	
322-A5	3	SOS SD VALVE	100/417H	65mm, drPN10, GS-C25, STRAIGHT PN40	STEEL	ECON	
322-A4	1	IMPULS UNIT	108	1/8inch, GG25, TRIPLE EX.	CAST IRON		
322-A3	2	PRESSURE INDICATOR	382	G 1/4inch, -1.0/+5.0 bar, D=63mm	BRASS	ECON	
322-A2	2	PRESSURE INDICATOR VALVE	369	1/4inch, plug type, pmax= 6bar	BRASS	ECON	
322-A1	1	PUMP SET ELECTR DRIVEN	AKHK 5101	AKHK 5101, 50mm, 24 m3/h, 1.5bar, 60Hz/5.5 kW, AN	BRONZE/C.I.	SIHI	440V/60HZ.

rev.	date	name	revision subject
D	05-01-94	H.J. Cornege"	Fillingline tank 4 added
C	04-01-94	H.J. Cornege"	Completely updated
B	01-12-93	W.F.L.van Roon	ITEM A6-7 REPLACED
A	19-11-93	W.F.L.van Roon	ITEM A1 CHANGED FROM 50 TO 60 Hz

inspected	date

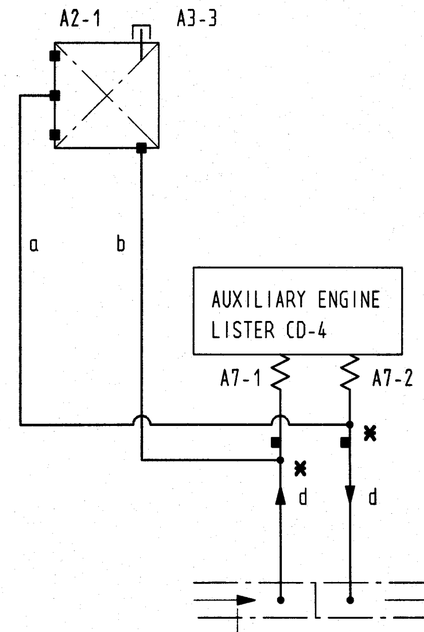
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 dame nl
Telefax: 31.1830.32189

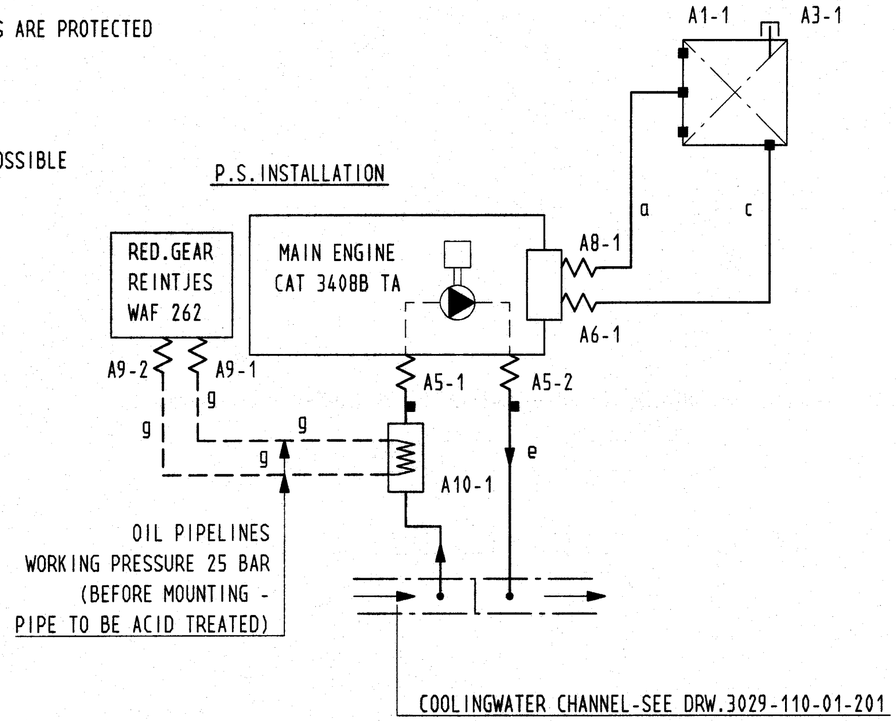
FUEL OIL CARGO/FILLING SYSTEM

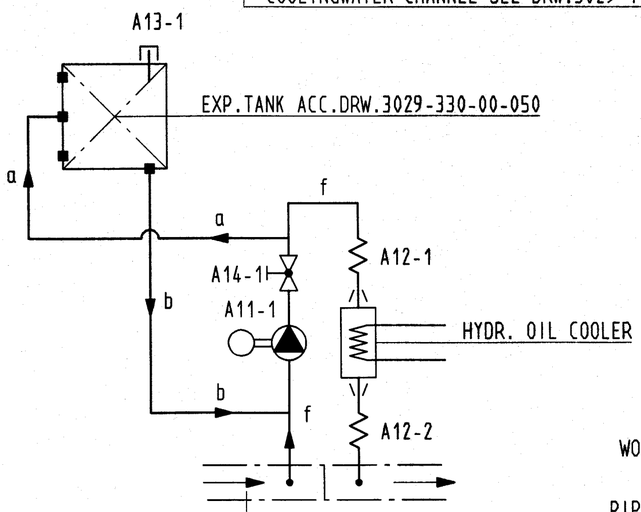
drawn: J.LEEUWIS	date: 02-11-93	scale:
checked:	ident. to drwg. no.:	
yard no.: 3029-3030	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-322-00-001	

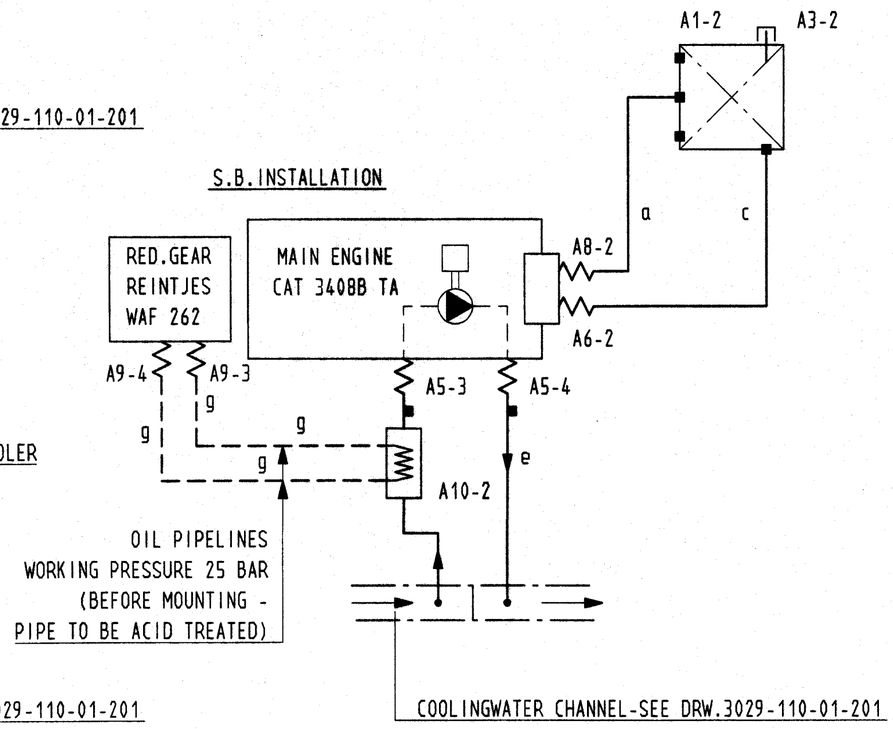
0cm 5 10cm

NOTE:
- ALL PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE
- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- ALL PIPING SHALL BE INSTALLED TENSION-FREE.
- ALL PIPING SHALL BE CLEAN BEFORE INSTALLING.
- ✖ CONNECTIONS FROM AND TO EXPANSION TANK AS HIGH AS POSSIBLE
- ▬■▬ 1/2' SOCKET









PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
330-a	10 x 1.5 DIN 1754 LRS and BV wall thickness	COPPER	
330-b	26.9 x 2.3 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
330-c	33.7 x 2.6 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
330-d	60.3 x 2.9 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
330-e	114.3 x 3.6 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
330-f	42.4 x 2.6 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
330-g	20 x 2 SEAML.PREC.DIN 2391 St.35.4 NBK	St 35.4 NBK	

ITEM	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS
330-A14	1	SD VALVE	268	DN 32 PN16, drilled PN10, GG25/RG5, STRAIGHT	CAST IRON	ECON	
330-A13	1	FILLER CAP	30kPa	Relief pressure 0.3bar			
330-A12	2	FLEXIBLE CONNECTION CPL		Compensator 3K DI= 42.4 with hose clamps			
330-A11	1	PUMP SET ELECTR DRIVEN	AOHA 3101	AOHA 3101,1 1/4Inch, 4.2m3/h,1,5bar,	CAST IRON	SIHI	440V/60HZ.
330-A10	2	HEAT EXCHANGER			STEEL		DEL. GEVEKE
330-A9	4	HOSE WITH FITTINGS		H.P. ID=16mm-OD=16mm l=650			DEL.GEVEKE
330-A8	2	HOSE WITH FITTINGS		H.P. ID=10mm-OD=10mm l=500			APPROVED TYPE
330-A7	2	FLEXIBLE CONNECTION CPL	DN 50	COMPL.WITH HOSE CLAMPS	RUBBER		DEL. D.B.R. APPROVED TYPE
330-A6	2	FLEXIBLE CONNECTION CPL	DN 25	COMPL.WITH HOSE CLAMPS	RUBBER		DEL.GEVEKE, APPROVED TYPE
330-A5	4	FLEXIBLE CONNECTION CPL	DN 100	COMPL.WITH HOSE CLAMPS	RUBBER		DEL.GEVEKE, APPROVED TYPE
330-A3	3	FILLER CAP	Caterpillar 330	For expansion tank coolingwater Cat	STEEL		DEL.GEVEKE
330-A2	1	EXPANSION TANK	30-1	Capacity 30 dm3	STEEL		DRW.330-0142-0003
330-A1	2	EXPANSION TANK	60-1	Capacity 60 dm3	STEEL		DRW.330-0142-0003

rev.	date	name	revision subject
C	24-01-94	MW vd Graaf	Pipe g changed to 20 x 2
B	22-11-93	W.F.L.van Roon	ITEM A11 CHANGED FROM 50 TO 60 Hz, ITEM A4 DELETED
A	15-11-93	W.F.L.van Roon	ITEM A9 UP TO A13 AND PIPE F & G ADDED

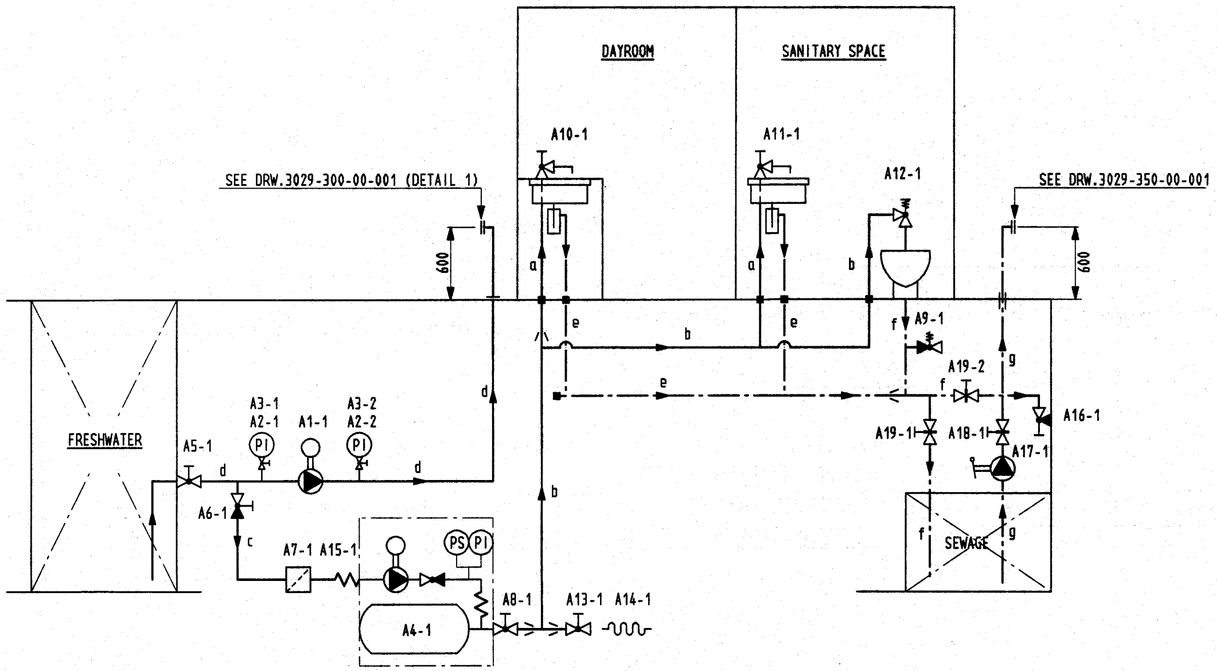
inspected	date

DAMEN SHIPYARDS
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P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 dame nl
Telefax: 31.1830.32189

COOLINGWATER SYSTEM

drawn: J.LEEUWIS	date: 27-10-93	scale:
checked:	ident. to drwg. no.:	
yard no.: 3029-3030	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-330-00-001	

0cm 5 10cm



PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
340-a	12 x 1 KIWA	COPPER	
340-b	22 x 1.1 KIWA	COPPER	
340-c	33.7 x 4 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	HOT DIP ZINC COATED
340-d	60.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	HOT DIP ZINC COATED
340-e	60.3 x 8 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
340-f	114.3 x 10 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
340-g	48.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	

NOTE:
- FOR SANITARY COMBINATIONS SEE SEPARATE LIST
- HOT DIP ZINC COATED ACC.TO NEN 1275 & 1332.
- ALL PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE
- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- ALL PIPING SHALL BE INSTALLED TENSION-FREE.
- ALL PIPING SHALL BE CLEAN BEFORE INSTALLING.
- ALL ECONOSTO VALVES DRILLED ACCORDING TO DIN 2501-PN10
- ■ ANBLOCKING PLUGS 1'

FRESH WATER LINES
DRAIN/DISCHARGE LINES

ITEM	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS
340-A19	2	GATE VALVE	317N	DN100 PN10, GG25, STRAIGHT	CAST IRON	ECON	
340-A18	1	GATE VALVE	290	1 1/2inch, PN30, RG5, STRAIGHT	BRONZE	ECON	
340-A17	1	PUMP MANUAL DRIVEN	Henderson MkV U	Cap 0.9dm3/stroke approx. 100dm3/min conn 38mm			
340-A16	1	STORM VALVE	1207	DN100 PN10, GS-C25, ANGLED, LOCKING DEVICE	STEEL	ECON	APPR.B.VERITAS
340-A15	1	FLEXIBLE CONNECTION CPL		DN 25 hose with 4 clamps L=150			
340-A14	1	HOSE MATERIAL	Multibar	DI= 13 Do= 20, plastic with crosslink synthetic	SYNTHETIC MAT.		L=10M.
340-A13	1	SANITARY COMBINATION	Combination S	Tap with cold freshwater, comb. S			
340-A12	1	SANITARY COMBINATION		For toilet, flushed by sea/fresh water			
340-A11	1	SANITARY COMBINATION		Washbowl 52cm with cold fresh water			
340-A10	1	SANITARY COMBINATION		For galley, with cold fresh water			
340-A9	1	NR VALVE	752329	1/2 INCH AERATION VALVE,	NICKEL-PL.BR.	T.U.	DEL. PIPE INSTALL
340-A8	1	SD VALVE	775130/774349	1 INCH, INCL.DRAINCOCK,	BRASS	T.U.	DEL.PIPE INSTALL.
340-A7	1	STRAINER	1017	DN 25 PN16, RG5, drilled PN10	BRONZE	ECON	
340-A6	1	SDNR VALVE	275LK	DN 25 PN16, drilled PN10, RG5, STRAIGHT,	BRONZE	ECON	
340-A5	1	SD VALVE	275	DN 50 PN16, drilled PN10, RG5, STRAIGHT,	BRONZE	ECON	
340-A4	1	HYDROPHORE	PM15	25 l/min, 110ltr, 24V,		SPECK	
340-A3	2	PRESSURE INDICATOR	382	G 1/4inch, -1.0/+5.0 bar, D=63mm	BRASS	ECON	
340-A2	2	PRESSURE INDICATOR VALVE	369	1/4inch, plug type, pmax= 6bar	BRASS	ECON	
340-A1	1	PUMP SET ELECTR DRIVEN	AKHK 5101	AKHK 5101, 50mm, 24 m3/h, 1.5bar, 60Hz/5.5 kW, BN	BRONZE	SIHI	440V/60HZ.

rev.	date	name	revision subject
B	19-11-93	W.F.L.van Raan	ITEM A4 changed to 110 ltr
A	19-11-93	W.F.L.van Raan	ITEM A17,18,19 & PIPE G ADDED A1 CHANGED IN 60 Hz

inspected	date

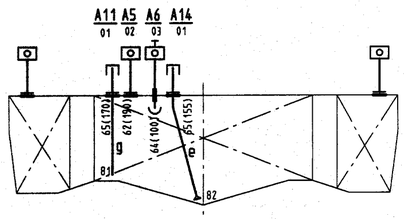
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damo nl
Telefax: 31.1830.32189

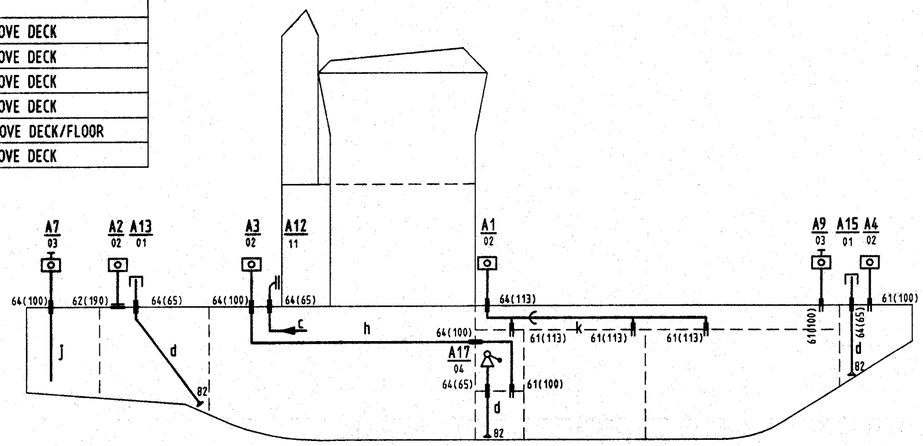
FRESHWATER CARGO AND SANITARY SYSTEM

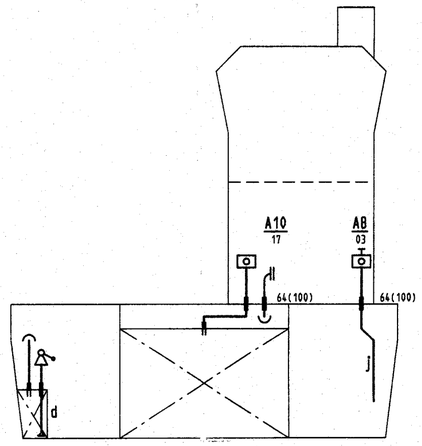
drawn: J.LEEUWIS	date: 02-11-93	scale:
checked:	ident. to dwg. no.:	
yard no.: 3029-3030	derived fr. dwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	dwg. no.: 3029-340-00-001	

SYMBOLS

◉	SOUNDING PIPE	
□	FILLING/DISCHARGE CONNECTION	600mm ABOVE DECK
⊡	DEAERATION CAP	760mm ABOVE DECK
⊡	DEAERATION CAP WITH CLOSING DEVICE	760mm ABOVE DECK
⌒	DEAERATION 'GOOSENECK TYPE'	760mm ABOVE DECK
⊼	SELFCLOSING SOUNDING COCK	1000mm ABOVE DECK/FLOOR
⊓	SOUNDING OR FILLING CAP	600mm ABOVE DECK

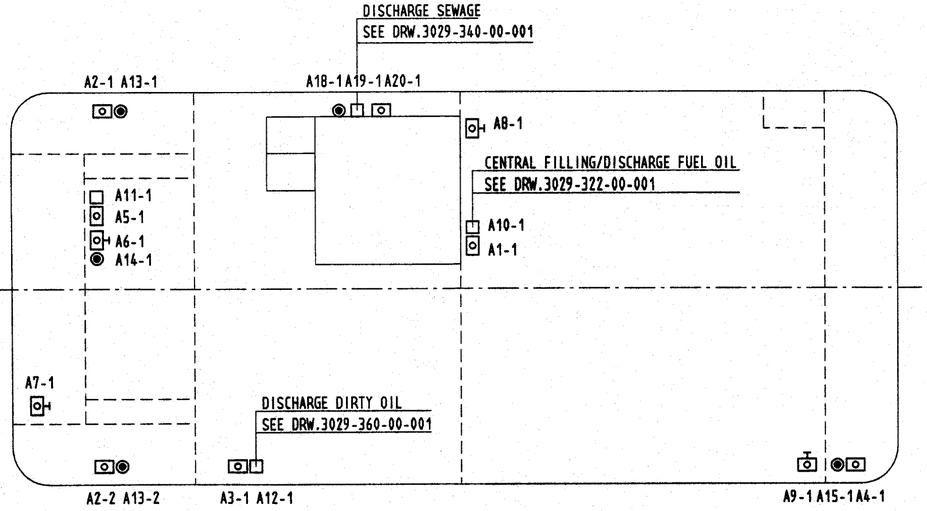






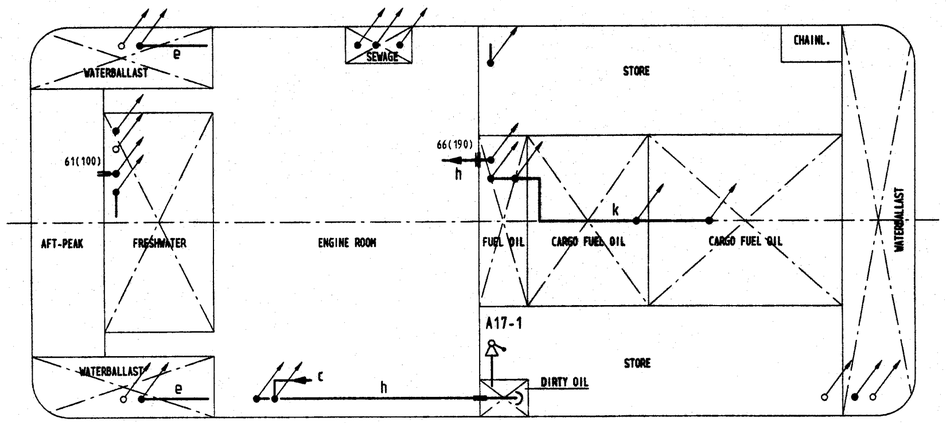
PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
350-a	10 x 1 KIWA	COPPER	
350-c	48.3 x 2.6 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-d	48.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-e	48.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	HOT DIP ZINC COATED
350-f	48.3 x 10 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-g	60.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	HOT DIP ZINC COATED
350-h	76.1 x 2.9 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-j	76.1 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	HOT DIP ZINC COATED
350-k	88.9 x 3.2 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-l	60.3 x 5 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-m	60.3 x 2.9 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	
350-n	60.3 x 10 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	

NOTE:
- FOR DECKPASSAGES CODES SEE DRW.000-350-01-001
- HOT DIP ZINC COATED ACC.TO NEN 1275 & 1332.
- ALL PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE
- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- ALL PIPING SHALL BE INSTALLED TENSION-FREE.
- ALL PIPING SHALL BE CLEAN BEFORE INSTALLING.



CRANKCASE DEAERATION MAIN-ENGINE (2x)

CRANKCASE DEAERATION GENERATORSET (1x)



ITEM	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS
350-A20	1	DE-AERATION DEVICE		DN 50, H=760, Code:02-64-61	STEEL		
350-A19	1	FILL/DISCHARGE DEVICE		DN 40, H=600, Code:12-64	STEEL		
350-A18	1	SOUNDING DEVICE		DN 40, H=600, Code:01-66-64-B2	STEEL		
350-A17	1	SOUNDING DEVICE		DN 40, H=1000, Code:04-64-B2	STEEL		
350-A15	1	SOUNDING DEVICE		DN 40, H=600, Code:01-64-B2	STEEL		
350-A14	1	SOUNDING DEVICE		DN 40, H=600, Code:01-65-B2	STEEL		HOT DIP ZINC COATED
350-A13	2	SOUNDING DEVICE		DN 40, H=600, Code:01-64-B2	STEEL		
350-A12	1	FILL/DISCHARGE DEVICE		DN 40, H=600, Code:11-64	STEEL		
350-A11	1	FILL/DISCHARGE DEVICE		DN 50, H=600, Code:01-65-B1	STEEL		HOT DIP ZINC COATED
350-A10	1	FILL/DISCHARGE DEVICE		DN 65, H=600, Code:17-64-66	STEEL		
350-A9	1	DE-AERATION DEVICE		DN 65, H=760, Code:03-61	STEEL		
350-A8	1	DE-AERATION DEVICE		DN 65, H=760, Code:03-64	STEEL		HOT DIP ZINC COATED
350-A7	1	DE-AERATION DEVICE		DN 65, H=760, Code:03-64	STEEL		HOT DIP ZINC COATED
350-A6	1	DE-AERATION DEVICE		DN 65, H=760, Code:03-64-61	STEEL		HOT DIP ZINC COATED
350-A5	1	DE-AERATION DEVICE		DN 65, H=760, Code:02-62	STEEL		
350-A4	1	DE-AERATION DEVICE		DN 65, H=760, Code:02-61	STEEL		
350-A3	1	DE-AERATION DEVICE		DN 65, H=760, Code:02-64-64-61	STEEL		
350-A2	2	DE-AERATION DEVICE		DN 65, H=760, Code:02-62	STEEL		
350-A1	1	DE-AERATION DEVICE		DN 80, H=760, Code:02-64-61-61-61	STEEL		

rev.	date	name	revision subject
B	01-02-94	MW vd Graaf	Cr.case deae.M.E changed to DN 50
A	18-11-93	W.F.L.van Roon	ITEM A18,A19,A20,PIPE L & SEWAGE TANK ADDED

inspected	date

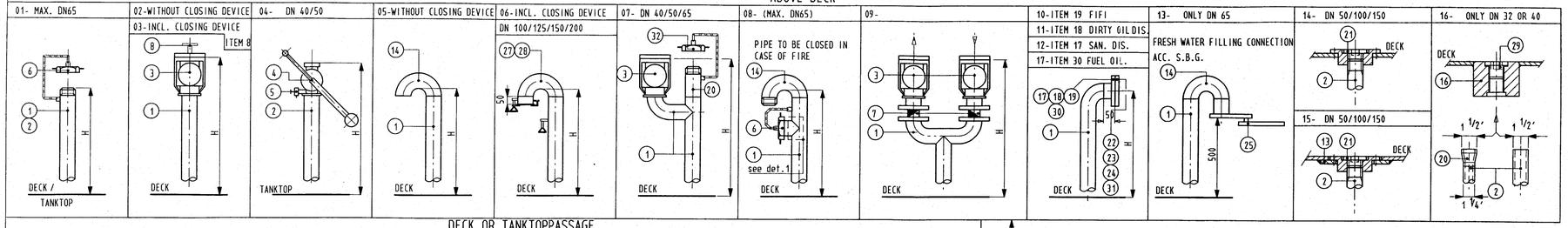
DAMEN SHIPYARDS
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Tel.: 31.1830.39911
Telex: 25335 dame nl
Telefax: 31.1830.32189

FILLING,SOUNDING AND DEAERATION SYSTEM

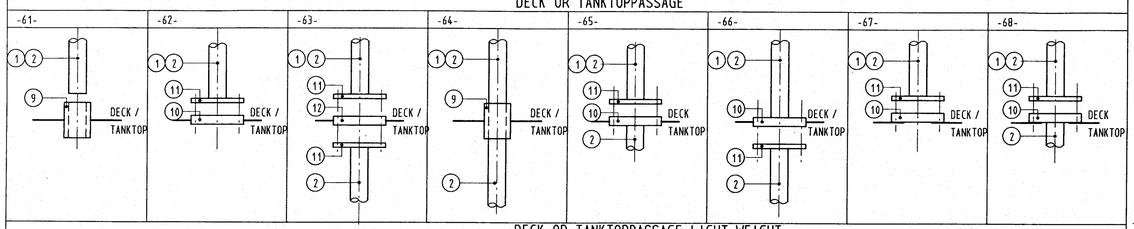
drawn: J.LEEUWIS	date: 03-11-93	scale:
checked:	ident. to drwg. no.:	
yard no.: 3029-3030	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-350-00-001	

0cm 5 10cm

ABOVE DECK

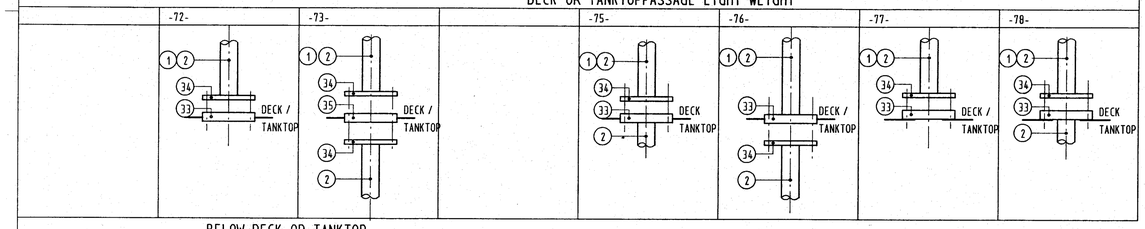


DECK OR TANKTOPPASSAGE



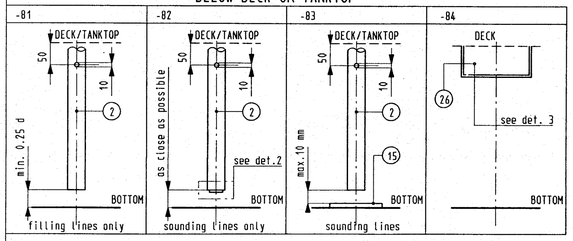
FOR NORMAL USE

DECK OR TANKTOPPASSAGE LIGHT WEIGHT



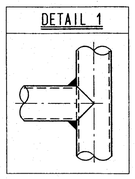
FOR LIGHT WEIGHT SHIPS ONLY

BELOW DECK OR TANKTOP

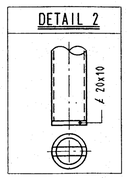


			ART. No.								REMARKS	MAKE
35	BULKHEADFLANGE DOUBLE (LIGHT WEIGHT)		1152.0093	1152.0094	1152.0095	1152.0096	1152.0097	1152.0098	1152.0099	1152.0100	1152.0101	double tapped	
			155-100-04	165.110.04	180.125.04	200.145.04	215.160.08	235.180.08	265.210.08	300.240.08	355.295.08		
34	WELDING FLANGE (LIGHT WEIGHT)		1151.0025	1151.0026	1151.0027	1151.0028	1151.0029	1151.0030	1151.0031	1151.0032	1151.0034	plain slip-on	
			140-100-04	150.110.04	165.125.04	185.145.04	200.160.08	220.180.08	250.210.08	285.240.08	340.295.08		
33	BULKHEADFLANGE SINGLE (LIGHT WEIGHT)		1152.0035	1152.0036	1152.0037	1152.0038	1152.0039	1152.0040	1152.0041	1152.0042	1152.0043	single tapped.	
			140-100-04	150.110.04	165.125.04	185.145.04	200.160.08	220.180.08	250.210.08	286.240.08	340.295.08		
32	COVER			0097.0005	0097.0006	0097.0007							
31	BLIND FLANGE fueloil			1150.0010	1150.0010	1150.0010	1150.0010	1150.0010					
				220.180.08	220.180.08	220.180.08	220.180.08	220.180.08					
30	FUEL OIL (SHORE CONN)			3534.0017	3534.0018	3534.0019	3534.0020	3534.0021				DRW. 350-3534-0001	
29	DRAIN PLUG			0127.0006									KROON 769006
28	DE-AERATOR H=900							1275.0031	1275.0032	1275.0033	1275.0034	DRW. 350-1275-0027	
27	DE-AERATOR H=760							1275.0027	1275.0028	1275.0029	1275.0030	DRW. 350-1275-0027	
26	SLOSHING DEVICE			0125.0001	0125.0001	0125.0001	0125.0001					DRW. 350-0121-0001	
25	FRESH WATER FILL FLANGE					0121.0001						DRW. 350-1150-0046	
24	BLIND FLANGE	FIFI		1150.0048	1150.0048	1150.0048	1150.0048					DRW. 350-1150-0046	
23		DIRTY OIL DISCHARGE		1150.0047	1150.0047	1150.0047	1150.0047	1150.0047	1150.0047			DRW. 350-1150-0046	
22		SANITARY DISCHARGE		1150.0046	1150.0046	1150.0046	1150.0046	1150.0046					
21	DECK DOWEL				0122.0001			0122.0002		0122.0003			
20	REDUCER		1170.0010	1170.0010	1170.0013	1170.0016	1170.0018	1170.0021				DIN 2616	
			48.3 / 42.4	48.3 / 42.4	60.3 /48.3	76.1 / 60.3	88.9 / 60.3	114.3 / 76.1					
19	FIFI FLANGE			3534.0012	3534.0013	3534.0014	3534.0015					DRW. 350-3534-0001	
18	DIRTY OIL DISCH.FLANGE			3534.0006	3534.0007	3534.0008	3534.0009	3534.0010	3534.0011			DRW. 350-3534-0001	
17	SANITARY DISCH.FLANGE			3534.0001	3534.0002	3534.0003	3534.0004	3534.0005				DRW. 350-3534-0001	
16	DECK DOWEL		0122.0004	0122.0004								DRW. 350-0122-0004	
15	STRIKINGPLATE			0124.0001	0124.0001	0124.0001	0124.0001					DRW. 350-0124-0001	
14	WELDING BEND		0017.0077	0017.0045	0017.0046	0017.0047	0017.0048	0017.0049	0017.0050	0017.0051	0017.0052	ANSI B 16.28	
			42.4 x 6.3	48.3 x 7.1	60.3 x 8.7	74 x 7	88.9 x 7.6	114.3 x 8.5	141.3 x 9.5	168.3 x 11	219.1 x 8.2		
13	DECK DOWEL FLANGE				0123.0001			0123.0002		0123.0003		DRW. 350-0123-0001	
12	BULKHEADFLANGE DOUBLE		1152.0020	1152.0021	1152.0022	1152.0023	1152.0024	1152.0025	1152.0026	1152.0027	1152.0028	double tapped	
			140-100-04	150.110.04	165.125.04	185.145.04	200.160.08	220.180.08	250.210.08	285.240.08	340.295.08		
11	WELDING FLANGE		1151.0005	1151.0006	1151.0007	1151.0008	1151.0009	1151.0010	1151.0011	1151.0012	1151.0014	plain slip-on	
			140-100-04	150.110.04	165.125.04	185.145.04	200.160.08	220.180.08	250.210.08	285.240.08	340.295.08		
10	BULKHEADFLANGE SINGLE		1152.0005	1152.0006	1152.0007	1152.0008	1152.0009	1152.0010	1152.0011	1152.0012	1152.0013	single tapped.	
			140-100-04	150.110.04	165.125.04	185.145.04	200.160.08	220.180.08	250.210.08	286.240.08	340.295.08		
9	DECKPASSAGE (PIPE)		0263.0019	0263.0003	0263.0004	0263.0005	0263.0006	0263.0007	0263.0008	0263.0009	0263.0010	Seaml.Steel. DIN 2448.	
			57 x 6.3	60.3 x 5	76.1 x 7.1	95 X 8.8	108 X 8	133 X 8	159 X 8	193.7 X 10	244.5 X 10		
8	DE-AERATOR (WITH CLOSURE)				1275.0014	1275.0017	1275.0018	1275.0021	1275.0022	1275.0024	1275.0026	DRW. 350-1275-0001	WIKO 5000-1
7	NR VALVE				3501.0236	3501.0237	3501.0238	3501.0239					ECON 2624
6	COVER		0097.0005	0097.0006	0097.0007	0097.0008							WINEL 211
5	TESTCOCK			0126.0001	0126.0001								ECON 727
4	SELFCL.SOUND.PIPE COCK			3522.0011	3522.0012								ECON 724
3	DEARERATION				1275.0001	1275.0004	1275.0005	1275.0008	1275.0009	1275.0011	1275.0013	DRW. 350-1275-0001	WIKO 5000-1
2	PIPES BELOW MAIN DECK		SEE CORRESPONDING SCHEMES										
1	PIPES ABOVE MAIN DECK		1140.0061	1140.0062	1140.0063	1140.0064	1140.0065	1140.0066	1140.0067	1140.0055	1140.0056	Seaml.Steel. DIN 2448.	
			42.4 x 6.3	48.3 x 6.3	60.3 x 6.3	76.1 x 6.3	88.9 x 7.1	114. X 8	139.7 X 8	168.3 X 10	219.1 X 10		
ITEM	DESCRIPTION / DN		32	40	50	65	80	100	125	150	200		

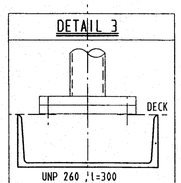
DETAIL 1



DETAIL 2



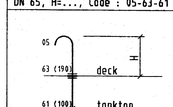
DETAIL 3



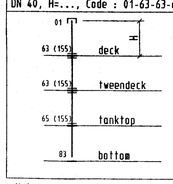
UNP 260 ,l=300
To be mounted in longitudinal direction of ship

EXAMPLES

DN 65, H=..., Code : 05-63-61



DN 40, H=..., Code : 01-63-63-83

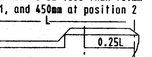


Note: measurments between brackets (...) are hole in plate

REMARKS

- G=hot dip zinc coated acc.NEN 1275
- DN=nominal diameter (de-aerators to be mounted with opening facing bkhd/bulwark
- deaerators size 50,65 and 80 to be provided with flame screens(only fuel oil)
- filling sounding and deaeration lines above deck to be provided with protection clamps
- cross sectional area of deaeration pipes to be 25% greater than the effective area of respective filling pipes
- sounding pipes are not to be less than 32mm bore
- deaeration pipes are not to be less than 50mm bore

- overflow height H for deaeration pipes for spaces, is not to be less than 900mm at position 1, and 760mm at position 2
- overflow height H for deaeration pipes for tanks, is not to be less than 760mm at position 1, and 450mm at position 2



position 1:exposed freeboard decks,exposed raised half decks,exposed decks of super structure within 0.25 L
position 2:other exposed decks of super-structure

See also relevant rules

rev.	date	name	revision subject
S	25-11-93	B.W.V	LIGHT WEIGHT CODES ADDED,DN 32 ADDED ,ITEM 16 UPDA
R	20-08-93	G.D	ITEM 29,ART CODE CHANGED
Q	15-06-93	G.D	ITEM 9,ART.NRS CHANGED ,ITEM 07 ITEMNR.
P	26-04-93	AvanDiest	ITEM 1,9,10,11,12,14,20.REVISED
N	05-06-92	G.D	CODE 67 AND 68 ADDED.
M	23-10-92	G.D	CODE 61,62,64,65 AND 66 REVISED

inspected	date
Lloyd's	23-01-92

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6200 AA Gorinchem/Holland
Tel.: 31.1830.39911a
Telex: 35335 damn nl
Telefax: 31.1830.32109

CODE LIST FILLING,SOUNDING AND DE-AERATION SCHEME.

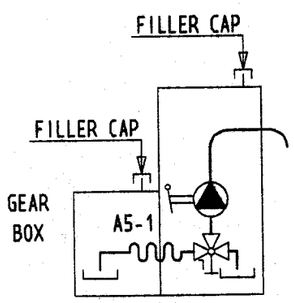
drawn: STANDATA	date: 27-04-92	scale: 1:5
checked:	ident. to dwg. no.:	
yard no.:	derived fr. dwg. no.: 1-0040	
COPYRIGHT BY DAMEN SHIPYARDS	dwg. no.: 000-350-01-001	

LET OP !
Wijzigingen altijd doorgeven aan Technische Automatisering

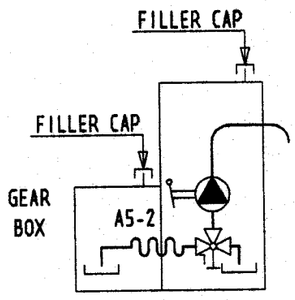
0 5 10cm

x6

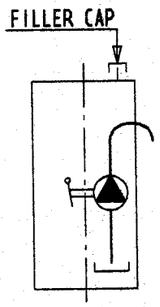
PIPE LETTER	SPECIFICATIONS	MATERIAL	REMARKS
360-a	16 x 2 SEAML.PREC.DIN 2391 St.35.4 NBK	St 35.4 NBK	
360-b	48.3 x 2.6 SEAMLESS DIN 2448/1629/3 St 37.0	St 37.0	



P.S. MAIN ENGINE



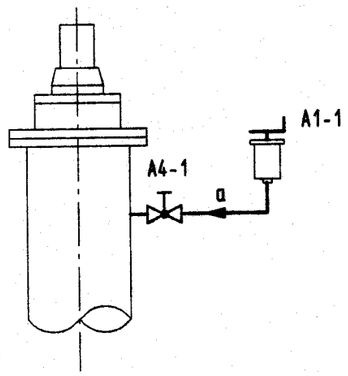
S.B. MAIN ENGINE



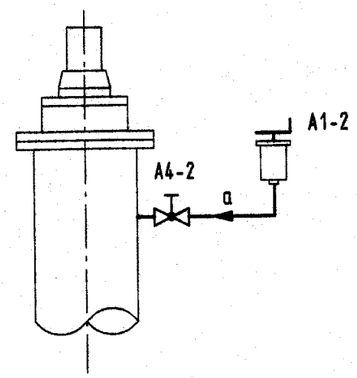
GENERATOR SET

NOTE:
- ALL PIPING SHALL BE ROUTED SUCH THAT IT IS PROTECTED AGAINST MECHANICAL DAMAGE
- ALL BRACKETING SHALL BE EXECUTED SUCH THAT THE PIPES ARE PROTECTED AGAINST DAMAGE BY VIBRATION.
- ALL PIPING SHALL BE INSTALLED TENSION-FREE.
- ALL PIPING SHALL BE CLEAN BEFORE INSTALLING.

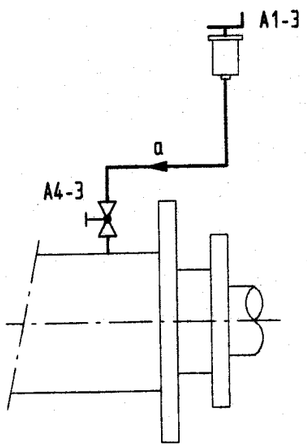
P.S.RUDDER INSTALLATION



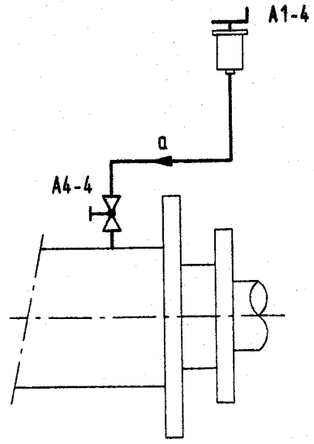
S.B.RUDDER INSTALLATION

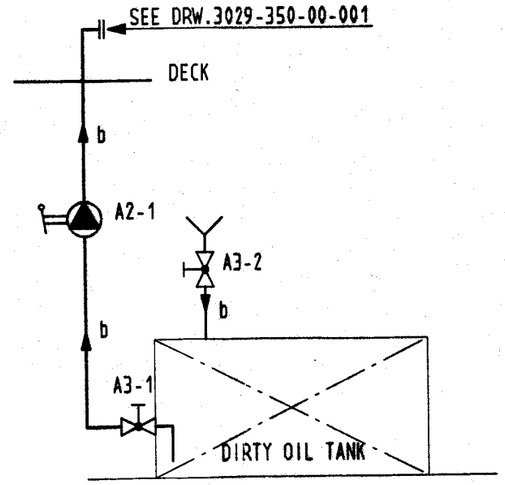


P.S.PROPELLER SHAFT INSTALLATION



S.B.PROPELLER SHAFT INSTALLATION





360-A5	2	HOSE MATERIAL		FOR DRAINING CRANKCASE			DEL.GEVEKE, APPROVED TYPE
360-A4	4	SD VALVE	250	1/2inch, PN16, RG5, , STRAIGHT	BRONZE	ECON	
360-A3	2	SD VALVE	1275	1 1/4inch, PN16, RG5, STRAIGHT, SEC.VALVEHEAD	BRONZE	ECON	
360-A2	1	PUMP MANUAL DRIVEN	950	1 1/2inch, SIZE 5	CAST IRON	ECON	
360-A1	4	GREASE DEVICE		Spindle type, manual	STEEL		
ITEM	QT	DESCRIPTION	TYPE	SPECIFICATIONS	MATERIAL	MANUF	REMARKS

rev.	date	name	revision subject

Inspected	date
B.V.	

DAMEN SHIPYARDS

Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland

Tel.: 31.1830.39911*
Telex: 25335 dame nl
Telefax: 31.1830.32189

LUBRICATION SYSTEM

drawn: J.LEEUWIS	date: 28-10-93	scale:
checked:	ident. to drwg. no.:	
yard no.: 3029-3030	derived fr. drwg. no.:	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no.: 3029-360-00-001	

0cm 5 10cm

Hatch Nr1 with Coaming
Drw: 131-1220-0001
Art 1220.0017

Coaming
Drw: 131-1229-0001
Art: 1229.0003

Hatch Nr: 5
Drw: 371 0056 0001
Art: 0056.0015

Hatch Nr:6
Drw: 371-0056-0001
Art: 0056.0016

Door: A 60
Drw: 710-0155-0001
Art: 0155.0018

Flush Hatch Nr:4
Drw:3029-131-00-001

Deckcrane
Typ: HEILA
Foundation
Drw:3029-540-00-001

Anchor Winch
Typ: R & H 16-2
Foundation
Drw: 3929-510-00-001

575 390
R25
L 50x50x5
600
425
3355
1500
2000
3500
550
1000
600
-15-
80
"P"
2
3
4
6
8
10
12

DOOR: B15
Drw. 710-0155-0001
Art. 0155.0003

Towrail
Drw. 3029-550-00-001

Railing verwijderen

Aanzicht "P"
Pipes 70x5/ 54x5
100
opmerkte
ϕ70x5
UNP80
90
20
vrijmaken T.b.v. monteren blok

Rollerhawse

Detail A

Towing bitt
Drw:3029-153-00-001

Bollard
Drw: 153-1232-0001
Art: 1232.0004

Bulwark after
Drw: 185-110-01-500

Hatch Nr: 2
Drw: 131-1220-0001
Art: 1220-0003

Coaming
Drw: 131-1229-0001
Art: 1229.0003

Eyeplate for tyre fender
drw. 000-151-01-003 (19x)

Hatch Nr: 7
Drw. 371-0056-0001
Art. 0056-0017

Door: 1440x740
Drw: 133-1250-0010
Art: 1250.0046

Flush Hatch Nr. 3
Drw 3029-131-00-001

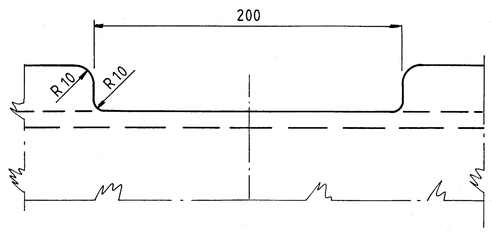
Towing Winch.
Typ: KA 40
Foundation
Drw: 3029-550-00-010

Bollard
Drw: 153-1232-0001
Art: 1232.0004

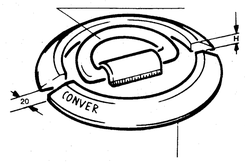
Bulwark fore
Drw:3029-110-00-500

Pushbow
Drw:3029-152-00-001

DETAIL A



LASHING EYE
ZA - 3 - 50



0cm 5 10cm

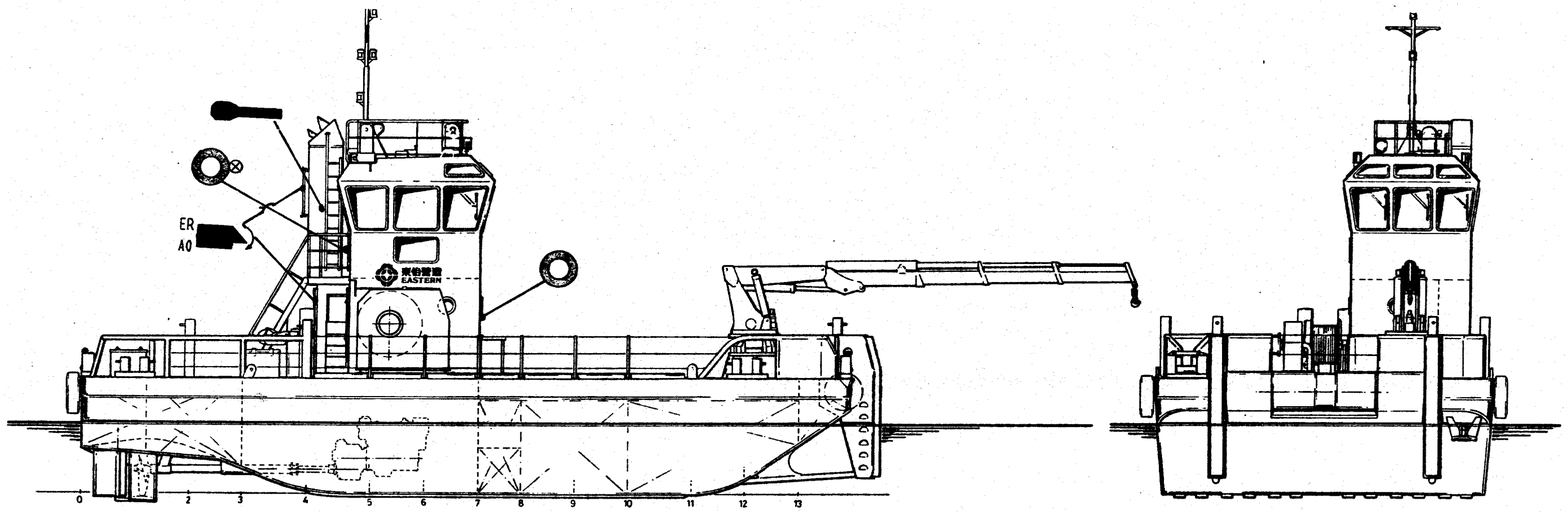
rev.	date	name	revision subject
D	11-01-'94	S.v.d.L.	Fender fr 4/11 only SB
C	14-12-'93	S.v.d.L.	Found. cradle liferaft
B	26-11-'93	S.v.d.Linden	Rollerhawse.
A	22-11-'93	F. Zillig	Towing Winch

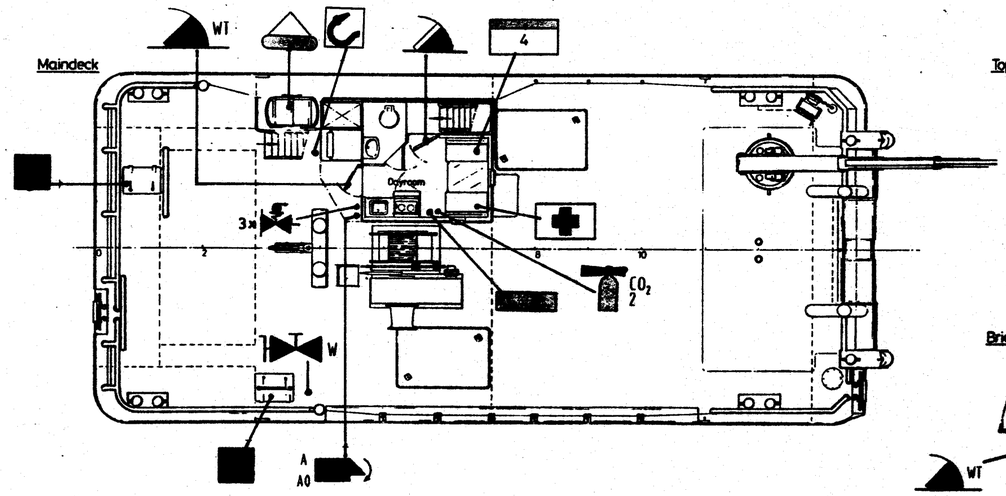
inspected	date
LLOYDS	23-04-'93
B.V.	25-10-'93

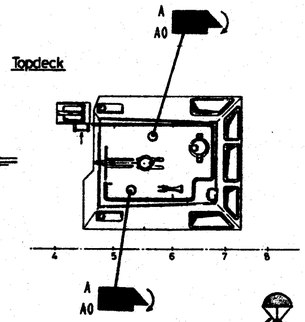
DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

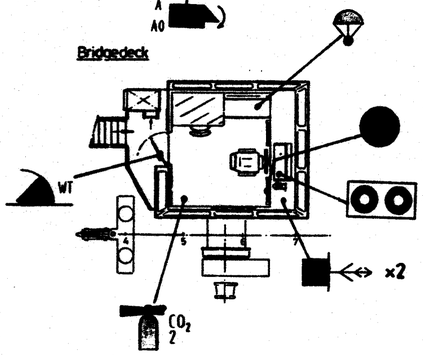
ARRANGEMENT MAINDECK

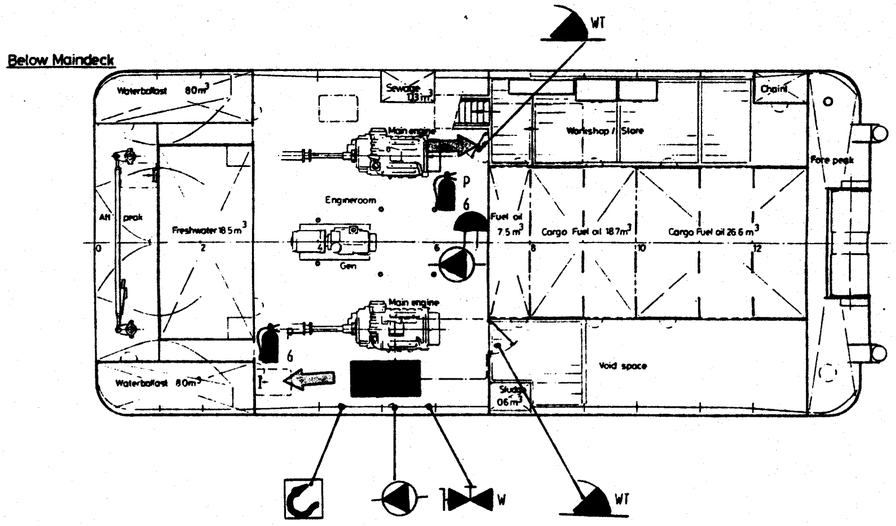
drawn: Idelenburg	date 15-03-93	scale 1:25	
checked	ident. to drwg. no.		
yard no. 3029/30	derived fr. drwg. no.		
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no 3029-500-00-001		



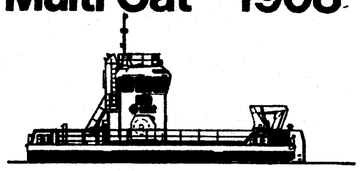








Damen Multi Cat 1908



Principal dimensions:

Length o.a.	18.70m
Beam o.a.	8.06m
Depth at sides (at half length)	2.75m

0cm 5 10cm

Symbol	Quant.	Description
		明牌
	1	Safety plan 救生滅火佈置圖
	1	Liferaft 6 persons 救生筏 6人份
	1	Lifebuoy 救生圈
	1	Lifebuoy with self igniting light 救生圈附自然燈
	4	Life jackets with whistle and light 救生衣附口笛和燈
	2	Portable VHF 手提發報机
	1	Emergency position indicating radio beacon (EPIRB) 緊急無線電指示位置
	1	Set of four rocket parachute flares 高空降落傘信號 4個
	3	Remote shut-off fuel oil valves 遙控燃油緊急切斷閥
A AO	3	Fire damper ventilation in- or outlet accommodation class AO 住艙区通風阻火擋板開閉裝置
ER AO	2	Fire damper ventilation in- or outlet engine room class AO 機艙区通風阻火擋板開閉裝置
	1	Fire door 滅火門
P 6	2	Powder type extinguisher 6 kg 乾粉滅火器 6公斤
CO_2	2	CO2 type extinguisher 2 kg 二氧化碳滅火器 2公斤
	2	Fire/general service pump 滅火，一般用水泵
W	2	Hydrant valve water 滅火栓
	2	Fire fighting hose with spray/jet fire nozzle length 10 m 滅火軟管及噴嘴，噴霧式，水柱式 10m
		Escape route 逃生通道
WT	4	Watertight door 水密門
	2	Watertight hatch 水密艙蓋
	1	General alarm push button 一般警鈴電鈕
	1	General alarm bell and light 一般警鐘和燈
	1	Emergency stop main engines 主机緊急停止裝置
	1	First aid kit 急救箱
	1	Battery set 電池箱

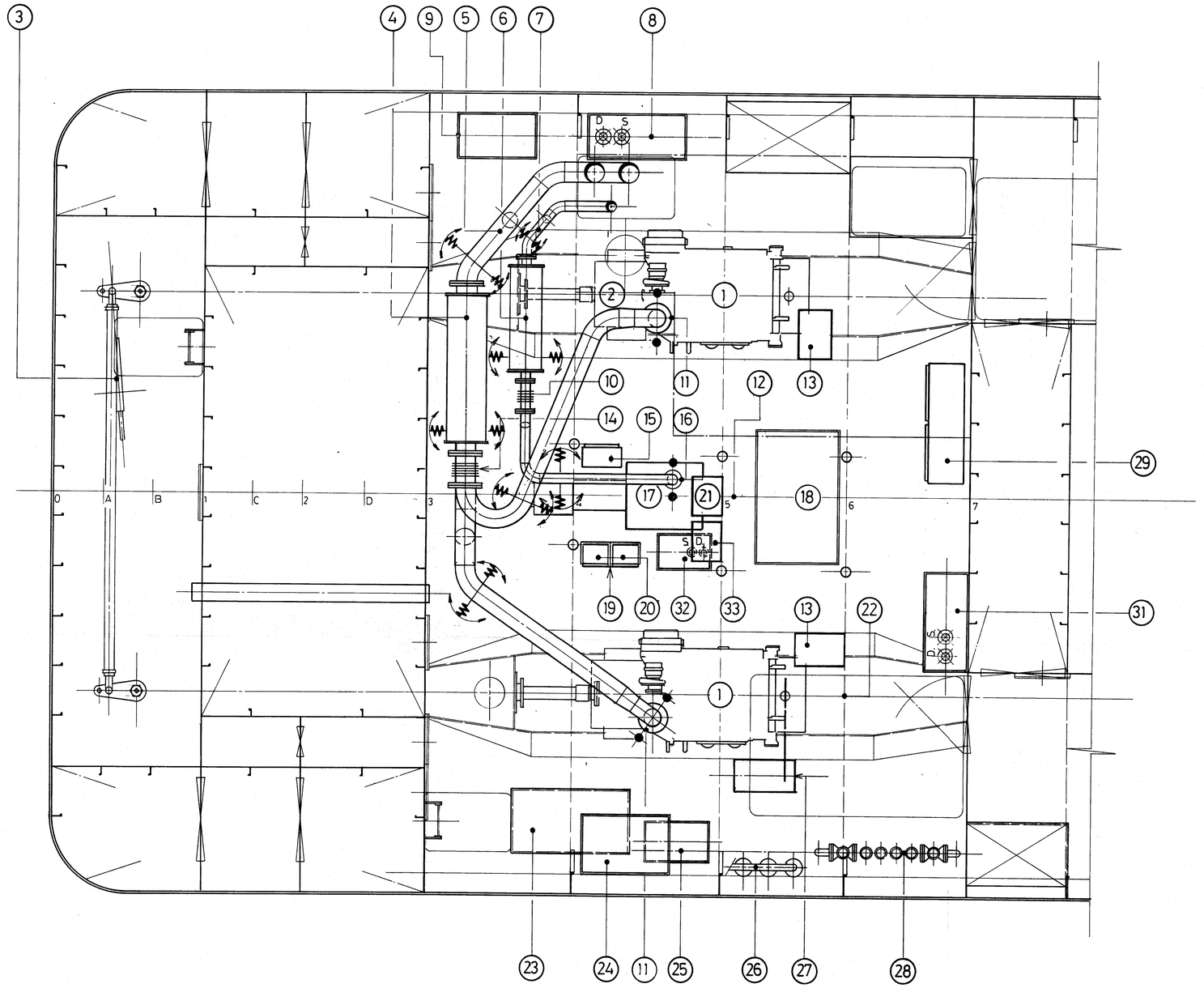
inspected	date
BV	16-11-'93

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Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

SAFETY PLAN

drawn F.A. van Leersum	date 11-2-'94	scale —
checked	ident. to drwg. no.	
yard no. 3029	derived fr. drwg. no.	
COPYRIGHT BY DAMEN SHIPYARDS	drwg. no. 3029 - 570 - 00 - 001	

Topview



ITEM	QUAN	DESCRIPTION	MAT.	DIMENSIONS / REMARKS
1	2	MAINENGINE		CAT. 3408 B _DI_TA
2	2	GEARBOX		REINTJES WAF 262 RED. 5.05 : 1
3	1	STEERINGGEAR INSTALLATION	ST	DRW. 3029_222_00_001
4	2	SILENCER	ST	TYPE AB_200
				DEL. MERCUREX
5	_	EXHAUST PIPE	ST	ø 219.1 x 4
6	1	SILENCER	ST	TYPE RA 25_100
				DEL. MERCUREX
7	_	EXHAUST PIPE	ST	ø 114.3 x 3.6
8	1	FRESHWATER CARGO PUMP	BRONZE/C.I.	TYPE AKHK 5101
				DEL. SIHI 19.7CUM/HR _ 1.5 BAR
9	1	HYDROPHORE		TYPE PM 15/55 _ 24 V
				DEL. SPECK
10	1	COMPENSATOR	ST.ST	TYPE DN 100
			AISI 321	DEL. R.D.
11	2	COMPENSATOR	ST.ST	ON MAINENGINE TYPE DN 150
			AISI 321	DEL. GEVEKE
12	1	HYDR. PUMP CRANE		ON AUX. SET
				DEL. TRANS_HYDRO
13	2	EXP. TANK	ST	DRW. 330_0142_0003
				ART. NO. 0142_0004
14	2	COMPENSATOR	ST.ST	TYPE DN 200
			AISI 321	DEL. R.D.
15	1	CONTROL BOX AUX. SET	ST	400 x 400 x 210
				DEL. DBR (LOOSE)
16	1	COMPENSATOR	ST.ST	ON AUX. SET
			AISI 321	DEL. DBR.
17	1	AUX. SET		LISTER
				STAMFORD
18	1	HYDR. TANK CRANE/WINCH	ST	
				DEL. TRANS_HYDRO
19	1	BATTERY CHANGE_OVER BOX		600 x 600 x 210
20	1	BATTERY CHARGER		
21	1	EXP. TANK	ST	DRW. 330_0142_0003
				ART. NO. 0142_0003
22	1	HYDR. PUMP WINCH		ON S.B MAINENGINE
				DEL. TRANS_HYDRO
23	1	BATTERY BOX	ST	DRW 410 4100_0001
				ART. NO. 4100_0003
24	1	HYDR. TANK STEERINGGEAR	ST	
				DEL. SPERRY
25	1	BILGE PUMP (ELECTR. DRIVEN)	BRONZE/C.I.	TYPE VW 20 S 3 A.c.
				DEL. SIHI 24 CUM/HR _ 10 BAR
26	1	BALLAST MANIFOLD	CAST IRON	DRW. 3029_310_00_001
27	1	BILGE PUMP (ENGINE DRIVEN)	BRONZE/C.I.	TYPE VW 20 S 3 A.c.
				DEL. SIHI 24 CUM/HR _ 10 BAR
28	1	BILGE MANIFOLD	CAST IRON	DRW. 3029_310_00_001
29	1	SWITCHBOARD		1200 x 1200 x 450
30				
31	1	F.O. CARGO PUMP	BRONZE/C.I.	TYPE AKHK 5101
				DEL SIHI 19.7 CUM/HR _ 1.5 BAR
32	1	COOLINGWATER PUMP HYDR. TANK	CAST IRON	TYPE AOHA 3101
				DEL SIHI 42 CUM/HR _ 1.5 BAR
33	1	EXP. TANK HYDR. TANK	ST	DRW. 3029_330_00_050
34				
35				

Remarks

- FOR CROSS SECTIONS ENGINE ROOM ARRANGEMENT SEE DRW. 3029_730_00_001
- ALL EXHAUST PIPES AND EXHAUST SILENCERS IN ENGINE ROOM TO BE LAGGED WITH 50 MM HEAT RESISTANT INSULATION
- ALL EXHAUST SILENCERS TO BE EXECUTED WITH PN 10 FLANGES (INCL. COUNTERFLANGES AND GASKETS)
- EACH EXHAUST PIPE LINE TO BE PROVIDED WITH TWO ½" SOCKETS (ART. NO 1180.0010) AS CLOSE AS POSSIBLE TO THE ENGINES
- FOR EXHAUST TEMP. MAINENGINES AND AUX SET SEE DRW. 185 _ 380_01_001
- SWIVEL SUPPORT
- FIXED POINT

rev.	date	name	revision subject
B	04_01_94	L.V.	ITEM NO 32 AND 33 ADDED ITEM NO 30 DELETED
A	19_11_'93	L.V.	UPDATED

inspected | date

DAMEN SHIPYARDS

Avelingen West 20
P.O. Box 1
4200 AA Gorinchem Holland

Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

Topview engine room arrangement

MULTI CAT 1908

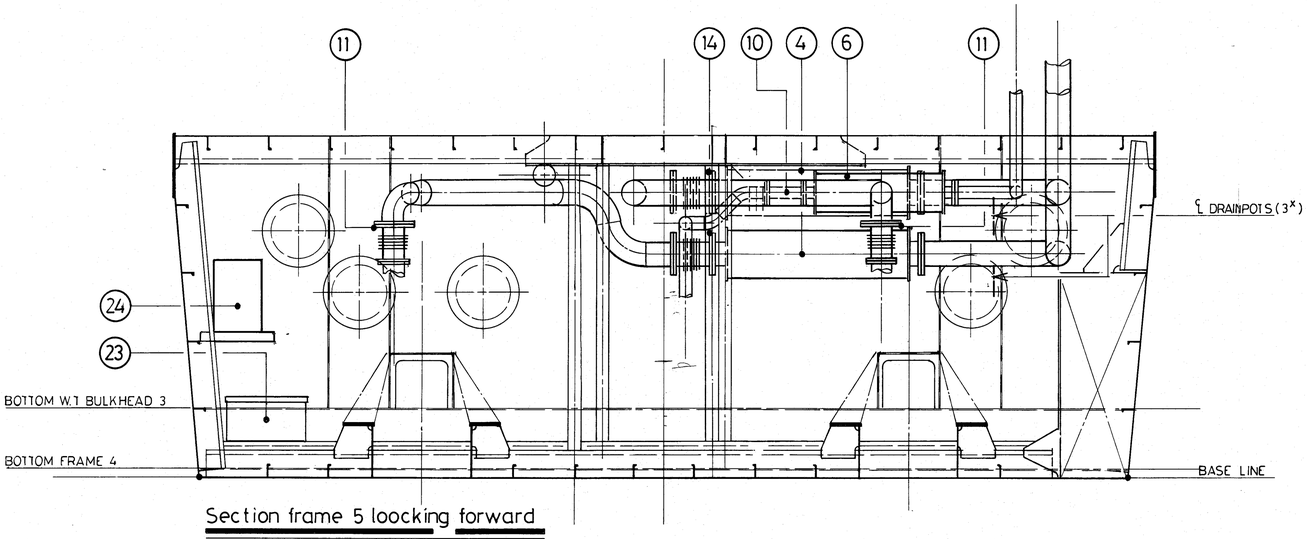
drawn	L.V.D.VLIET	date 09_11_'93	scale 1:25
checked		ident. to drwg. no.	
yard no.	3029 / 3030	derived fr. drwg. no.	

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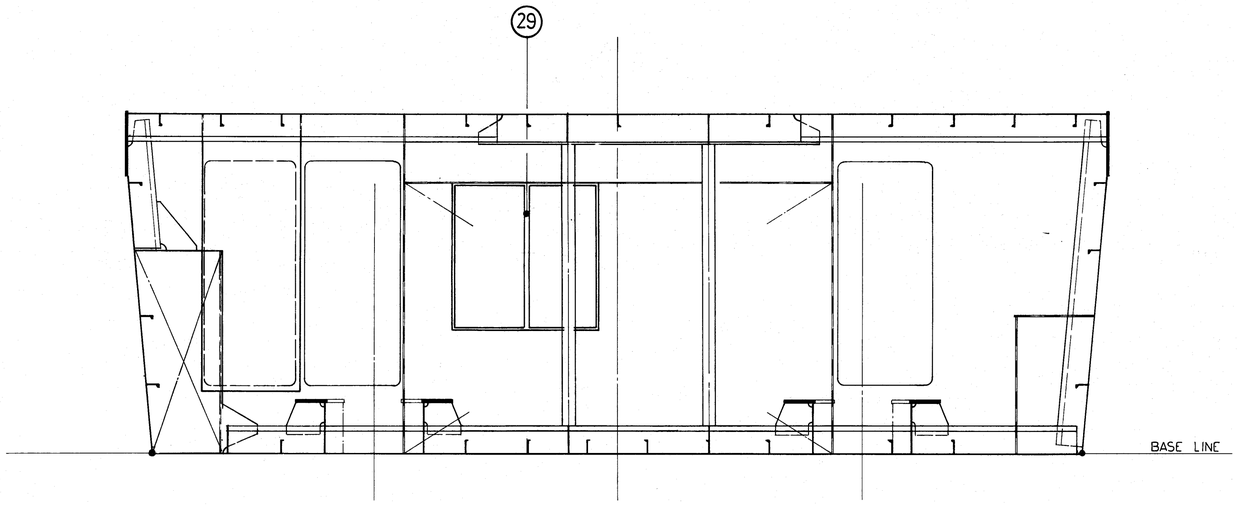
drwg. no. 3029_730_00_001

0cm 5 10cm

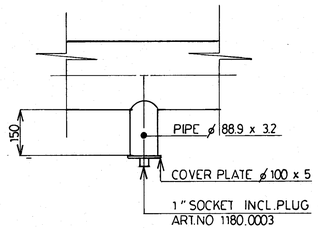
Section frame 5 loocking aft



Section frame 5 loocking forward



Detail drainpot (Typical)



Remarks

— FOR TOPVIEW ENGINE ROOM ARRANGEMENT SEE DRW. 3029_730_00_001

— FOR OTHER REMARKS SEE DRW. 3029_730_00_001

rev	date	name	revision subject
A	19_11_'93	L.V.	UPDATED

inspected	date

DAMEN SHIPYARDS

Avelingen West 20
P.O. Box 1
4200 AA Gorinchem/Holland

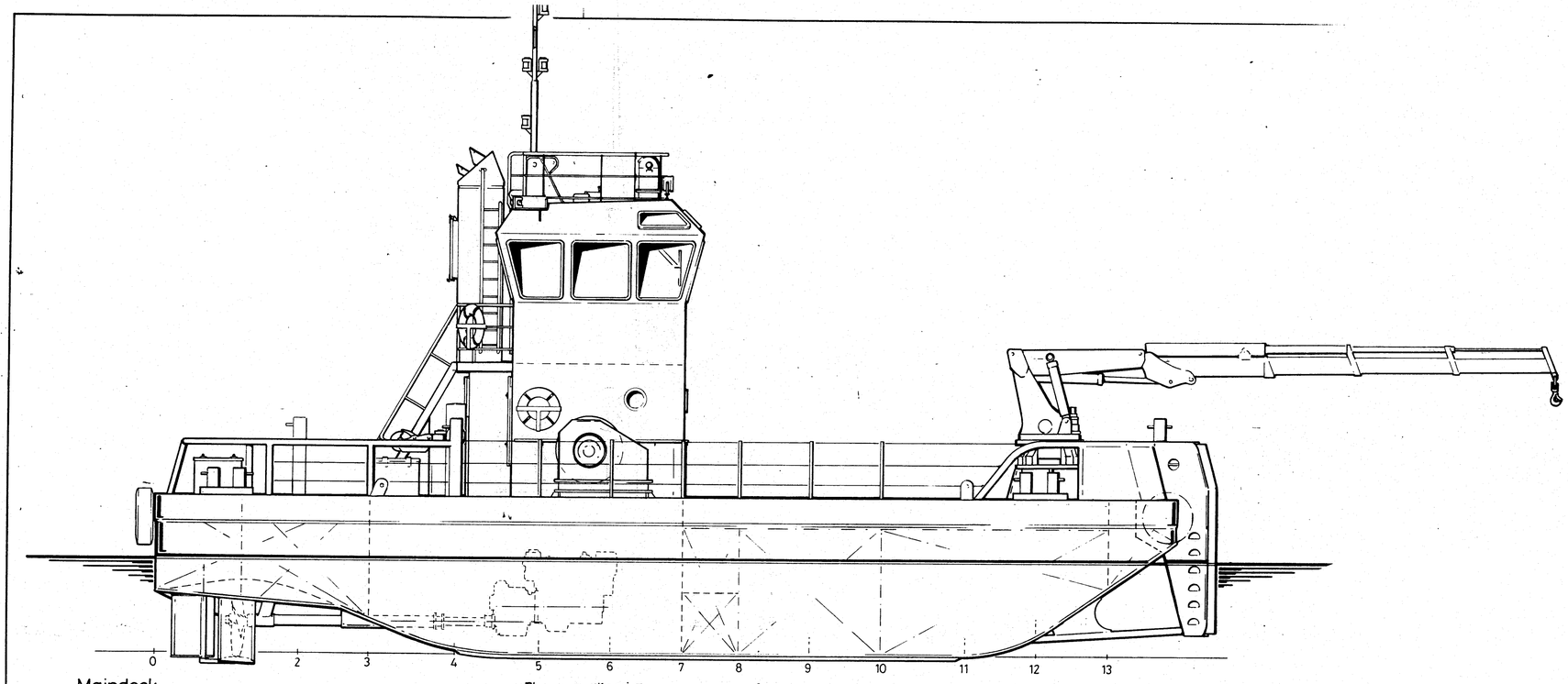
Tel.: 31.1830.39911*
Telex: 25335 damen nl
Telefax: 31.1830.32189

Cross sections engine room arrangement

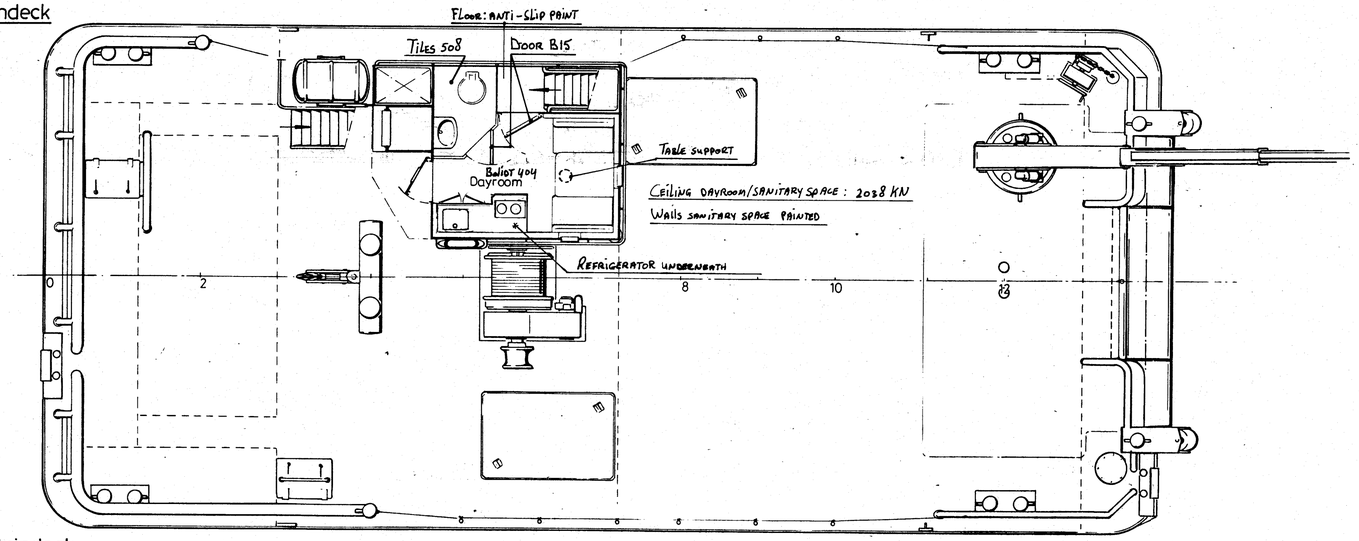
MULTI CAT 1908

drawn	L.V.D.VLIET	date 09_11_'93	scale 1 : 25
checked		ident. to drwg. no.	
yard no.	3029 / 3030	derived fr. drwg. no.	
COPYRIGHT BY DAMEN SHIPYARDS		drwg. no.	3029_730_00_002

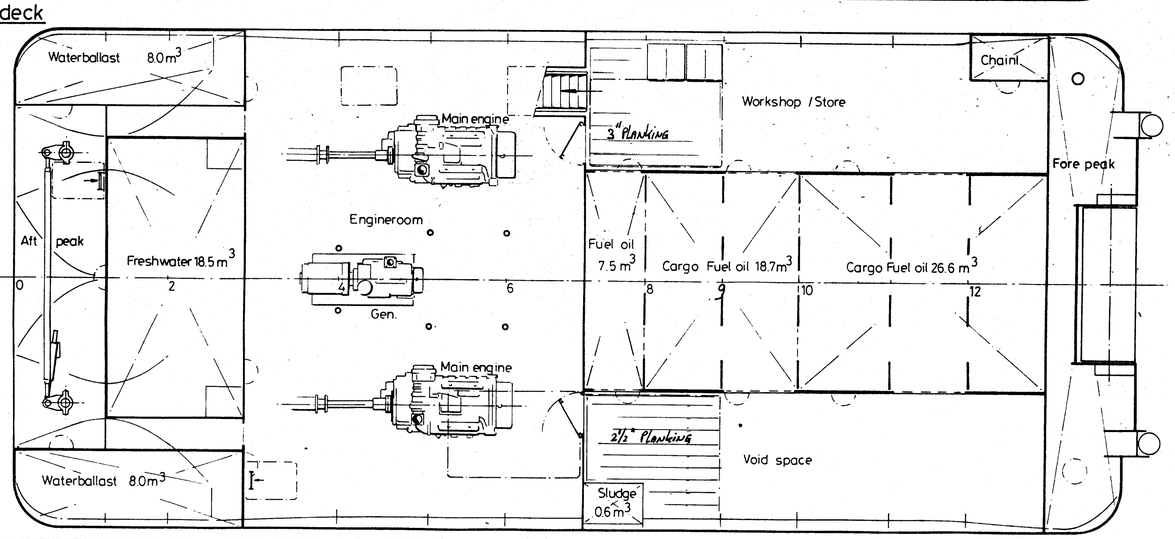
0cm 5 10cm



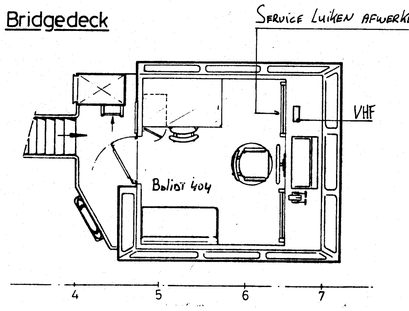
Maindeck



Below Maindeck



Bridgedeck



Scale
0 1 2 3 4 5 m.

SEE ALSO JOB LIST

inspected	date
rev.	date

DAMEN SHIPYARDS
Avelingen West 20
P.O. Box 1
4200 AA Gorinchem Holland
Tel 31 1830 39911
Telex 25335 Damen NL
Telefax 31 1830 32189

Arrangement accommodation / stores

date 6-1-'94	drawn EAL	checked	scale 1:50
ident. to drwg. no		derived fr. drwg. no	
yard no 3029/30		drwg. no 3029-750-00-001	

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